Single-Cell Foundation Models (scFMs): A 2025 Guide to Applications, Benchmarks, and Future Directions

Generated at: 2026-03-24
Source URL: https://www.cellbiotek.com/posts/singlecell-foundation-models-scfms-a-2025-guide-to-applications-benchmarks-and-future-directions

Single-Cell Foundation Models (scFMs): A 2025 Guide to Applications, Benchmarks, and Future Directions

Abstract

Single-cell foundation models (scFMs) represent a paradigm shift in computational biology, leveraging large-scale deep learning on massive single-cell datasets to create versatile tools for analyzing cellular heterogeneity. This article provides a comprehensive overview for researchers and drug development professionals, covering the core concepts of transformer-based architectures and tokenization strategies that allow these models to interpret the 'language of cells'. We explore their methodological applications in key tasks like cell type annotation, batch integration, and perturbation prediction, while critically addressing current limitations revealed by rigorous zero-shot evaluations. A detailed comparative analysis of leading models like scGPT, Geneformer, and scFoundation offers practical guidance for model selection, balancing performance across biological relevance, computational efficiency, and task-specific requirements. The article concludes by synthesizing the path toward more robust, interpretable, and clinically impactful scFMs, highlighting their potential to transform disease modeling and therapeutic development.

What Are Single-Cell Foundation Models? Decoding the AI Revolution in Cell Biology

The emergence of single-cell foundation models (scFMs) represents a paradigm shift in computational biology, drawing powerful analogies from natural language processing (NLP). The core conceptual framework—**cells as sentences and genes as words**—has enabled researchers to repurpose transformer-based architectures that have revolutionized artificial intelligence. This linguistic analogy provides a mathematical foundation for representing cellular identity and function, where individual cells constitute coherent "documents" composed of gene "vocabulary" arranged in specific patterns that convey biological meaning. The transcriptome of each cell can be viewed as a sentence, with the expression levels of approximately 20,000 human genes forming a rich vocabulary whose combinatorial patterns encode cellular states, functions, and developmental trajectories [1] [2].

This whitepaper explores the technical foundations, implementation challenges, and research applications of this core analogy within the rapidly evolving field of single-cell foundation models. By framing biological data through a linguistic lens, researchers can leverage sophisticated NLP techniques to uncover previously inaccessible relationships within complex biological systems. The conversion of single-cell RNA sequencing (scRNA-seq) data into a grammatical structure enables the application of self-supervised learning approaches that capture the fundamental "syntax" and "semantics" of gene regulation [2]. This approach has demonstrated remarkable success in diverse applications including cell type annotation, perturbation response prediction, and drug discovery, establishing scFMs as powerful tools for extracting biological insights from high-dimensional omics data.

The Analogical Framework: From Linguistics to Biology

Conceptual Mapping Between Language and Genomics

The cells-sentences/genes-words analogy establishes a precise mathematical correspondence between linguistic elements and biological components, enabling the direct application of transformer architectures to single-cell data. This mapping extends beyond superficial similarity to capture deep structural parallels in how information is encoded and processed in both domains.

Table: Conceptual Mapping Between Linguistic and Biological Domains

Linguistic Concept	Biological Equivalent	Computational Representation
Vocabulary	Gene repertoire	Dictionary of ~20,000 protein-coding genes
Words	Individual genes	Gene tokens with embedding vectors
Sentences	Individual cells	Ranked gene expression profiles
Documents	Cell populations or samples	Collections of single-cell measurements
Grammar	Gene regulatory programs	Patterns of gene co-expression
Semantics	Cellular identity and function	Biological meaning encoded in expression patterns

Linguistic Concept	Biological Equivalent	Computational Representation
Language modeling	Learning cellular states	Pre-training on scRNA-seq datasets

This analogical framework transforms how we conceptualize cellular identity, moving from static taxonomic classifications toward dynamic, context-dependent interpretations based on transcriptional "narratives." Just as words gain meaning from their contextual usage in sentences, genes derive functional significance from their expression patterns across cellular contexts [2]. A gene like TP53 may play dramatically different "semantic roles" in different cell types, analogous to how the word "bank" carries different meanings in different sentences. This contextual understanding enables scFMs to capture nuanced biological relationships that are obscured by traditional analytical approaches.

Technical Implementation of the Analogy

The practical implementation of the linguistic analogy requires solving several technical challenges, primarily centered on how to convert continuous gene expression values into discrete tokens suitable for transformer architectures. Unlike natural language with its inherently discrete vocabulary, gene expression presents as a continuous measurement that must be strategically discretized to create effective "sentences" for model input.

The leading approaches for this conversion include:

- **Rank-based tokenization** (employed by Geneformer and cell2sentence): Genes are sorted by expression level within each cell, creating an ordered sequence where positional information encodes expression magnitude [2]. This approach preserves relative expression relationships while normalizing for technical variation in sequencing depth.
- **Bin-based tokenization** (employed by scGPT): Expression values are discretized into bins representing different expression levels, creating a vocabulary that captures both gene identity and expression intensity [1]. This method preserves more quantitative information but increases vocabulary size.
- **Hybrid approaches** that combine gene identity with categorical expression levels (e.g., low, medium, high) to create composite tokens that capture both qualitative and quantitative information.

The embedding layer of scFMs must then represent these gene tokens in a continuous vector space where biological relationships can be captured through geometric relationships. Gene embeddings are typically initialized randomly and learned during pre-training, eventually positioning functionally related genes closer in the embedding space [1] [2]. For example, genes involved in oxidative phosphorylation naturally cluster together, while immune response genes form separate clusters, effectively creating a "semantic space" for biological function.

Technical Architectures of Single-Cell Foundation Models

Model Architectures and Tokenization Strategies

Current scFMs employ diverse architectural implementations of the core analogy, each with distinct advantages for specific biological applications. While all leverage transformer architectures, they differ significantly in their tokenization schemes, pre-training objectives, and fine-tuning approaches.

Table: Architectural Comparison of Major Single-Cell Foundation Models

Model	Architecture Type	Tokenization Strategy	Pre-training Dataset	Key Innovations
Geneformer	Encoder-only	Gene ranking by expression	30 million cells from mouse and human atlas	Rank-based attention mechanism; context-aware embeddings
scGPT	Encoder-decoder	Binned expression values	33 million cells from human cell atlas	Multi-task learning; perturbation prediction
cell2sentence (C2S)	Decoder-only	Natural language tokenization of gene ranks	57 million human and mouse cells + biological texts	Integration of scientific literature; biological knowledge grounding
scBERT	Encoder-only	Expression threshold-based	15 million cells from human immune atlas	BERT-style masked token prediction; cell type annotation focus
scFoundation	Encoder-only	Proportional expression encoding	50 million cells from multiple tissues	Scale-efficient attention; large-batch training

The transformer architecture processes these tokenized sequences through multiple layers of self-attention, enabling the model to learn context-dependent relationships between genes. In practice, this means the model can learn that genes A and B are co-expressed only in specific cellular

contexts, but not others—mirroring the way attention mechanisms in NLP capture how word meanings shift based on surrounding context [2]. The self-attention mechanism computes weighted sums of all genes in the "sentence" (cell), allowing the model to identify which genes are most relevant for understanding each particular gene's expression pattern in that specific cellular context.

Advanced Technical Implementation: Positional Encodings and Multi-Modal Extensions

Beyond basic tokenization, sophisticated implementations incorporate additional linguistic elements to enhance model performance. **Positional encodings** represent a particularly challenging aspect, as the natural ordering of genes in the genome may not reflect functional relationships. Some models use learned positional embeddings based on genomic coordinates, while others employ expression-level sorting that creates unique orderings for each cell [2].

The field is rapidly evolving toward **multi-modal models** that extend the linguistic analogy to incorporate multiple data types. Recent architectures like CAPTAIN and SCARF jointly model single-cell RNA and ATAC sequencing data, effectively creating "multilingual" models that can understand relationships across different omics languages [3]. These approaches tokenize different data types using modality-specific vocabularies while learning shared embedding spaces that capture complementary biological information.

For spatial transcriptomics, models like Nicheformer and SToFM incorporate spatial coordinates as additional "punctuation" in the cellular language, enabling the model to learn how physical proximity influences transcriptional patterns [3]. This represents a significant extension of the core analogy, adding geographical context to the linguistic framework.

Experimental Evaluation and Benchmarking

Comprehensive Benchmarking Framework

Evaluating the effectiveness of the cells-sentences analogy requires rigorous benchmarking across diverse biological tasks. A comprehensive 2025 study assessed six major scFMs against traditional baselines using twelve metrics spanning unsupervised, supervised, and knowledge-based approaches [1]. The evaluation framework encompassed two gene-level tasks (tissue specificity prediction and Gene Ontology term prediction) and four cell-level tasks (batch integration, cell type annotation, cancer cell identification, and drug sensitivity prediction) across multiple datasets with varying biological conditions and technical artifacts.

The benchmarking revealed that while scFMs are robust and versatile tools for diverse applications, no single model consistently outperforms others across all tasks [1] [4]. This emphasizes the need for task-specific model selection based on factors including dataset size, task complexity, and computational resources. The study introduced novel biology-driven evaluation metrics including scGraph-OntoRWR, which measures the consistency of cell type relationships captured by scFMs with prior biological knowledge, and the Lowest Common Ancestor Distance (LCAD) metric, which assesses the ontological proximity between misclassified cell types to evaluate the severity of annotation errors [1].

Quantitative Performance Analysis

Performance evaluation demonstrates that scFMs pre-trained using the linguistic analogy significantly outperform traditional methods on tasks requiring contextual understanding and transfer learning, while simpler baseline models remain competitive on dataset-specific tasks with limited data [1].

Table: Performance Comparison Across Cell-Level Tasks (Based on Genome Biology 2025 Benchmark)

Task Category	Best Performing scFM	Traditional Baseline	Performance Gap	Key Insights
Batch integration	scGPT	Harmony	+7.3% (kBET metric)	scFMs better preserve biological variation while removing technical artifacts
Cell type annotation	scBERT	Seurat + HVGs	+12.1% (accuracy)	Larger gains for rare cell types and cross-species annotation
Cancer cell identification	Geneformer	Random Forest	+15.7% (F1 score)	scFMs effectively capture subtle transcriptional shifts in malignancy
Drug sensitivity prediction	scGPT	Linear regression	+9.4% (Pearson correlation)	Improved generalization across cell lines and compounds
Perturbation response	cell2sentence	Linear baseline	+5.8% (MSE)	Context-aware prediction of combinatorial effects

The benchmarking results indicate that the primary advantage of scFMs emerges in scenarios requiring generalization across diverse cellular contexts, where their pre-training on massive datasets enables robust performance. However, for tasks with limited data or narrow experimental

conditions, traditional machine learning approaches often provide more efficient adaptation [1]. This suggests that the linguistic analogy provides the greatest value for exploratory analysis and hypothesis generation across diverse biological systems, while targeted analysis of specific experimental conditions may not always justify the computational overhead of large foundation models.

Experimental Protocols for Model Evaluation

Protocol for Gene Embedding Evaluation

Objective: Evaluate the biological relevance of gene embeddings learned by scFMs through the linguistic analogy.

Materials:

- Pre-trained scFM (e.g., Geneformer, scGPT, cell2sentence)
- Reference gene sets with known biological relationships (e.g., GO terms, KEGG pathways)
- Gene similarity benchmarks (e.g., tissue-specific co-expression networks)

Procedure:

- Extract gene embedding matrix from the input layer of the pre-trained scFM.
- Compute cosine similarity between all pairs of gene embeddings to construct model-derived gene relationship network.
- Retrieve known biological relationships from reference databases (GO, KEGG, Reactome).
- Calculate precision-recall curves for the ability of embedding similarities to recover known biological relationships.
- Compare against baseline methods (random embeddings, sequence-based embeddings, network-based embeddings).
- Perform functional enrichment analysis on gene clusters identified in the embedding space.

Validation Metrics:

- Area Under Precision-Recall Curve (AUPRC)
- Functional coherence of embedding neighborhoods
- Tissue specificity prediction accuracy
- Gene Ontology term prediction performance

This protocol revealed that scFM gene embeddings capture complementary biological information compared to sequence-based and network-based approaches, particularly excelling at context-specific functional relationships [1].

Protocol for Cell Type Annotation Benchmarking

Objective: Assess zero-shot cell embedding quality for cell type annotation across diverse tissues and species.

Materials:

- Pre-trained scFMs in zero-shot configuration (no fine-tuning)
- Reference annotated datasets with held-out cell types
- Traditional baselines (Seurat, SCANPY, scVI)
- Evaluation metrics (accuracy, F1 score, LCAD)

Procedure:

- Generate cell embeddings for target dataset using each scFM in zero-shot mode.
- For traditional methods, process data according to standard pipelines (HVG selection, normalization, PCA).
- Train identical classifier architectures (e.g., k-NN, random forest) on all embedding types.
- Evaluate performance using stratified cross-validation with held-out cell types.
- Calculate ontology-based metrics (LCAD) to assess biological meaningfulness of errors.
- Perform cross-dataset generalization tests to evaluate robustness.

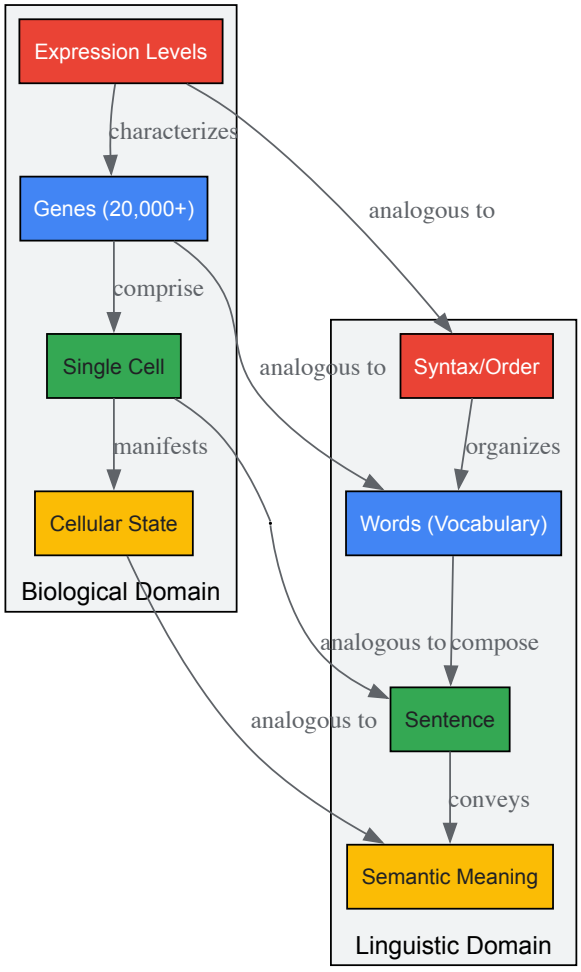
Validation Metrics:

- Cell type annotation accuracy
- Weighted F1 score
- Lowest Common Ancestor Distance (LCAD) for misclassifications
- Cross-dataset generalization performance

This protocol demonstrated that scFMs achieve superior performance for novel cell type identification and cross-species annotation, with errors that are biologically more plausible (closer in ontology space) [1].

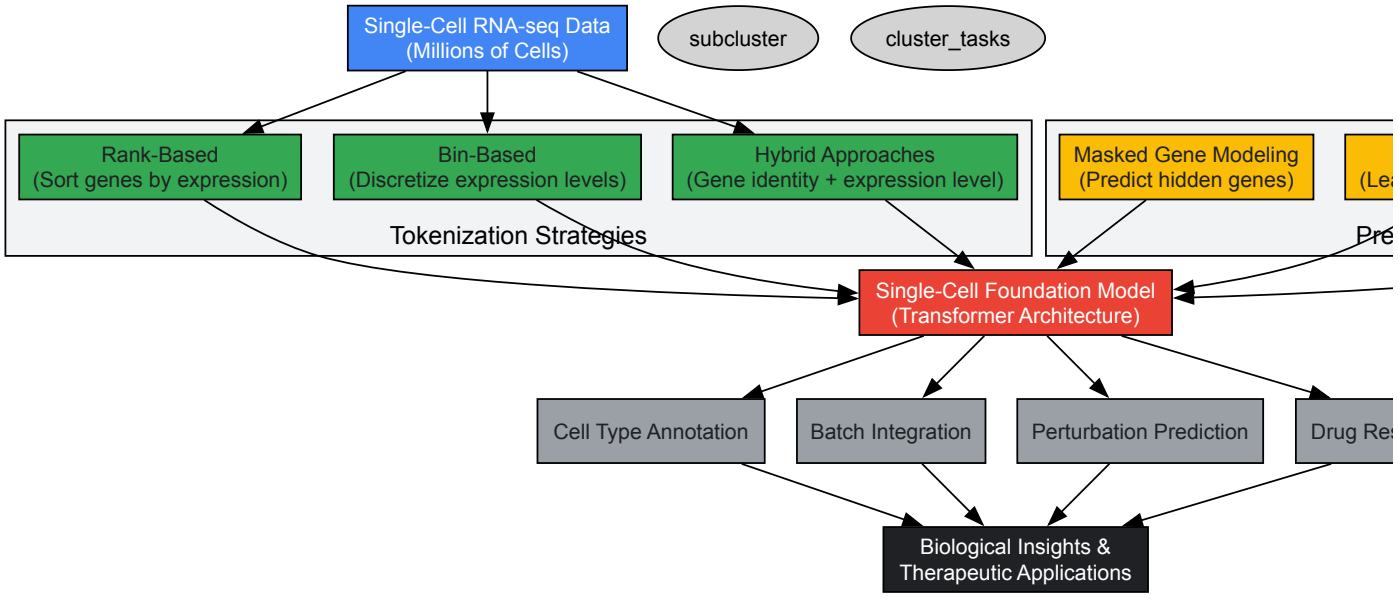
Visualization of Core Concepts and Workflows

The Linguistic Analogy in Single-Cell Foundation Models



The Core Linguistic Analogy in Single-Cell Biology

Single-Cell Foundation Model Training and Evaluation Workflow



scFM Training and Application Pipeline

The Scientist's Toolkit: Essential Research Reagents

Implementing and evaluating the cells-sentences analogy requires specialized computational tools and biological resources. The following table details essential components of the research pipeline for developing and applying single-cell foundation models.

Table: Essential Research Reagents and Computational Tools

Category	Resource	Specification	Application in scFM Research
Pre-training Data	CellXGene Atlas	~50M human cells across tissues	Large-scale pre-training corpus for learning fundamental biology
	Tabula Sapiens	500K cells across 24 human tissues	Cross-tissue reference for evaluating model generalization
	Asian Immune Diversity Atlas (AIDA)	Diverse population for bias assessment	Testing performance across genetic backgrounds [1]
Evaluation Datasets	Heart Cell Atlas v2	90K cardiac cells with detailed annotation	Benchmarking cell type annotation and rare population identification [2]
	Cancer Cell Atlas	1M+ cells across cancer types	Evaluating malignancy detection and tumor heterogeneity modeling
	Perturbation Datasets	CRISPR-based gene knockout screens	Testing perturbation response prediction accuracy
Software Libraries	Transformer Libraries (PyTorch, TensorFlow)	GPU-optimized deep learning frameworks	Model architecture implementation and training
	Single-Cell Toolkits (Scanpy, Seurat)	Standardized preprocessing pipelines	Data normalization, HVG selection, and baseline comparisons
	Mechanistic Interpretability Tools	Sparse autoencoders, transcoders	Circuit analysis and model dissection [2]
Evaluation Metrics	scGraph-OntoRWR	Cell ontology-informed metric	Assessing biological consistency of learned representations [1]
	Lowest Common Ancestor Distance (LCAD)	Ontological error severity measure	Evaluating biological plausibility of misclassifications [1]
	Roughness Index (ROGI)	Landscape smoothness quantification	Predicting model adaptability to new datasets [1]

Advanced Applications and Future Directions

Interpretability and Mechanistic Analysis

A significant challenge in scFMs is the "black box" nature of deep neural networks, which complicates biological interpretation. Recent advances in mechanistic interpretability, particularly transcoder-based circuit analysis, have enabled researchers to extract internal decision-making circuits from scFMs and map them to biologically plausible pathways [2].

Transcoders—sparse autoencoders trained to approximate transformer MLP layers—decompose model computations into interpretable components by resolving the polysemanticity problem where individual neurons encode multiple distinct biological concepts [2]. When applied to the cell2sentence model, transcoders successfully identified circuits corresponding to known biological pathways, demonstrating that scFMs internally organize knowledge in biologically meaningful ways despite being trained solely on expression data without explicit pathway annotation.

This approach enables researchers to move beyond correlative predictions to mechanistic understanding, tracing how information about specific gene expression patterns flows through the model to generate predictions about cellular state or function. For drug development applications, this interpretability is crucial for building confidence in model predictions and identifying potential therapeutic targets.

Emerging Frontiers and Clinical Translation

The linguistic analogy continues to evolve toward more sophisticated implementations, including multi-modal models that integrate transcriptomics with epigenomics, proteomics, and spatial information [3]. These "multilingual" models create unified representations that capture complementary

biological information, analogous to how multilingual language models learn shared representations across different natural languages.

Clinical translation represents the most promising frontier, with scFMs demonstrating remarkable performance in cancer cell identification, drug sensitivity prediction, and patient stratification [1]. By framing clinical questions in linguistic terms—e.g., "What is the transcriptional 'sentence' that distinguishes responsive from non-responsive tumors?"—researchers can leverage the full power of transformer architectures to address challenging biomedical problems.

Future developments will likely focus on scaling laws, efficiency improvements, and integration with emerging experimental technologies. As the field matures, the cells-sentences analogy promises to fundamentally transform how we extract meaning from the complex language of cellular biology, ultimately accelerating therapeutic development and precision medicine.

The emergence of single-cell foundation models (scFMs) represents a paradigm shift in computational biology, enabling researchers to decipher the complex language of cellular systems at unprecedented scale and resolution. This revolution is powered by an unexpected architectural backbone: the transformer network. Originally developed for natural language processing (NLP) in the landmark 2017 paper "Attention Is All You Need" [5], transformer architecture has become the fundamental engine driving modern single-cell AI research. Unlike traditional analysis methods that process data sequentially, transformers employ self-attention mechanisms to simultaneously analyze entire sets of genomic features, capturing complex relationships across thousands of genes and millions of cells [6]. This capability has positioned scFMs as indispensable tools for researchers and drug development professionals seeking to unravel cellular heterogeneity, identify novel cell types, and understand disease mechanisms at single-cell resolution.

The adaptation of transformer architecture to biological data represents one of the most significant computational advancements in single-cell genomics. By treating individual cells as "sentences" and genes or genomic features as "words," researchers have successfully applied transformer-based models to massive single-cell transcriptomics datasets, creating systems that learn fundamental biological principles generalizable to new datasets and downstream tasks [6]. This technical guide explores the core architectural components of transformer networks, their implementation in single-cell foundation models, and the experimental protocols validating their performance in biological and clinical contexts.

Core Architectural Components of Transformer Networks

Self-Attention Mechanism: The Fundamental Innovation

The self-attention mechanism represents the foundational innovation that distinguishes transformers from previous neural architectures. Unlike recurrent neural networks (RNNs) that process sequential data word-by-word, self-attention enables the model to examine all elements of a sequence simultaneously and determine how each element relates to every other element [5]. In biological terms, this allows a transformer-based scFM to understand not just individual gene expressions, but the complex web of interactions and dependencies between them.

The mathematical implementation of self-attention involves three critical components for each input element: the Query (Q), Key (K), and Value (V) vectors. These are created through linear transformations of the input embeddings [7]. The mechanism computes attention scores by taking the dot product of the query vector of one element with the key vectors of all elements in the sequence, followed by scaling and softmax normalization to create a probability distribution. The output is a weighted sum of value vectors, where weights are determined by these attention scores [7]. The complete calculation is expressed as:

$$\mathbf{Attention(Q, K, V) = softmax((Q \times K^T) / \sqrt{d_k}) \times V}$$

where d_k represents the dimension of the key vectors, and the scaling factor $\sqrt{d_k}$ prevents gradient vanishing issues during training [7].

Table: Self-Attention Components and Their Biological Interpretations in scFMs

Component	Technical Function	Biological Interpretation in scFMs
Query (Q)	Represents the "question" being asked about a specific position	What biological state or function does this gene help define?
Key (K)	Represents what each element "offers" or "contains"	What biological processes is this gene involved in?
Value (V)	Represents the actual content to be weighted and summed	The specific expression pattern and functional impact of the gene
Attention Weights	Determine how much focus to place on other elements	The strength of functional relationship or co-regulation between genes

Multi-Head Attention: Multi-Perspective Biological Analysis

Transformers enhance this basic attention mechanism through multi-head attention, which allows the model to simultaneously attend to information from different representation subspaces [5] [7]. In practical terms, each attention "head" can learn to focus on different types of biological

relationships—some heads might specialize in identifying cell-type specific gene programs, while others might focus on stress response pathways, metabolic processes, or signaling cascades [6]. The outputs of all attention heads are concatenated and linearly transformed to produce the final multi-head attention output.

For scFMs, this multi-head capability is particularly valuable for capturing the multifaceted nature of biological systems. A gene can participate in multiple pathways and processes simultaneously, and multi-head attention provides the architectural capacity to represent these complex, overlapping biological functions [6]. For example, in analyzing tumor microenvironments, different attention heads might independently focus on immune cell signatures, stromal interactions, and malignant cell characteristics, together providing a comprehensive view of the tumor ecosystem.

Positional Encoding: Addressing the Non-Sequential Nature of Genomics Data

A significant challenge in applying transformers to biological data is that gene expression data lacks natural sequential ordering—unlike words in a sentence, genes have no inherent positional relationship [6]. To address this, researchers have developed various positional encoding strategies specifically for single-cell data. The original transformer architecture used sinusoidal functions of different frequencies to encode position information [7], but scFMs have adopted more biologically-relevant approaches.

Common strategies include ranking genes within each cell by expression levels and using this ordered list as the input "sentence" [6]. Other models partition genes into bins based on expression values or simply use normalized counts with learned positional embeddings [6]. These approaches create a deterministic structure that enables the transformer to process the non-sequential genomic data effectively while preserving the model's ability to capture gene-gene interactions regardless of their positional encoding.

Feed-Forward Networks and Layer Normalization

Beyond attention mechanisms, transformers incorporate position-wise feed-forward networks (FFNs) that apply identical fully connected layers to each position separately [8]. Recent research has revealed that in biological applications, these FFNs play a crucial role in maintaining the diversity of cell representations, preventing the collapse of distinct cell types into a single embedding space—a phenomenon known as representation collapse [8].

Each sub-layer (both self-attention and FFN) in the transformer is surrounded by residual connections and followed by layer normalization, which stabilizes training and enables deeper networks [7]. This "Add & Norm" approach allows gradients to flow more effectively through the network during training and has proven essential for scaling transformers to the sizes needed for effective foundation models in biology.

Transformer Adaptation to Single-Cell Foundation Models

Architectural Variations in scFMs

Single-cell foundation models have adapted the core transformer architecture in several specialized ways to address the unique challenges of biological data. The two primary architectural approaches are encoder-based models (inspired by BERT) and decoder-based models (inspired by GPT), each with distinct advantages for biological analysis.

Encoder-based models like scBERT employ bidirectional attention, meaning they can process all genes in a cell simultaneously and understand each gene in the context of all other genes [6]. This approach is particularly valuable for classification tasks such as cell type annotation, where comprehensive context leads to more accurate predictions. Decoder-based models like scGPT use a unidirectional masked self-attention mechanism that iteratively predicts masked genes conditioned on known genes [6]. This architecture excels at generative tasks and perturbation prediction, where the goal is to forecast cellular responses to genetic or environmental changes.

Table: Transformer Architectural Variants in Single-Cell Foundation Models

Model Type	Representative Examples	Key Characteristics	Ideal Biological Applications
Encoder-based	scBERT, UCE	Bidirectional attention; processes all inputs simultaneously	Cell type annotation, batch integration, knowledge extraction
Decoder-based	scGPT, Geneformer	Masked self-attention; generative capabilities	Perturbation prediction, hypothesis generation, trajectory inference
Hybrid Architectures	scFoundation, scCello	Combine encoder and decoder components; custom modifications	Multi-task learning, complex predictive tasks requiring both encoding and decoding

Tokenization Strategies for Single-Cell Data

Tokenization—the process of converting raw biological data into discrete units processable by transformer models—represents a critical design decision in scFM development. Unlike NLP where tokens are naturally occurring words, scFMs must define what constitutes a "token" from single-cell omics data [6]. The most common approach treats individual genes as tokens, with their expression values incorporated into the token representation.

Advanced tokenization strategies may include special tokens representing cell-level metadata, experimental conditions, or batch information [6]. Multi-modal scFMs incorporate tokens indicating different data modalities (e.g., RNA expression, ATAC accessibility, protein abundance) to create unified representations across measurement types. Some models additionally incorporate gene metadata such as Gene Ontology terms or chromosomal locations to provide richer biological context [6].

Pretraining Strategies and Objectives

The power of scFMs emerges from their pretraining on massive, diverse single-cell datasets—often encompassing tens of millions of cells from various tissues, species, and experimental conditions [6] [1]. During pretraining, models learn through self-supervised objectives, most commonly through masked language modeling approaches where random portions of the input gene expression profile are masked and the model must predict the missing values based on context [6].

This pretraining enables scFMs to develop a fundamental understanding of cellular biology that can be transferred to various downstream tasks with minimal task-specific training. The scale of pretraining corpora is crucial—models trained on larger, more diverse datasets generally demonstrate better performance across multiple applications, highlighting the importance of data diversity and volume in building effective biological foundation models [1].

Experimental Validation and Benchmarking

Performance Across Biological Tasks

Recent comprehensive benchmarking studies have evaluated scFMs across diverse biological tasks to assess their capabilities and limitations. These evaluations typically compare multiple scFMs against traditional computational methods under realistic conditions. A landmark 2025 benchmark study evaluated six prominent scFMs against established baselines across two gene-level and four cell-level tasks [1].

The findings revealed that while scFMs are robust and versatile tools for diverse applications, no single model consistently outperforms others across all tasks [4] [1]. This emphasizes the importance of task-specific model selection rather than seeking a universal best model. The study also found that simpler machine learning models can sometimes outperform complex foundation models, particularly in scenarios with limited data or computational resources [4] [1].

Novel Evaluation Metrics for Biological Relevance

Traditional computational metrics alone are insufficient for evaluating scFMs, as they may not capture biologically meaningful patterns. To address this limitation, researchers have developed novel evaluation approaches specifically designed to assess the biological relevance of model outputs [1]. These include:

- **scGraph-OntoRWR**: Measures the consistency between cell type relationships captured by scFMs and established biological knowledge from cell ontologies [1].
- **Lowest Common Ancestor Distance (LCAD)**: Assesses the severity of cell type annotation errors by measuring the ontological proximity between misclassified cell types [1].
- **Roughness Index (ROGI)**: Quantifies the smoothness of cell-property landscapes in the latent space, with smoother landscapes generally indicating better model performance [1].

These biologically-grounded metrics provide crucial insights beyond traditional performance measures, ensuring that scFMs capture scientifically valid patterns rather than merely optimizing mathematical objectives.

Table: Benchmark Performance of scFMs Across Key Biological Tasks

Task Category	Specific Tasks	Top-Performing Models	Key Findings
Gene-Level Tasks	Tissue specificity prediction; GO term prediction	scGPT, Geneformer	Functionally similar genes cluster in embedding space; models capture known biological relationships
Cell-Level Tasks	Batch integration; cell type annotation; cancer cell identification	scBERT, UCE, scFoundation	Effective batch correction while preserving biological variation; accurate annotation of novel cell types

Task Category	Specific Tasks	Top-Performing Models	Key Findings
Clinical Applications	Drug sensitivity prediction; treatment response modeling	scGPT, LangCell	Predictive of patient-specific drug responses; potential for personalized treatment strategies

Implementation and Practical Considerations

Computational Ecosystems and Tools

The implementation of transformer-based scFMs relies on several well-established computational ecosystems. The three primary frameworks for single-cell analysis include Seurat (R-based), Bioconductor (R-based), and scverse (Python-based) [9]. Each ecosystem offers distinct advantages, with selection often depending on researcher preference, existing infrastructure, and specific analytical needs.

Seurat provides a comprehensive toolkit for single-cell analysis with extensive documentation and regular updates, making it particularly accessible for researchers new to computational biology [9]. The Bioconductor ecosystem offers highly interoperable packages following consistent design principles, while scverse—centered around scanpy—provides scalability and strong interoperability for Python users [9]. Each of these ecosystems supports the implementation and fine-tuning of transformer-based scFMs, with varying levels of customization and computational efficiency.

The Scientist's Toolkit: Essential Research Reagents

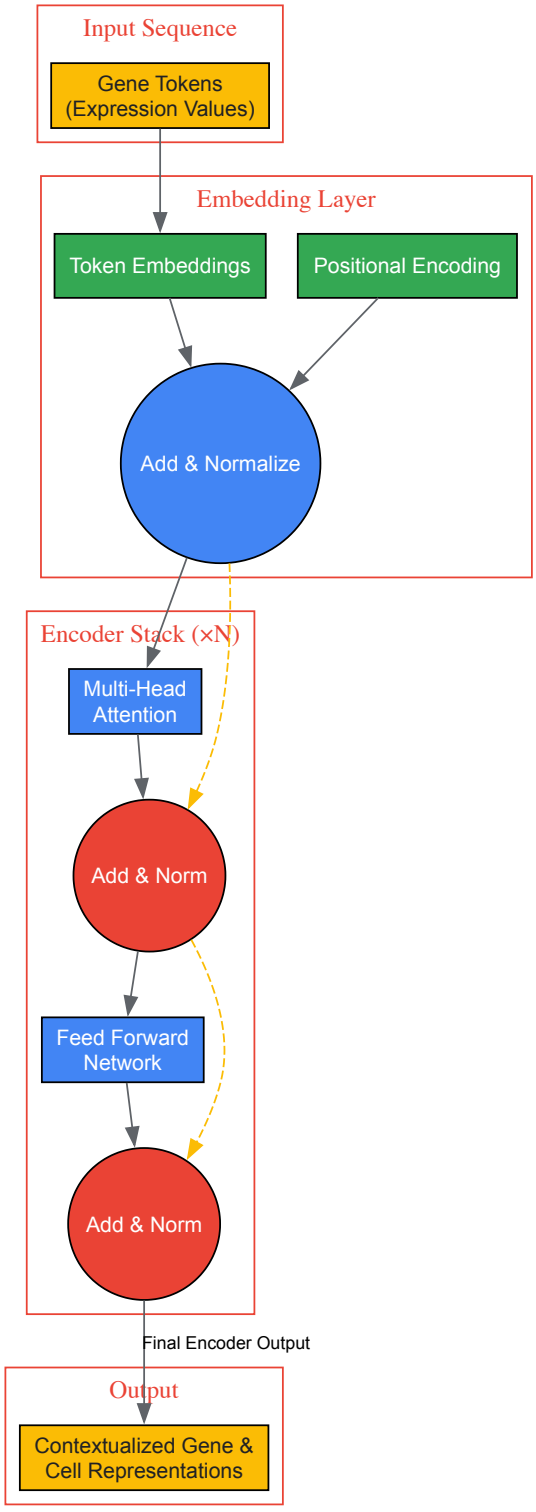
Implementing and applying scFMs requires both computational tools and biological data resources. The following table outlines key components of the modern computational biologist's toolkit for transformer-based single-cell analysis.

Table: Essential Research Reagents and Computational Tools for scFM Research

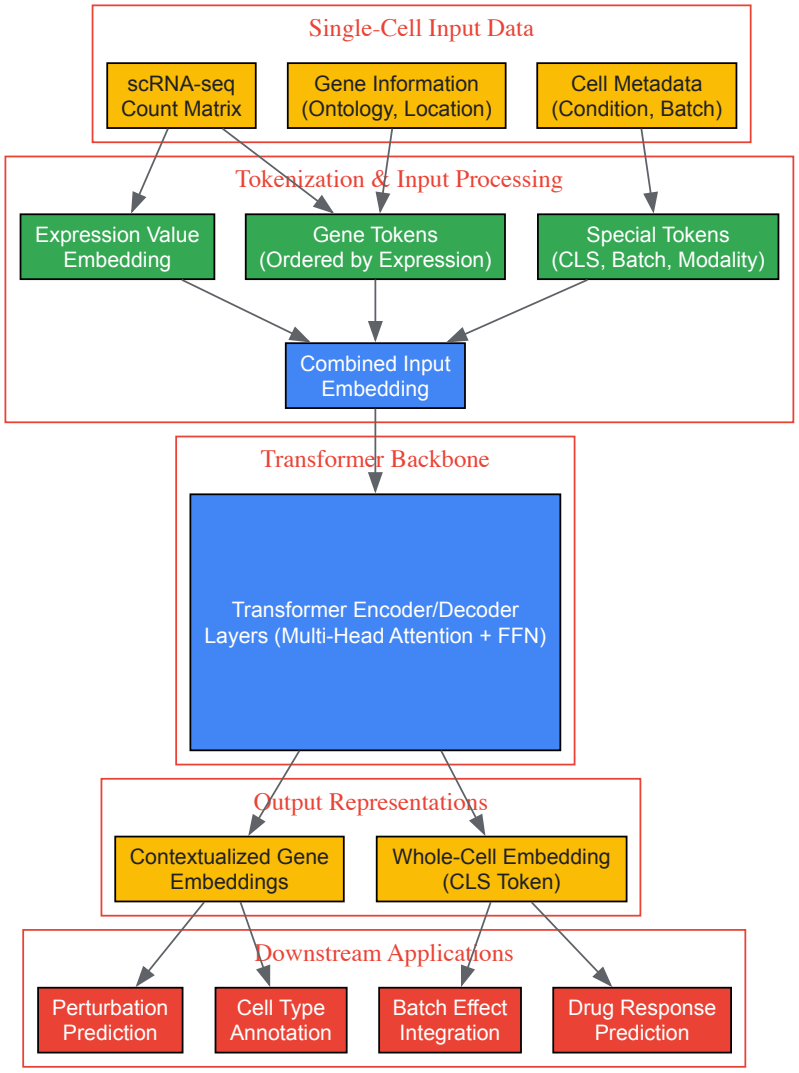
Resource Category	Specific Tools/Resources	Function and Application
Analysis Ecosystems	Seurat, Bioconductor, scverse (scanpy)	Primary computational frameworks for implementing scFMs and conducting downstream analysis
Data Resources	CZ CELLxGENE, Human Cell Atlas, PanglaoDB	Curated single-cell datasets for model training, fine-tuning, and validation
Pretrained Models	scGPT, scBERT, Geneformer, scFoundation	Foundation models that can be adapted to specific research questions without pretraining from scratch
Benchmarking Tools	scGraph-OntoRWR, LCAD metrics, ROGI	Specialized metrics for evaluating model performance and biological relevance
Visualization Tools	UMAP, t-SNE, custom attention visualizers	Methods for interpreting model outputs and understanding biological patterns

Architectural Visualizations

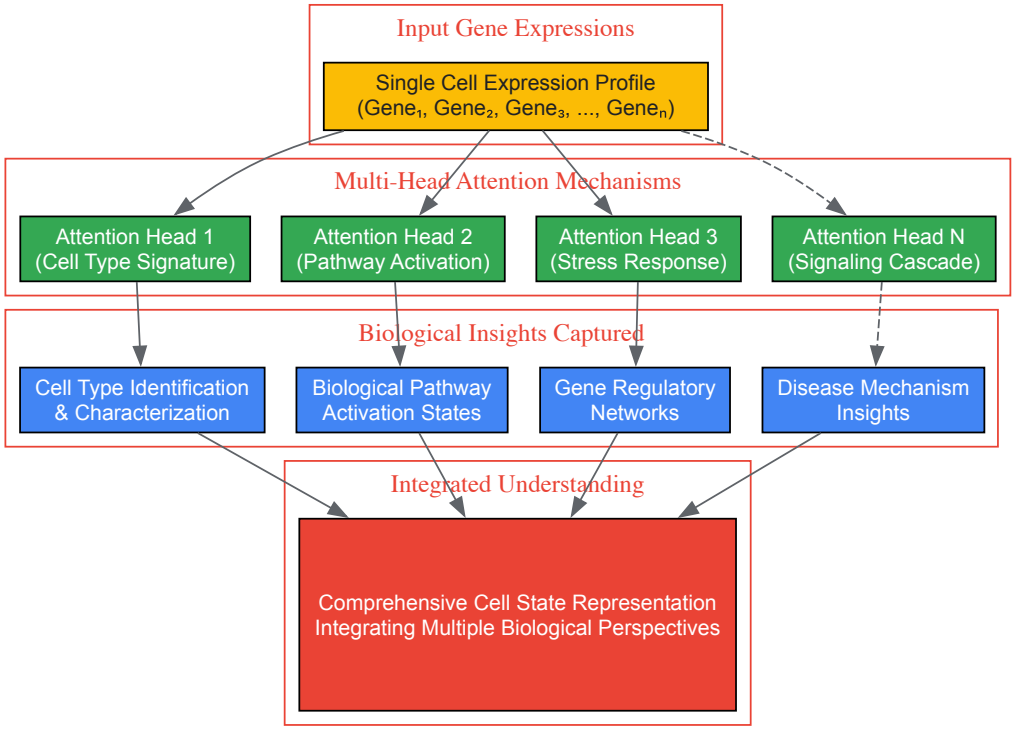
Core Transformer Architecture



Single-Cell Foundation Model Adaptation



Multi-Head Attention in Biological Context



The integration of transformer networks into single-cell biology represents a fundamental shift in how researchers approach biological data analysis. The attention mechanism's capacity to capture complex, long-range dependencies in genomic data has enabled the development of foundation models that learn generalizable biological principles from massive datasets [6]. As these models continue to evolve, several promising directions emerge for future development.

Future scFMs will likely incorporate more diverse data modalities—including spatial transcriptomics, proteomics, and epigenetics—to create more comprehensive representations of cellular states [6]. Architectural innovations such as the Parallel Attention and Feed-Forward Net (PAF) design may improve model efficiency and performance [8]. Additionally, enhanced interpretability methods will be crucial for extracting biologically meaningful insights from these complex models and building trust within the research community.

For researchers and drug development professionals, transformer-based scFMs offer powerful new approaches to understanding disease mechanisms, identifying novel therapeutic targets, and predicting treatment responses. However, successful implementation requires careful consideration of task requirements, data characteristics, and computational resources, as no single model architecture dominates all applications [4] [1]. As the field matures, transformer networks will undoubtedly remain the key architectural backbone enabling increasingly sophisticated analysis of single-cell data and accelerating discoveries in biomedical research.

The emergence of single-cell foundation models (scFMs) represents a paradigm shift in computational biology, enabling unprecedented analysis of cellular heterogeneity and function. These large-scale deep learning models, pretrained on vast single-cell datasets, leverage self-supervised learning to develop generalized representations that can be adapted to diverse downstream tasks including cell type annotation, perturbation prediction, and disease mechanism investigation [10]. The performance and generalizability of scFMs are fundamentally constrained by the quality, scale, and diversity of their pretraining data. Consequently, major biological data sources have become critical infrastructure for advancing this field, with platforms like CZ CELLxGENE, the Human Cell Atlas, and public repositories providing the essential raw material for model development [10] [1]. This technical guide examines the core data sources powering scFM research, providing detailed quantitative comparisons, standardized access protocols, and practical frameworks for their utilization in model training and validation.

Table 1: Core Characteristics of Major scFM Pretraining Data Sources

Data Source	Primary Content	Scale (Cells)	Key Organisms	Data Format	Access Method
CZ CELLxGENE Discover	Curated single-cell transcriptomics	93.6M+ unique cells (93.6M human, 16M mouse) [11] [12]	Human, Mouse	AnnData (h5ad), TileDB-SOMA	GUI, REST API, Census API
Human Cell Atlas (HCA)	Multi-omic single-cell data	63.3M+ cells [13]	Human	Loom, H5AD, Matrix	Data Portal, AWS S3, Azul API
GEO/SRA	Heterogeneous omics data	Variable (4,000+ scRNA-seq datasets) [14] [15]	Multiple	FASTQ, Count Matrices	Web interface, SRA Toolkit, eUtils
Single Cell Expression Atlas	Annotated scRNA-seq	Variable	Multiple	Expression Matrices	Web portal, REST API
Single Cell Portal	Analyzed single-cell data	Variable	Human, Mouse	H5AD, LOOM	Web interface, Download

Biological and Technical Coverage

Table 2: Biological Context and Technical Metadata Coverage

Data Source	Tissues/Cell Types	Disease States	Experimental Factors	Standardization Level	Metadata Richness
CZ CELLxGENE	Comprehensive (50+ tissues) [12]	Healthy, Disease, Treatment [11]	Age, Sex, Ancestry, Protocol [12]	High (minimal schema + ontologies) [12]	High (11 required fields + extensibility) [12]
HCA	Organ-focused atlases [13]	Primarily healthy reference	Developmental stage, Tissue origin	Medium (project-specific standards)	Variable (consortium-dependent)
GEO/SRA	Extremely diverse	Highly diverse	Highly diverse	Low (investigator-defined)	Highly variable
Single Cell Expression Atlas	Tissue-focused	Healthy vs. Disease comparisons	Experimental conditions	Medium (curated baseline/differential)	Standardized experimental factors
Allen Brain Cell Atlas	Brain regions	Healthy, Some disease	Brain region, Cell class	High (standardized taxonomy)	Consistent hierarchical annotations

Technical Architectures and Data Models

Data Standardization and Schema Implementation

A critical differentiator among data sources is their approach to standardization. CZ CELLxGENE enforces a **minimal schema** with 11 required fields curated using established ontologies, ensuring interoperability across datasets [12]. This schema encompasses essential biological covariates strongly correlated with gene expression variation, including organism, sex, tissue, cell type, and assay type, all validated against community ontologies such as Cell Ontology (CL), Uberon, and Experimental Factor Ontology (EFO) [12]. The platform employs a **collaborative curation model** where curators work directly with data contributors during submission rather than retrospectively interpreting metadata, ensuring accurate representation and avoiding ambiguous interpretations [12].

In contrast, the Human Cell Atlas operates through a **federated model** where individual consortia maintain their data standards while adhering to overarching HCA metadata frameworks. This balances flexibility with sufficient standardization for cross-project integration [13] [16]. Public repositories like GEO and SRA impose minimal standardization, resulting in heterogeneous metadata quality that necessitates extensive preprocessing before use in scFM training [14] [15].

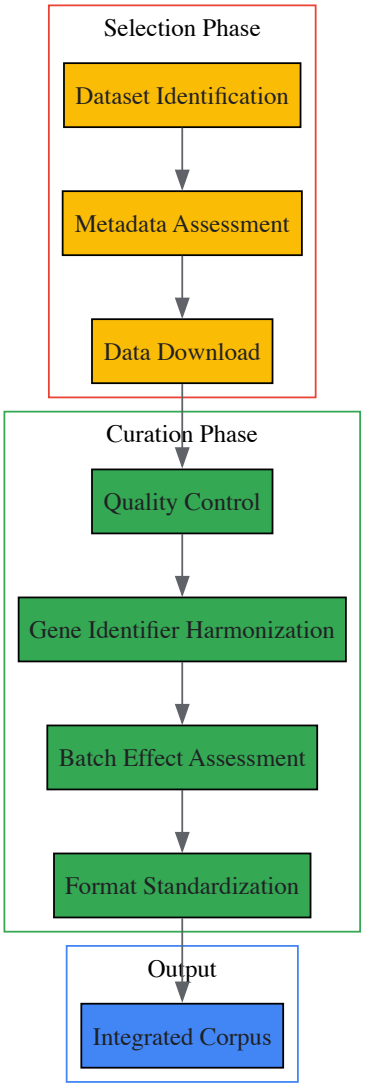
Data Access Architectures and Computational Interfaces

Table 3: Computational Access Methods and Infrastructure

Data Source	Primary Access Methods	Computational Interfaces	Bulk Download Options	Cloud Integration
CZ CELLxGENE	GUI, REST API, Census API [11] [12]	Python (cellxgene_census), R	Partial dataset download	Hosted on CZI infrastructure
HCA	Data Portal, Azul API, AWS S3 [16]	CLI, HCA-CLI, DCP Client	Full project downloads	AWS Public Dataset Program
GEO/SRA	Web interface, e-utilities, SRA Toolkit [15]	Programmatic via e-utils, SRA Toolkit	Study-level downloads	NCBI cloud resources
Single Cell Portal	Web interface, direct download [14]	Manual download with subsequent processing	Dataset-level downloads	Limited cloud integration

Experimental Protocols for scFM Pretraining Data Curation

Standardized Data Retrieval and Processing Workflow



Protocol 1: CELLxGENE Census API Integration

Purpose: Programmatic access to standardized single-cell data for large-scale scFM pretraining.

Materials:

- CELLxGENE Census Python package
- Computational environment with ≥16GB RAM
- TileDB-SOMA compatibility layer

Procedure:

- **Environment Setup**:
- **Data Corpus Discovery**:
- **Filtered Data Extraction**:
- **AnnData Export for Model Training**:
- **Quality Control Metrics**:
  - Calculate cells with >500 genes detected
  - Exclude cells with >20% mitochondrial reads
  - Verify concordance with original publication metrics

Validation: Cross-reference cell type annotations with independent sources using marker gene expression profiles.

Protocol 2: HCA Data Integration via Azul API

Purpose: Aggregation of multi-project data from Human Cell Atlas for specialized tissue-specific scFMs.

Materials:

- HCA Azul CLI or Python client
- AWS S3 access credentials
- Computational storage for terabyte-scale data

Procedure:

- **Project Catalog Query**:
- **Metadata Harmonization**:
  - Map project-specific terms to Cell Ontology
  - Standardize developmental stage annotations
  - Resolve protocol method discrepancies
- **Bulk File Manifest Generation**:
- **Distributed Download**:
- **Matrix Consolidation**:
  - Convert Loom to AnnData format
  - Apply consistent gene annotation (ENSEMBL v105)
  - Resolve duplicate cell barcodes across projects

Validation: Assess integration quality using dataset-specific mixing metrics and biological conservation scores.

Protocol 3: GEO/SRA Bulk Processing Pipeline

Purpose: Mining heterogeneous public repositories for maximal pretraining data diversity.

Materials:

- SRA Toolkit (prefetch, fasterq-dump)
- Computational cluster environment
- Bulk RNA-seq processing pipeline (ALEIGN or similar)

Procedure:

- **Study Identification**:
  - GEO Advanced Search: "Expression profiling by high throughput sequencing" + "single cell" [15]
  - SRA Strategy filter: "RNA-Seq" + Source: "Transcriptomic single cell" [15]
- **Metadata Extraction**:
  - Download SOFT format files from GEO
  - Parse sample attributes with custom parsers
  - Map to schema-compliant structure
- **Raw Data Processing**:
- **Uniform Alignment and Quantification**:
  - Implement STARsolo or CellRanger pipeline
  - Apply consistent reference genome (GRCh38)
  - Generate count matrices with identical gene sets
- **Quality Harmonization**:
  - Apply uniform QC thresholds across studies
  - Remove datasets with poor mapping rates (<60%)
  - Exclude studies with insufficient metadata

Validation: Compare clustering results with original publications to verify processing fidelity.

The Scientist's Toolkit: Essential Research Reagents

Table 4: Critical Computational Tools and Platforms for scFM Data Curation

Tool/Platform	Primary Function	Application in scFM Research	Access Method
CELLxGENE Census	Standardized data access	Programmatic retrieval of curated single-cell data for pretraining	Python API (cellxgene_census)
SRA Toolkit	Sequence data management	Bulk download and processing of raw sequencing data from public repositories	Command-line interface
Scanpy	Single-cell analysis	Data preprocessing, quality control, and integration of multiple datasets	Python library
TileDB-SOMA	Sparse array storage	Efficient storage and querying of massive single-cell datasets	Computational backend
Azul API	HCA metadata search	Project discovery and manifest generation for HCA data retrieval	REST API
CellTypist	Automated cell annotation	Validation and standardization of cell type labels across datasets	Python model inference
scArches	Reference mapping	Integration of new data with existing references for continuous pretraining	Python package
OmicsPlayground	Comparative analysis	Benchmarking scFM performance against traditional methods	Web interface or platform

Downstream Applications and Validation Frameworks

Benchmarking scFM Performance on Biological Tasks

The true test of pretraining data quality emerges in downstream applications. Recent benchmarking studies evaluate scFMs across diverse tasks including batch integration, cell type annotation, cancer cell identification, and drug sensitivity prediction [1]. These evaluations employ biologically-informed metrics such as scGraph-OntoRWR, which measures consistency between model-derived cell relationships and established biological knowledge encoded in ontologies [1]. The **Lowest Common Ancestor Distance (LCAD)** metric quantifies the severity of cell type misclassification by measuring ontological proximity between predicted and actual cell types, providing more nuanced evaluation than simple accuracy [1].

Data Quality Assessment Metrics

Table 5: scFM Pretraining Data Quality Evaluation Framework

Metric Category	Specific Metrics	Target Threshold	Evaluation Method
Technical Quality	Median genes per cell, Mitochondrial percentage, Doublet score	>500 genes/cell, <20% MT reads, Doublets <5%	Cell-level QC filtering
Biological Coverage	Cell type diversity, Tissue representation, Donor heterogeneity	Balanced organ representation, Multiple biological conditions	Metadata analysis and clustering
Annotation Quality	Ontology compliance, Marker gene concordance, Manual validation rate	100% CL ontology compliance, Marker AUC >0.7	Cross-reference with independent atlases
Integration Potential	Batch effect severity, Integration LISI score, Biological conservation	LISI >0.7, Cell type purity >80%	Benchmarking with Harmony/Seurat

Future Perspectives and Emerging Data Paradigms

The trajectory of scFM development points toward increasingly multimodal foundation models incorporating spatial transcriptomics, single-cell ATAC-seq, and proteomic data [10] [3]. Emerging platforms are addressing this need through unified data models that maintain modality-specific information while enabling cross-modal inference. The CELLxGENE ecosystem is evolving toward support for spatial transcriptomics and multiome assays, while specialized foundation models like EpiFoundation (for scATAC-seq) and CAPTAIN (for RNA-protein co-assay) are creating new data requirements [3].

A critical challenge remains the development of standardized evaluation frameworks that can objectively assess the biological relevance of scFM embeddings beyond technical metrics. The introduction of cell ontology-informed metrics represents progress in this direction, enabling quantification of how well models capture established biological relationships [1]. As the field matures, we anticipate increased emphasis on data provenance tracking, federated learning approaches that respect data privacy, and specialized foundation models pretrained on disease-specific corpora for targeted therapeutic applications.

The advent of single-cell RNA sequencing (scRNA-seq) has revolutionized biological research by enabling the investigation of gene expression at an unprecedented resolution, revealing cellular heterogeneity, identifying novel cell populations, and illuminating developmental trajectories. However, this powerful technology generates complex datasets fraught with technical challenges that can confound biological interpretation if not properly addressed. Large-scale single-cell transcriptomic datasets are typically compiled from multiple experiments conducted at different times, by different personnel, using different reagent lots, equipment, and even technology platforms. These variations introduce systematic technical artifacts known as **batch effects**, which present significant obstacles to data integration and analysis [17].

The single-cell research community is now at a pivotal juncture, with the emergence of **single-cell foundation models (scFMs)** offering promising new approaches for data integration and interpretation. These large-scale deep learning models, pretrained on vast datasets, have the potential to revolutionize how we handle batch effects, quality control, and standardization in single-cell genomics [10] [1]. However, to effectively leverage these sophisticated tools, researchers must first grasp the fundamental data challenges inherent to single-cell technologies. This technical guide examines the core data challenges in single-cell research – batch effects, quality control, and standardization – within the context of scFM development and application, providing researchers with both established best practices and insights into next-generation computational approaches.

Understanding and Addressing Batch Effects

The Nature and Impact of Batch Effects

Batch effects represent systematic technical variations introduced when samples are processed in different batches, potentially obscuring biological signals of interest. In scRNA-seq data, these effects arise from multiple sources including differences in capturing times, handling personnel, reagent lots, equipment, and sequencing technologies [17]. The highly multiplexed nature of single-cell experiments, where data is often aggregated across multiple laboratories and platforms, makes them particularly susceptible to these technical artifacts.

The challenge of batch effect correction is particularly nuanced in single-cell data due to characteristic features like **"drop-out" events** (an excessive number of zeros in the data resulting from stochastic gene expression or failures in RNA capture or amplification during sequencing) and the potential for biological differences to be mistakenly removed as technical artifacts [18]. Effective batch correction must therefore carefully distinguish between technical variations and genuine biological differences, preserving the latter while removing the former.

Established Batch Correction Methods

Numerous computational methods have been developed to address batch effects in single-cell data. A comprehensive benchmark study evaluating 14 different batch correction methods across diverse scenarios provides valuable insights into their relative performance [17] [19]. The study tested these methods on ten datasets encompassing various tissue types and sequencing technologies, evaluating them based on computational runtime, ability to handle large datasets, and efficacy in correcting batch effects while preserving biological variation.

Table 1: Performance Overview of Select Batch Correction Methods

Method	Key Algorithm	Strengths	Considerations
Harmony	Iterative clustering in PCA space with dataset integration	Fast runtime; good batch mixing	Recommended as first choice due to speed [17]
Seurat 3	CCA combined with MNN "anchors"	High cell type purity preservation	Established, widely-used platform [17]
LIGER	Integrative non-negative matrix factorization (NMF)	Separates technical from biological variation	Assumes not all inter-dataset differences are technical [17]
MNN Correct	Mutual nearest neighbors in high-dimensional space	Handles non-identical cell type compositions	Computationally intensive in original form [20]
fastMNN	MNN in PCA subspace	Improved speed and accuracy over MNN	Requires similar cell type distributions [17]
Scanorama	MNN in dimensionally reduced spaces	Similarity-weighted integration	Panoramic stitching of datasets [17]
BBKNN	MNN in reduced spaces	Fast batch balancing for visualization	Preserves local relationships [17]
scGen	Variational autoencoder (VAE)	Predicts cellular responses to perturbation	Requires reference dataset for training [17]

Based on comprehensive benchmarking, **Harmony, LIGER, and Seurat 3** emerged as the generally recommended methods for batch integration, with Harmony particularly recommended as the first method to try due to its significantly shorter runtime [17]. The performance of these methods

was evaluated using multiple metrics including k-nearest neighbor batch-effect test (kBET), local inverse Simpson's index (LISI), average silhouette width (ASW), and adjusted rand index (ARI), which collectively assess both batch mixing and biological structure preservation.

Experimental Protocol: Batch Effect Correction with Harmony

For researchers implementing batch correction, the following protocol outlines the key steps for using Harmony, one of the top-performing methods identified in benchmark studies:

- **Preprocessing**: Begin with a normalized, scaled, and log-transformed single-cell count matrix. Identify highly variable genes (HVGs) using standard methods (e.g., Seurat's `FindVariableFeatures` or Scanpy's `pp.highly_variable_genes`).
- **Dimensionality Reduction**: Perform principal component analysis (PCA) on the HVGs to obtain a low-dimensional representation of the data. Typically, the first 20-50 principal components are used as input for Harmony.
- **Harmony Integration**: Apply Harmony to the PCA embedding, specifying the batch covariate(s) (e.g., sequencing run, donor, technology). The algorithm works by:
  - Clustering cells in the PCA space while maximizing batch diversity within each cluster.
  - Calculating a correction factor for each cell based on its cluster assignment and batch origin.
  - Iteratively repeating this process until convergence.
- **Downstream Analysis**: Use the Harmony-corrected embeddings for downstream analyses such as clustering, visualization (UMAP/t-SNE), and trajectory inference. The corrected data should show improved mixing of batches while maintaining separation of distinct cell types.

The entire process can be implemented using the Harmony package in R or Python, with detailed tutorials available in the package documentation.

Quality Control Best Practices

Fundamentals of Single-Cell QC

Quality control represents a critical first step in single-cell RNA-seq analysis, aiming to distinguish high-quality cells from those affected by technical artifacts or cell death. Proper QC is essential because low-quality cells can distort downstream analyses, including clustering, differential expression, and trajectory inference. Single-cell data presents unique QC challenges due to its characteristic high sparsity, high dimensionality, and low signal-to-noise ratio [1].

Cell QC is typically performed based on three primary metrics, each capturing different aspects of data quality [21] [22] [23]:

- **Number of counts per barcode (count depth)**: The total number of UMIs (unique molecular identifiers) or reads associated with a cell. unusually high counts may indicate multiplets (multiple cells captured together), while low counts may represent empty droplets or low-quality cells.
- **Number of genes per barcode**: The number of genes with detectable expression in a cell. This metric often correlates with count depth and can help identify multiplets (high gene counts) or poor-quality cells (low gene counts).
- **Fraction of mitochondrial counts**: The percentage of reads mapping to mitochondrial genes. Elevated levels often indicate compromised cell viability, as dying cells may release cytoplasmic RNA while retaining mitochondrial RNA.

Table 2: Quality Control Metrics and Their Interpretation

QC Metric	Low Value Interpretation	High Value Interpretation	Common Thresholding Approach
Count Depth	Empty droplet, low-quality cell, or quiescent cell	Multiplet (multiple cells)	MAD-based outlier detection; permissive lower limit [21] [23]
Genes Detected	Empty droplet, low-quality cell	Multiplet or high transcriptional activity	Correlated with count depth; consider joint distribution [22]
Mitochondrial %	Viable cell	Dying cell, broken membrane	Tissue-dependent (e.g., >5-20%); consider cell type [21] [23]
Ribosomal %	Varies by cell type	Potential indicator of metabolic activity	Usually not filtered but monitored
Hemoglobin %	Varies by cell type	Potential indicator of red blood cell contamination	Relevant in blood/marrow datasets

It is crucial to consider these QC metrics jointly rather than in isolation, as cells with particular biological functions may naturally exhibit extreme values for certain metrics. For example, cells involved in respiratory processes may have higher mitochondrial content, while quiescent cells or specific cell types like neutrophils may naturally have lower RNA content [23]. Overly stringent filtering based on single metrics risks removing biologically meaningful cell populations.

Experimental Protocol: Systematic Quality Control

A robust QC workflow involves both computational and biological considerations:

- **Metric Calculation**: Compute QC metrics from the count matrix using standard tools like `sc.pp.calculate_qc_metrics` in Scanpy or `PercentageFeatureSet` in Seurat. Define gene sets for mitochondrial, ribosomal, and hemoglobin genes (appropriate for your species; "MT-" for human, "mt-" for mouse).
- **Visual Assessment**: Create visualizations to explore QC metric distributions:
  - Violin plots or density plots for each metric across samples
  - Scatter plots comparing metrics (e.g., total counts vs. genes detected, colored by mitochondrial percentage) These visualizations help identify thresholds and detect unexpected patterns in the data.
- **Threshold Determination**: Establish filtering thresholds using either:
  - **Manual thresholding**: Based on visual inspection of distributions and knowledge of expected biology.
  - **Automatic thresholding**: Using robust statistical methods like Median Absolute Deviation (MAD), where cells exceeding a certain number of MADs (e.g., 5 MADs) from the median are flagged as outliers [21].
- **Iterative Filtering**: Apply filters and proceed with downstream analysis, but remain open to revisiting filtering parameters if analysis results are difficult to interpret. In some cases, performing preliminary cell type annotation before final filtering can help preserve rare cell populations that might otherwise be removed.
- **Doublet Detection**: Employ specialized doublet detection tools (e.g., DoubletFinder, Scrublet, Solo) that generate artificial doublets and compare gene expression profiles to identify potential multiplets [23].
- **Ambient RNA Removal**: Consider using tools like SoupX, DecontX, or CellBender to address contamination from ambient RNA – a common issue in droplet-based protocols where RNA released from dead cells can be captured in droplets containing intact cells [23].

Standardization and Integration with Foundation Models

The Rise of Single-Cell Foundation Models

Single-cell foundation models represent a paradigm shift in how we approach single-cell data analysis. These large-scale deep learning models are pretrained on massive, diverse collections of single-cell datasets using self-supervised learning objectives, enabling them to learn fundamental biological principles that can be transferred to various downstream tasks [10]. The public domain now contains tens of millions of single-cell omics datasets, spanning numerous cell types, states, and conditions, providing the raw material for training these models.

Inspired by transformer architectures that revolutionized natural language processing, scFMs treat individual cells as analogous to sentences and genes or genomic features as words or tokens [10] [1]. By exposing models to millions of cells across diverse tissues and conditions, scFMs can learn a unified representation of single-cell data that captures underlying biological structure while being robust to technical variations. Early scFMs like scBERT and scGPT have demonstrated promising capabilities in tasks such as cell type annotation, batch integration, and perturbation response prediction [10].

Benchmarking scFMs Against Traditional Methods

Recent comprehensive benchmark studies have evaluated scFMs against established methods under realistic conditions, providing insights into their relative strengths and limitations. One such study evaluated six scFMs against well-established baselines across two gene-level and four cell-level tasks, using 12 metrics spanning unsupervised, supervised, and knowledge-based approaches [1].

The benchmarking reveals that while scFMs are robust and versatile tools for diverse applications, simpler machine learning models can sometimes be more efficient for specific datasets, particularly under resource constraints. Notably, no single scFM consistently outperforms others across all tasks, emphasizing the need for tailored model selection based on factors such as dataset size, task complexity, biological interpretability, and computational resources [1].

For batch integration specifically, scFMs show particular promise in handling complex batch effects arising from multiple sources (inter-patient, inter-platform, inter-tissue), while preserving subtle biological variations that might be lost with traditional methods. The ability of scFMs to learn

from massive datasets enables them to recognize cell states and types across diverse contexts, potentially overcoming limitations of methods that assume consistent cell type compositions across batches.

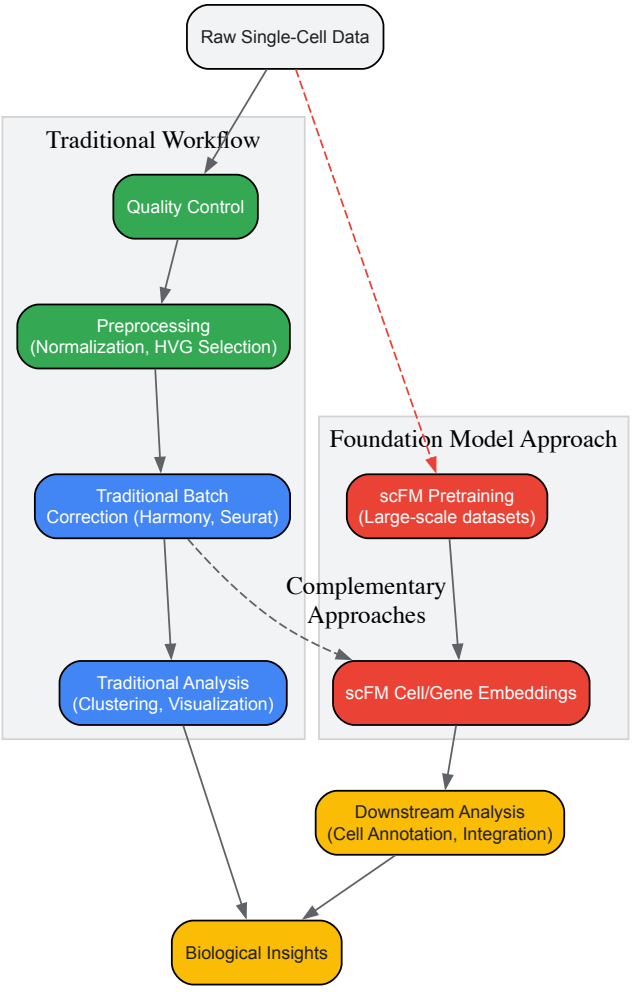
MASI: A Model-Free Approach to Standardization

An alternative approach to data integration and standardization is exemplified by MASI (Marker-Assisted Standardization and Integration), a fast model-free method that relies on cell-type marker genes from reference data to uniformly annotate and integrate query datasets [18]. Unlike model-based approaches that require extensive training, MASI converts gene expression matrices into cell-type score matrices using prior knowledge of marker genes, effectively condensing biological information from high-dimensional gene space into a lower-dimensional cell-type feature space.

Benchmarking studies demonstrate that MASI can compete with well-established model-based annotation and integration methods while offering significantly reduced computational requirements – it can annotate approximately one million cells on a personal laptop, making large-scale single-cell data integration more accessible to researchers with limited computational resources [18].

Integrated Workflow and Visualization

The relationship between traditional single-cell analysis steps and the emerging approach using foundation models can be visualized through the following workflow:



Single-Cell Analysis: Traditional vs. Foundation Model Approaches

This diagram illustrates how traditional analysis pipelines and foundation model approaches can complement each other in addressing single-cell data challenges. While traditional methods provide established, interpretable workflows for standard analyses, foundation models offer an alternative pathway that leverages large-scale pretraining to generate biologically meaningful embeddings resistant to batch effects.

The Scientist's Toolkit: Essential Research Reagents and Computational Solutions

Table 3: Key Resources for Single-Cell Data Analysis

Resource Type	Specific Tools/Sources	Primary Function	Application Context
Batch Correction Software	Harmony, Seurat 3, LIGER, fastMNN	Remove technical batch effects	Data integration across experiments/technologies [17]
Quality Control Tools	Scanpy, Seurat, Scater	Calculate QC metrics, filter cells	Initial data preprocessing [21] [22]
Doublet Detection	DoubletFinder, Scrublet, Solo	Identify multiplets	QC for droplet-based protocols [23]
Ambient RNA Removal	SoupX, DecontX, CellBender	Remove contamination from ambient RNA	QC for droplet-based protocols [23]
Marker Gene Databases	CellMarker, PanglaoDB, ScType	Provide cell-type-specific markers	Cell annotation, MASI integration [18]
Single-Cell Foundations Models	Geneformer, scGPT, scBERT, UCE	Learn universal representations from large data	Multiple downstream tasks [1]
Benchmarking Platforms	pipeComp, scGraph-OntoRWR	Evaluate method performance	Objective comparison of tools/models [1]
Data Repositories	CZ CELLxGENE, Human Cell Atlas, GEO/SRA	Provide standardized, annotated datasets	Model training, benchmarking [10]

The field of single-cell genomics continues to evolve rapidly, with batch effects, quality control, and standardization remaining central challenges as datasets grow in size and complexity. Traditional computational methods like Harmony, Seurat, and LIGER have established strong foundations for addressing these issues, with comprehensive benchmarks guiding researchers toward appropriate tool selection based on their specific data characteristics and analytical needs [17] [19].

The emergence of single-cell foundation models represents a promising new frontier, offering the potential for more biologically aware integration and standardization that preserves subtle but meaningful biological variations [10] [1]. However, current benchmarking indicates that these models have not yet consistently outperformed simpler alternatives across all tasks, suggesting that traditional methods will remain relevant for the foreseeable future.

As the field progresses, successful single-cell research will require thoughtful application of both established and emerging approaches, with careful attention to the specific biological questions, dataset characteristics, and computational resources at hand. By combining rigorous quality control, appropriate batch correction strategies, and emerging foundation models, researchers can overcome the data challenges inherent in single-cell genomics and unlock the full potential of this transformative technology.

How scFMs Work: Tokenization, Architecture, and Real-World Applications

In single-cell biology, the advent of single-cell foundation models (scFMs) represents a transformative approach to analyzing cellular heterogeneity and complex regulatory networks. These large-scale deep learning models, pretrained on vast single-cell genomics datasets, have revolutionized data interpretation through self-supervised learning with capacity for various downstream tasks [6]. A critical technical challenge in developing these models lies in how to convert raw, non-sequential gene expression data into structured input that deep learning architectures can process—a procedure known as tokenization [6]. Tokenization serves as the foundational bridge that standardizes raw, often unstructured single-cell data into a structured format that models can understand, process, and learn from, thereby enabling the application of transformer-based architectures that have revolutionized natural language processing to biological data [6] [2]. The effectiveness of this process directly impacts a model's ability to capture meaningful biological patterns and relationships.

This technical guide examines the current tokenization strategies employed in scFMs, focusing on their conceptual frameworks, methodological implementations, and practical considerations for researchers. Within the broader thesis of scFM research, tokenization represents more than merely a data preprocessing step—it constitutes a fundamental design choice that determines how biological information is encoded and ultimately interpreted by artificial intelligence systems. As the field progresses toward more unified frameworks capable of integrating and comprehensively analyzing rapidly expanding single-cell data repositories, standardized and biologically-informed tokenization approaches will be crucial for advancing single-cell genomics and unlocking deeper insights into cellular function and disease mechanisms [6].

Fundamental Concepts of Tokenization in scFMs

What is Tokenization and Why Does It Matter?

In computational terms, tokenization refers to the process of converting raw input data into a sequence of discrete units called tokens [6]. For single-cell RNA sequencing (scRNA-seq) data, which naturally exists as high-dimensional vectors of gene expression counts per cell, tokenization transforms this continuous, non-sequential data into structured sequences amenable to processing by transformer architectures [6] [1]. This

transformation is particularly crucial because gene expression data lacks the inherent ordering found in natural language—unlike words in a sentence, genes in a cell have no natural sequence [6] [1].

The tokenization process in scFMs typically treats individual cells analogously to sentences, while genes or other genomic features along with their expression values become the words or tokens [6]. This conceptual framing allows researchers to leverage advanced neural architectures developed for natural language processing, but requires careful consideration of how to impose meaningful structure on the inherently unordered set of genes expressed in a single cell. The premise is that by exposing a model to millions of cells encompassing diverse tissues and conditions through an appropriate tokenization scheme, the model can learn fundamental principles of cellular biology that generalize to new datasets and downstream tasks [6].

Comparative Analysis of Tokenization Approaches

Table 1: Key Tokenization Strategies in Single-Cell Foundation Models

Strategy	Core Methodology	Gene Ordering Principle	Representative Models	Key Advantages	Major Limitations
Expression Ranking	Ranks genes by expression level within each cell	Expression magnitude (highest to lowest)	Geneformer, scGPT, cell2sentence [6] [2]	Deterministic; preserves most highly expressed genes	Arbitrary sequence; may lose low-expression signals
Value Binning	Partitions genes into bins by expression values	Expression value ranges	scBERT [6] [24]	Reduces dimensionality; handles technical noise	Coarse-grained; may obscure subtle expression differences
Fixed Gene Order	Uses consistent gene ordering across all cells	Predefined gene sequence	xTrimoGene, scFoundation [24]	Consistent positional encoding; efficient processing	May not reflect cell-specific expression patterns
Natural Language Tokenization	Applies NLP tokenizers to gene sequence strings	Gene rank order converted to text	cell2sentence (C2S) [2]	Leverages pretrained NLP components; captures biological knowledge from text	Additional complexity of string conversion

Technical Implementation of Tokenization Strategies

Core Tokenization Components

The tokenization process in scFMs typically incorporates multiple embedding components that work in concert to represent the rich information contained in single-cell data:

- **Gene Embeddings**: Analogous to word embeddings in natural language processing, these embeddings represent the identity of each gene, potentially capturing biological functions and relationships [1]. These are typically learned during pretraining and allow functionally similar genes to be embedded in close proximity in the latent space [1].
- **Value Embeddings**: These components represent the expression level of each gene in a given cell, encoding quantitative information that is crucial for understanding cellular states [6] [1]. Implementation approaches vary, with some models using separate value embeddings while others incorporate expression information directly into the token representation.
- **Positional Embeddings**: Since transformer architectures lack inherent notion of sequence order, positional embeddings provide information about each token's position in the input sequence [6]. This presents a particular challenge for scFMs due to the non-sequential nature of gene expression data, necessitating various gene ordering strategies [6] [1].

Additional special tokens may be incorporated to enrich the input representation, including tokens representing cell identity and metadata, modality indicators for multi-omics approaches, and batch information tokens to account for technical variations [6].

Gene Ordering Strategies

A fundamental challenge in scFM tokenization is imposing sequence order on inherently unordered gene expression data. Several approaches have emerged:

- **Expression-Based Ordering**: The most common strategy ranks genes within each cell by their expression levels, feeding the ordered list of top genes as the input sequence [6]. This provides a deterministic approach that prioritizes highly expressed genes, though the ranking is

arbitrary from a biological perspective.

- **Binning Approaches**: Some models partition genes into bins by their expression values and use these rankings to determine positional encoding [6]. This can reduce the impact of technical noise in expression measurements.
- **Fixed Ordering**: Alternative approaches employ a fixed gene order across all cells, often based on chromosomal location or other biological priors [6]. While computationally efficient, this method may not reflect cell-specific expression patterns.

Notably, several models report no clear advantages for complex ranking strategies and simply use normalized counts with minimal preprocessing [6].

Table 2: Advanced Tokenization Features in Modern scFMs

Feature Type	Implementation Purpose	Technical Approach	Example Model Usage
Cell Identity Tokens	Prepend token representing cell's own identity and metadata	Special classification token added to sequence start	scGPT, UCE [6]
Modality Indicators	Incorporate multiple omics data types	Tokens indicating data modality (e.g., RNA, ATAC)	Multi-ome sequencing models [6]
Biological Context Tokens	Incorporate gene metadata	Gene ontology, chromosome location information	scFoundation, LangCell [6]
Batch Effect Tokens	Account for technical variations	Batch information as special tokens	scGPT [6]

Experimental Protocols for Tokenization Evaluation

Benchmarking Framework for Tokenization Strategies

Evaluating the effectiveness of tokenization strategies requires carefully designed experimental protocols that assess performance across multiple biological tasks. A comprehensive benchmarking framework should encompass both gene-level and cell-level tasks to evaluate how well the tokenization approach captures biological relationships [1].

For **gene-level evaluation**, researchers can extract gene embeddings from the input layers of scFMs and use them to predict known biological relationships, including tissue specificity and Gene Ontology (GO) terms [1]. This evaluation tests whether functionally similar genes are embedded in close proximity in the latent space, analogous to how word embeddings capture semantic relationships in natural language models.

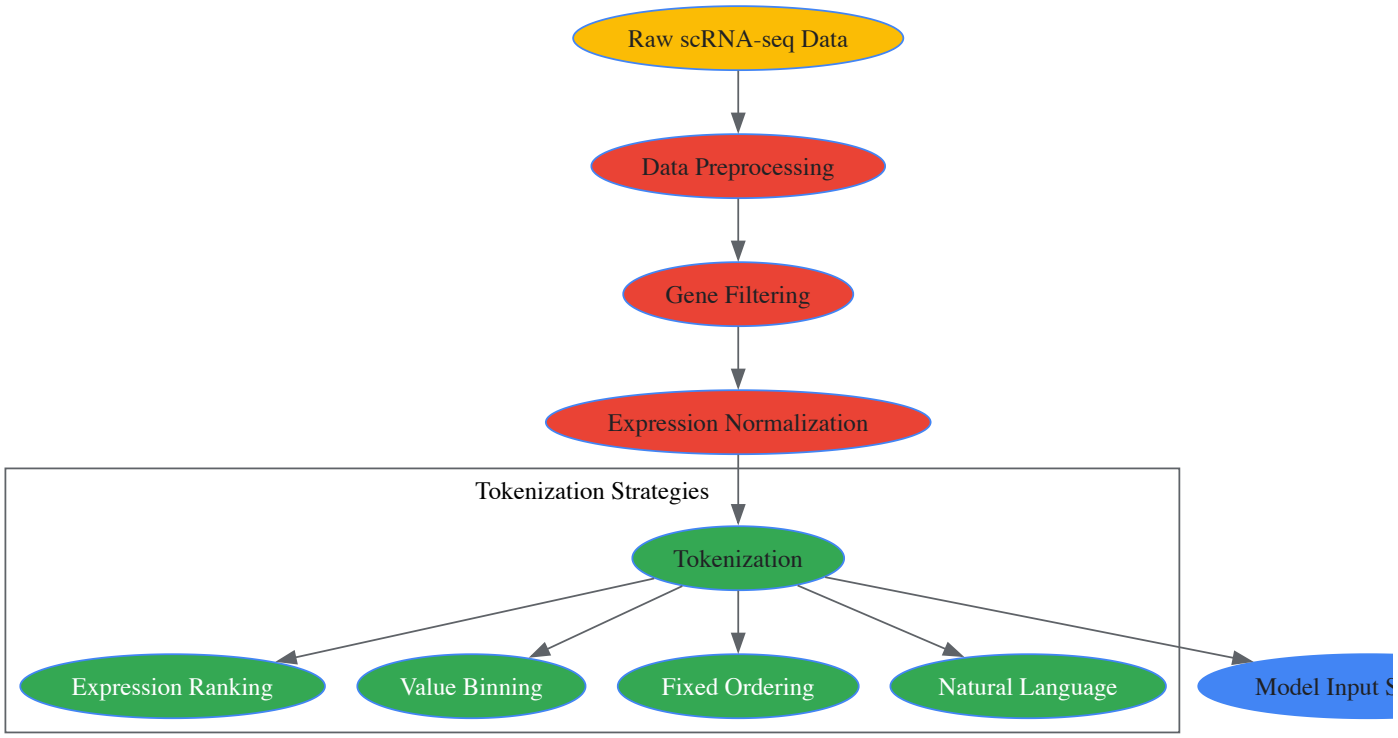
For **cell-level evaluation**, standard protocols assess the efficiency of zero-shot scFM cell embeddings in core analytical tasks including dataset integration and cell type annotation [1]. These evaluations typically employ high-quality datasets with manual annotations that vary in size and diversity while containing multiple sources of batch effects (inter-patient, inter-platform, and inter-tissue variations) to test the robustness of the representation learning [1].

Implementation Protocol: Tokenization for Cell Type Annotation

The following detailed methodology outlines a practical implementation for cell type annotation using tokenization in scale-free and unbiased transformers:

- **Data Preprocessing**:
  - For human datasets, follow preprocessing procedures established by scBERT to ensure comparable evaluation [24].
  - For mouse expression matrices, apply quality control by retaining samples with over 200 genes expressed.
  - Perform log-normalization with a library size of 10,000.
  - Filter out noise genes expressed in three or fewer cell samples using the Scanpy package [24].
  - Notably, avoid highly variable gene (HVG) selection when using models designed to handle full gene sets [24].
- **Tokenization Process**:
  - Segment each cell sample into dimensionally reduced, information-dense sub-vectors using a fixed window size [24].
  - Apply sequential tokenization and 1D-convolution to expand the attention receptive field of gene tokens [24].
  - Implement an innovative gene embedding algorithm that enables bias-free attention computation in transformers.
- **Model Training**:
  - Incorporate a self-supervised mask reconstruction task into an efficient encoder-decoder framework [24].
  - Jointly optimize with a class-imbalanced annotation task to improve latent representations [24].

- Leverage precision-preserving attention mechanisms for end-to-end annotation across the full gene length.



Tokenization Workflow: Single-cell data undergoes preprocessing before conversion to tokens via different strategies.

Table 3: Essential Research Resources for scFM Tokenization Development

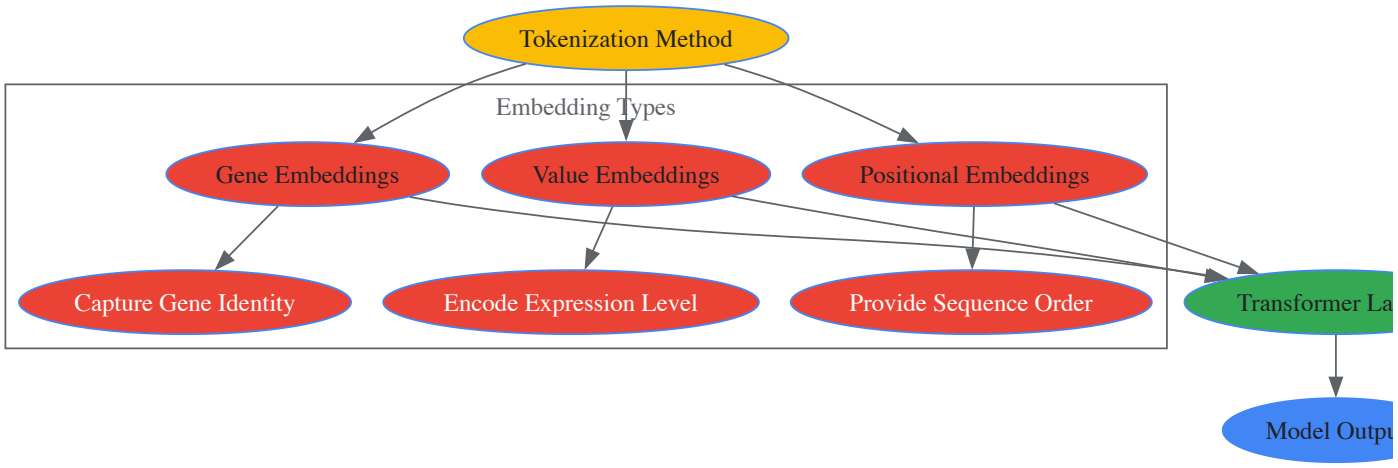
Resource Category	Specific Tools & Platforms	Primary Function in Tokenization Research	Key Features
Data Repositories	CZ CELLxGENE [6], Human Cell Atlas [6], NCBI GEO [6]	Provide standardized, annotated single-cell datasets for training and evaluation	Curated collections with quality controls; CELLxGENE contains >100M unique cells [6]
Benchmark Datasets	Heart Cell Atlas v2 [2], Asian Immune Diversity Atlas (AIDA) v2 [1]	Enable evaluation of tokenization strategies across diverse biological conditions	High-quality labels; multiple sources of batch effects; tissue and species diversity
Computational Frameworks	Scanpy [24], Hugging Face [2]	Data preprocessing and model implementation	Standardized pipelines; pretrained model access; interoperability
Evaluation Metrics	scGraph-OntoRWR [1], LCAD [1]	Assess biological relevance of learned representations	Cell ontology-informed metrics; measure consistency with prior biological knowledge

Emerging Challenges and Future Directions

Despite significant progress in tokenization strategies for scFMs, several challenges remain unresolved. The non-sequential nature of omics data continues to present fundamental questions about optimal gene ordering approaches [6]. While current strategies based on expression ranking provide practical solutions, they lack strong biological justification for the imposed sequence structure. Future research may explore adaptive ordering mechanisms that dynamically adjust gene sequence based on biological context.

Additional challenges include inconsistency in data quality across datasets and the computational intensity required for training and fine-tuning scFMs with various tokenization strategies [6]. Furthermore, interpreting the biological relevance of latent embeddings and model representations remains nontrivial, necessitating continued development of interpretability methods tailored to biological applications [6] [2].

Future directions for tokenization research include developing multimodal tokenization approaches that seamlessly integrate diverse data types such as scATAC-seq, spatial transcriptomics, and single-cell proteomics [6]. There is also growing interest in transfer learning approaches that leverage pretrained tokenization schemes across related biological domains, potentially reducing the resource burden for applying scFMs to new research questions.



Tokenization Components: Gene, value, and positional embeddings create input for transformer layers.

The emergence of single-cell foundation models (scFMs) represents a paradigm shift in computational biology, enabling unprecedented understanding of cellular heterogeneity and function. Within this rapidly evolving field, a fundamental architectural schism has emerged between **encoder-focused designs** and **decoder-focused designs**, each with distinct capabilities, performance characteristics, and application suitability. These architectural differences mirror the evolution seen in natural language processing but are uniquely adapted to the complexities of biological systems, where "genes as words" and "cells as documents" provide a powerful analytical framework [25].

Encoder-dominant models like scBERT and scRobust employ **bidirectional attention mechanisms** to create compressed, informative cellular representations, excelling in classification and embedding tasks. In contrast, decoder-centric models like scGPT leverage **causal masking and generative pretraining** to predict gene expressions, demonstrating superior performance in generative tasks and multi-omic integration. This architectural spectrum reflects a broader thesis in scFM research: that model design decisions fundamentally shape biological insight extraction, with significant implications for drug discovery, therapeutic development, and precision medicine [26] [27].

Architectural Principles: Core Design Philosophies

Encoder-Dominant Architectures

Encoder-focused models in single-cell analysis build upon the transformer encoder architecture, which processes all input genes simultaneously through **self-attention mechanisms**. This design enables the model to capture complex, bidirectional relationships across the entire genomic landscape of a cell. Models like scBERT [27] and scRobust [28] exemplify this approach, treating gene expression profiles as unordered sets where global dependencies matter more than sequential order.

The pretraining objectives for encoder models typically include **masked gene modeling** and **contrastive learning**. In masked gene modeling, random subsets of genes have their expressions hidden, and the model must reconstruct these values based on the remaining genomic context. Contrastive learning, as implemented in scRobust, creates augmented views of individual cells and trains the model to identify representations originating from the same cellular source while distinguishing them from others [28]. This approach forces the encoder to learn robust, noise-invariant representations that capture essential biological signals despite the sparsity characteristic of scRNA-seq data.

Decoder-Dominant Architectures

Decoder-focused models adopt an **autoregressive approach** to modeling cellular systems, processing gene expressions in a defined sequential order. scGPT [25] [29], the prominent model in this category, employs causal masking in its attention mechanism, ensuring that each position in the gene sequence can only attend to previous positions. This architecture mirrors the design principles of large language models like GPT, but adapted for biological sequences.

The pretraining strategy for decoder models centers on **next-gene prediction**, where the model learns to predict each gene's expression level based on previously encountered genes in the sequence. This training objective encourages the model to develop a comprehensive understanding of gene-gene interactions and regulatory networks. scGPT's generative approach has demonstrated remarkable flexibility across diverse downstream applications, including **perturbation response prediction** and **multi-omic integration** [25]. By learning the underlying "language" of cellular biology, these models can generate realistic in-silico profiles of cellular states under various conditions, providing a powerful tool for hypothesis generation and experimental design.

Comparative Analysis: Performance Across Biological Tasks

Quantitative Benchmarking

Table 1: Performance comparison of encoder vs. decoder models across key tasks in single-cell analysis

Task Type	Model Architecture	Representative Model	Key Performance Metrics	Comparative Advantage
Cell Type Annotation	Encoder	scRobust	Macro F1: 0.84-0.91 across 9 benchmarks [28]	Excels in rare cell type identification (28% accuracy for CD4+ T Helper 2 vs. <10% for others) [28]
	Decoder	scGPT	73.4% accuracy (outperformed scBERT & SingleCellNet) [30]	Superior generalization across diverse tissue types
Batch Integration	Encoder	scBERT	Improved batch correction while preserving biological signals [30]	Effective technical noise reduction
	Decoder	scGPT	State-of-the-art in multi-batch and multi-omic integration [25]	Preserves fine-grained biological variation
Drug Response Prediction	Encoder	scFoundation	Mean F1: 0.971 (pooled-data evaluation) [27]	Best performance with abundant training data
	Decoder	scGPT	Mean F1: 0.858 (zero-shot setting) [27]	Superior cross-data generalization with limited samples
Handling Data Sparsity	Encoder	scRobust	Maintains >80% accuracy with 50% additional dropout [28]	Robust to extreme sparsity through unique gene selection
	Decoder	scGPT	Gene expression binning reduces impact of dropouts [29]	Generative imputation capabilities

Table 2: Architectural properties and their biological implications

Architectural Property	Encoder Models (scBERT, scRobust)	Decoder Models (scGPT)
Attention Mechanism	Bidirectional (full gene-gene attention)	Causal masking (autoregressive)
Pretraining Objective	Masked gene modeling, contrastive learning	Next-gene prediction, expression binning
Information Flow	Global context integration	Sequential, unidirectional
Handling Sparsity	Strategic unique gene selection [28]	Expression value binning and categorization [29]
Computational Requirements	Moderate (attention scales with gene set size)	High (sequential processing)
Interpretability	Attention weights reveal gene associations	Generation patterns show regulatory dependencies
Ideal Application Scope	Cell classification, embedding generation, rare cell identification	Perturbation modeling, multi-omic integration, generative tasks

Task-Specific Performance Patterns

The benchmarking data reveals a consistent pattern of **architectural specialization** across biological tasks. Encoder models demonstrate particular strength in **discriminative tasks** requiring comprehensive cellular representation. For example, scRobust achieves remarkable performance in identifying rare cell populations—a critical capability for understanding tumor heterogeneity and immune microenvironment composition [28]. This advantage stems from the encoder's ability to integrate global gene expression patterns into dense, informative embeddings that capture subtle biological differences.

Decoder models excel in **generative and predictive tasks** that benefit from sequential reasoning about cellular states. scGPT's strong performance in zero-shot drug response prediction highlights its capacity for generalizing to unseen cellular contexts [27]. This capability is particularly valuable in drug discovery, where predicting responses for novel therapeutic compounds or rare cell types can significantly accelerate research. The autoregressive nature of decoder models appears better suited for modeling temporal processes and perturbation effects, making them ideal for studying disease progression and treatment responses.

Experimental Protocols and Methodologies

Benchmarking Frameworks and Evaluation Strategies

Comprehensive evaluation of scFMs employs standardized benchmarking frameworks that assess model performance across multiple dimensions. The **scDrugMap** framework provides a representative example of rigorous model assessment, incorporating both pooled-data evaluation and cross-data evaluation scenarios [27]. In pooled-data evaluation, models are trained and tested on aggregated data from multiple studies, assessing performance under ideal data availability conditions. Cross-data evaluation tests model generalization by training on one set of studies and evaluating on completely independent datasets, mimicking real-world application scenarios.

Transfer learning methodologies form a critical component of scFM evaluation. Studies typically employ two fine-tuning approaches: **layer freezing** (where pretrained weights remain fixed while training only task-specific heads) and **full fine-tuning** (often using parameter-efficient methods like Low-Rank Adaptation). The performance gap between these approaches reveals the balance between preserving pretrained knowledge and adapting to new tasks [27]. For decoder models like scGPT, zero-shot evaluation provides additional insights into the fundamental biological knowledge captured during pretraining, without any task-specific fine-tuning [29].

Data Processing and Augmentation Strategies

Table 3: Key research reagents and computational tools in scFM experimentation

Resource Type	Specific Examples	Function in Experimental Pipeline
Pretraining Datasets	CellXGene (33M+ cells) [29], Primary collections (326,751 cells) [27]	Large-scale foundational data for pretraining scFMs
Benchmark Datasets	Baron Human, Muraro, Segerstolpe, TM, Zheng 68K [28]	Standardized evaluation across protocols and tissues
Data Augmentation	Artificial dropout (30%, 50% additional masking) [28], Cell augmentation [28]	Testing robustness to sparsity and improving generalization
Evaluation Metrics	Macro F1, Accuracy, AUC, AUPR [27] [28] [31]	Quantifying performance across classification tasks
Transfer Learning Methods	Layer freezing, LoRA (Low-Rank Adaptation) [27]	Adapting foundation models to specific downstream applications
Domain Adaptation	SSDA4Drug [31], Adversarial training [31]	Transferring knowledge from bulk to single-cell data

Data preprocessing pipelines significantly impact model performance, with different architectures employing specialized strategies to handle scRNA-seq sparsity. Encoder models like scRobust implement **unique gene selection** strategies that prioritize rarely expressed but biologically informative genes, effectively mitigating information loss from dropout events [28]. Decoder models like scGPT employ **expression value binning**, categorizing continuous expression values into discrete ranges that are more amenable to token-based processing [29]. These preprocessing decisions reflect fundamental differences in how each architecture conceptualizes and processes biological data.

For robustness evaluation, researchers systematically introduce **artificial dropout** (30-50% additional masking) to simulate extreme sparsity conditions [28]. This approach tests model resilience to data quality issues commonly encountered in real experimental data. Data augmentation techniques like **cell augmentation** (creating multiple embeddings from random gene subsets) further enhance model robustness by encouraging learning of redundant biological representations [28].

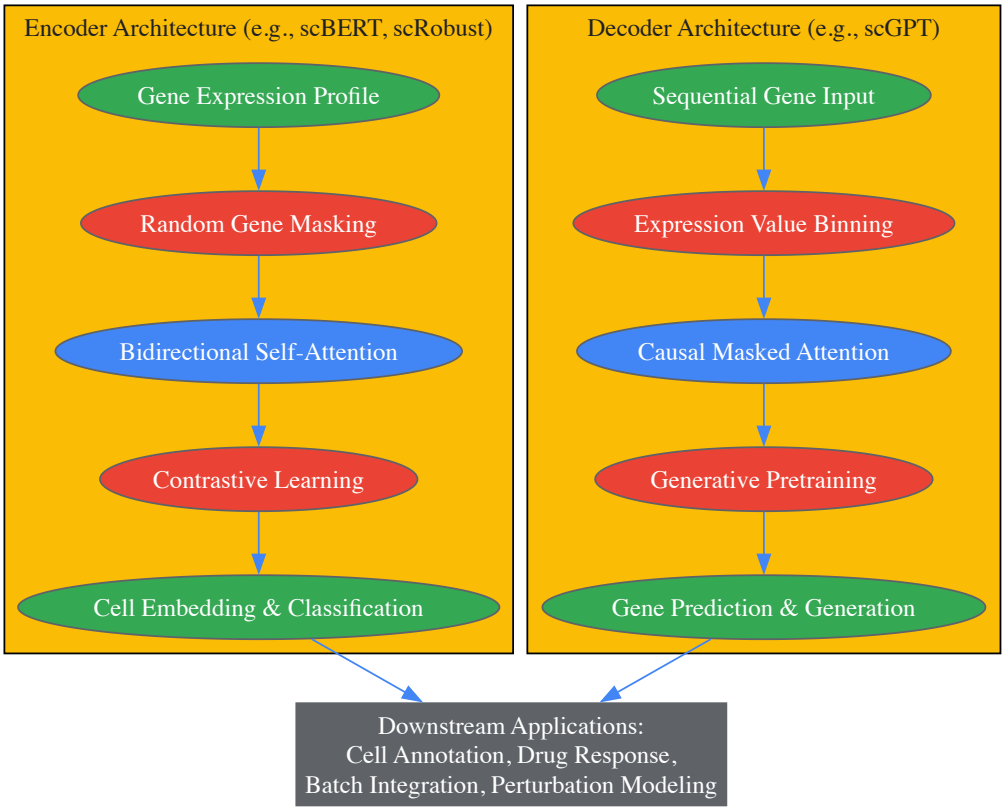


Diagram 1: Comparative workflow of encoder vs. decoder architectures in single-cell foundation models, highlighting distinct processing strategies and shared application domains.

Applications in Drug Discovery and Development

The architectural differences between encoder and decoder models translate to distinct advantages in pharmaceutical applications. Encoder models have demonstrated exceptional performance in **drug response prediction** when substantial training data is available, with scFoundation achieving remarkable F1 scores of 0.971 in pooled-data evaluation scenarios [27]. This capability enables precise identification of therapeutic responders and non-responders at single-cell resolution, revealing resistant subpopulations within seemingly homogeneous tumors.

Decoder models excel in **zero-shot prediction and cross-domain generalization**, achieving strong performance (F1: 0.858) even without exposure to target domain data during training [27]. This capability is particularly valuable for predicting responses to novel therapeutic compounds or rare cellular states where labeled data is scarce. Frameworks like CRISP leverage foundation models for predicting drug perturbation responses in unseen cell types, enabling drug repurposing through cross-domain prediction—such as translating insights from solid tumors to blood cancers [32].

Both architectures contribute significantly to understanding **drug resistance mechanisms**. Encoder models help identify characteristic gene expression patterns associated with treatment resistance, while decoder models can simulate cellular responses to various perturbation conditions, generating hypotheses about resistance pathways [31]. These complementary strengths create a powerful ecosystem for pharmaceutical research, enabling both deep characterization of known therapeutic responses and exploration of novel treatment spaces.

Future Directions and Architectural Convergence

The evolving landscape of scFM architectures points toward several promising research directions. **Hybrid architectures** that combine bidirectional context understanding with generative capabilities may overcome limitations of both approaches, potentially offering state-of-the-art performance across diverse task types. Initial benchmarking studies have demonstrated that no single architecture dominates across all applications, suggesting that task-specific optimal designs will continue to emerge [27].

Multi-modal integration represents another frontier, with models like scGPT already demonstrating capabilities in combining gene expression with chromatin accessibility data [25]. Future architectures will likely expand to incorporate protein expression, spatial context, and metabolic information, requiring sophisticated architectural adaptations to handle diverse data types and resolutions. The development of **specialized attention mechanisms** that incorporate biological priors—such as gene network relationships or chromosomal proximity—may further enhance model efficiency and biological plausibility.

As the field matures, we anticipate increased emphasis on **interpretability and biological insight extraction**. Current attention mechanisms provide some visibility into model decision processes, but more sophisticated interpretation frameworks are needed to translate model insights into

testable biological hypotheses. The integration of scFMs into larger **experimental design frameworks** will close the loop between computational prediction and experimental validation, accelerating the cycle of scientific discovery in single-cell biology and therapeutic development.

The architectural spectrum between encoder and decoder designs in single-cell foundation models represents a rich design space with significant implications for biological discovery and therapeutic development. Encoder models offer superior performance in discriminative tasks like cell annotation and rare population identification, while decoder models excel in generative applications and cross-domain generalization. This division of capabilities creates a complementary ecosystem rather than a competitive landscape, with each approach illuminating different aspects of cellular biology.

The broader thesis of scFM research suggests that architectural decisions fundamentally shape the types of biological questions that can be effectively addressed. As the field progresses, the development of task-aware architectural selection and hybrid approaches will enable researchers to match model capabilities to scientific objectives more precisely. This architectural diversity, coupled with rigorous benchmarking frameworks and standardized evaluation methodologies, provides a solid foundation for advancing single-cell biology and transforming drug discovery through more predictive, interpretable, and actionable computational models.

Masked gene prediction has emerged as a foundational self-supervised learning task for training single-cell foundation models (scFMs). By learning to reconstruct randomly obscured portions of single-cell transcriptomic data, scFMs develop powerful latent representations that capture fundamental biological principles. This whitepaper provides an in-depth technical examination of masked gene prediction methodologies, architectural implementations, and evaluation frameworks. We detail how this pretraining paradigm enables models to learn rich, transferable representations of cellular states and functions without explicit supervision, facilitating their application to diverse downstream biological tasks from cell type annotation to drug sensitivity prediction. The technical guidelines presented herein equip computational biologists and drug development professionals with the essential knowledge for implementing and leveraging these transformative approaches in biomedical research.

Single-cell RNA sequencing (scRNA-seq) technologies have generated vast amounts of transcriptomic data, creating unprecedented opportunities for understanding cellular heterogeneity at scale. Single-cell foundation models (scFMs) represent a paradigm shift in analyzing this data by leveraging self-supervised learning on massive, diverse datasets before being adapted to specific downstream tasks [33]. The core premise involves training large-scale deep learning models on extensive single-cell omics corpora to learn fundamental biological principles that generalize across tissues, conditions, and species [33] [1].

These models typically employ transformer architectures that process single-cell data by treating individual cells as analogous to sentences and genes or genomic features as words or tokens [33]. This conceptual framework enables the application of successful pretraining strategies from natural language processing, particularly masked prediction tasks, to biological data. The resulting models capture intricate gene-gene relationships and cellular states that form a foundational understanding of cell biology, which can be fine-tuned for specific applications with relatively few labeled examples [33] [1].

Conceptual Foundations of Masked Gene Prediction

Self-Supervised Learning Paradigm

Self-supervised learning enables models to learn from unlabeled data by creating supervisory signals from the data itself. For single-cell transcriptomics, this approach is particularly valuable due to the scarcity of meticulously labeled datasets and the inherent complexity of biological systems [33]. The model learns to capture the underlying data distribution and intrinsic structure of single-cell omics data without manual annotation, developing a comprehensive understanding of gene interactions and co-expression patterns that reflect biological reality.

Masked prediction represents a particularly powerful self-supervised approach where the model learns by predicting intentionally obscured portions of the input data [33]. This task forces the model to develop a robust understanding of contextual relationships between genes and their expression patterns across diverse cellular contexts. By learning to reconstruct missing information based on surrounding context, the model develops a deep understanding of transcriptional regulation and cellular states.

Biological Interpretation of Masked Learning

When a scFM performs masked gene prediction, it effectively learns the complex conditional dependencies between genes—how the expression of certain genes implies probable expression levels of other genes [33]. These learned relationships often correspond to biologically meaningful patterns such as coregulated gene modules, pathway memberships, and functional associations. The model develops an implicit understanding of transcriptional networks that govern cellular identity and function, encoded within its parameters through the pretraining process.

The attention mechanisms in transformer architectures enable the model to weight the importance of different genes when making predictions about masked tokens, effectively learning which genes are most informative for inferring cellular state [33]. This process results in rich latent representations at both the gene and cell levels that capture biological semantics analogous to how language models capture word meanings and sentence structure.

Technical Implementation of Masked Gene Prediction

Data Preparation and Tokenization

A critical first step in implementing masked gene prediction is converting raw single-cell expression data into a structured format suitable for transformer models. This tokenization process defines the fundamental units the model will process:

Table 1: Tokenization Strategies for Single-Cell Data

Token Type	Description	Implementation Examples
Gene Tokens	Represent gene identifiers	Embedding vectors for each gene [33]
Value Tokens	Encode expression levels	Binned expression values or normalized counts [33] [1]
Positional Tokens	Provide sequence context	Gene rank order or chromosomal position [33]
Special Tokens	Add biological context	Cell type, batch, or modality indicators [33]

A fundamental challenge in applying transformers to single-cell data is that gene expression data lacks natural sequential ordering. Unlike words in a sentence, genes have no inherent sequence. To address this, several ordering strategies have been developed:

- **Expression-ranked sequencing**: Genes are ordered by their expression levels within each cell [33]
- **Expression binning**: Genes are partitioned into bins based on expression values [33]
- **Biological ordering**: Genes are ordered by genomic coordinates or functional groupings [33]
- **Arbitrary ordering**: Some models report minimal impact from specific orderings and use simple normalized counts [33]

After tokenization, all tokens are converted to embedding vectors that combine information about gene identity and expression level, often supplemented with positional encodings to represent the chosen gene ordering [33].

Model Architectures for Masked Prediction

Most scFMs utilize transformer architectures, which employ self-attention mechanisms to model relationships between all genes in a cell simultaneously [33]. The specific architectural implementations vary:

- **Encoder-based models** (e.g., scBERT): Use bidirectional attention mechanisms that learn from all genes in a cell simultaneously, similar to BERT in natural language processing [33]
- **Decoder-based models** (e.g., scGPT): Employ unidirectional masked self-attention that iteratively predicts masked genes conditioned on known genes, inspired by GPT architectures [33]
- **Hybrid architectures**: Combine encoder and decoder components for specialized tasks [33]

The attention mechanisms in these architectures enable the model to learn and weight relationships between any pair of input tokens (genes), effectively determining which genes are most informative for predicting masked elements based on the cellular context [33].

Masking Strategies and Objectives

The core of the pretraining task involves strategically masking portions of the input data and training the model to reconstruct them:

Table 2: Masking Strategies in scFMs

Strategy	Method	Advantages
Random Masking	Random selection of genes to mask	Simple implementation, broad coverage
Progressive Masking	Increasing masking ratio during training	Encourages robust feature learning
Strategic Masking	Targeting biologically related gene sets	Enhances learning of functional relationships
Multi-modal Masking	Extending across data types (RNA, ATAC, etc.)	Enables integrated representation learning

The masking ratio (percentage of genes masked) typically ranges from 15% to 40%, balancing the difficulty of the reconstruction task with preserving sufficient context for meaningful predictions [33]. The model is trained to minimize the discrepancy between the predicted and actual expression values of masked genes, often using mean squared error or similar reconstruction loss functions.

Experimental Protocols and Methodologies

Standard Pretraining Workflow

The following diagram illustrates the complete masked gene prediction pretraining workflow:

Implementation Protocol

Phase 1: Data Curation and Preprocessing

- **Data Collection**: Compile diverse single-cell datasets from public repositories (CELLxGENE, GEO, SRA, Human Cell Atlas) encompassing multiple tissues, conditions, and species [33]
- **Quality Control**: Filter cells based on quality metrics (mitochondrial content, number of detected genes) and remove low-quality datasets [33]
- **Normalization**: Apply appropriate normalization methods to account for technical variation (sequencing depth, batch effects) while preserving biological signal [33]
- **Gene Selection**: Curate a standardized gene vocabulary, typically focusing on highly variable genes to reduce dimensionality [33]

Phase 2: Model Configuration

- **Architecture Selection**: Choose appropriate transformer configuration (encoder-based, decoder-based, or hybrid) based on intended applications [33]
- **Hyperparameter Tuning**: Optimize model dimensions (embedding size, number of layers, attention heads) based on available computational resources and data scale [33]
- **Tokenization Scheme**: Implement chosen tokenization strategy with appropriate value discretization and positional encoding [33]

Phase 3: Pretraining Execution

- **Training Initialization**: Initialize model parameters using standard strategies (Xavier uniform, etc.)
- **Masked Training**: Implement iterative training with dynamic masking patterns
- **Validation Monitoring**: Track reconstruction loss on held-out validation cells to prevent overfitting
- **Checkpointing**: Save model checkpoints at regular intervals for subsequent fine-tuning

Evaluation Frameworks and Benchmarking

Performance Metrics

Comprehensive evaluation of scFMs requires multiple metrics assessing different aspects of model performance:

Table 3: scFM Evaluation Metrics

Metric Category	Specific Metrics	Biological Interpretation
Reconstruction Quality	Mean Squared Error, Mean Absolute Error	Precision in predicting masked gene expressions
Gene Embedding Quality	Gene function prediction, Tissue specificity	Capturing functional gene relationships [1]
Cell Embedding Utility	Cell type annotation accuracy, Batch correction	Preserving biological identity while removing technical artifacts [1]
Biological Consistency	scGraph-OntoRWR, LCAD metrics	Alignment with established biological knowledge [1]

Benchmarking Insights

Recent comprehensive benchmarks reveal several key insights about scFMs trained with masked gene prediction:

- **Task-dependent performance**: No single scFM consistently outperforms others across all tasks, emphasizing the need for tailored model selection [1]
- **Biological relevance**: Models capture meaningful biological relationships, with gene embeddings reflecting functional similarities and cell embeddings preserving ontological relationships [1]
- **Comparison to baselines**: While scFMs show robust performance across diverse tasks, simpler machine learning methods can be more efficient for specific dataset-limited scenarios [1]
- **Zero-shot capabilities**: Pretrained models demonstrate impressive zero-shot performance on some tasks without fine-tuning, indicating genuine learning of biological principles [1]

The novel evaluation metrics such as scGraph-OntoRWR (which measures consistency of cell type relationships with biological knowledge) and LCAD (which assesses ontological proximity of misclassified cells) provide biologically-grounded assessment beyond traditional technical metrics [1].

Research Reagent Solutions

Implementing masked gene prediction requires both computational resources and biological data assets:

Table 4: Essential Research Resources for scFM Development

Resource Category	Specific Resources	Role in scFM Development
Data Repositories	CZ CELLxGENE, NCBI GEO, EBI Expression Atlas	Provide diverse training corpora [33]
Reference Atlases	Human Cell Atlas, PanglaoDB, Tabula Muris	Offer comprehensive cell type benchmarks [33]
Annotation Resources	CellMarker, Gene Ontology, MSigDB	Enable biological interpretation of learned representations [33]
Software Frameworks	Scanpy, Seurat, SCVI	Facilitate data preprocessing and baseline comparisons [1]
Computational Resources	GPU clusters, High-memory servers	Enable training of large transformer models [33]

Technical Challenges and Limitations

Despite their promising capabilities, scFMs trained with masked gene prediction face several significant challenges:

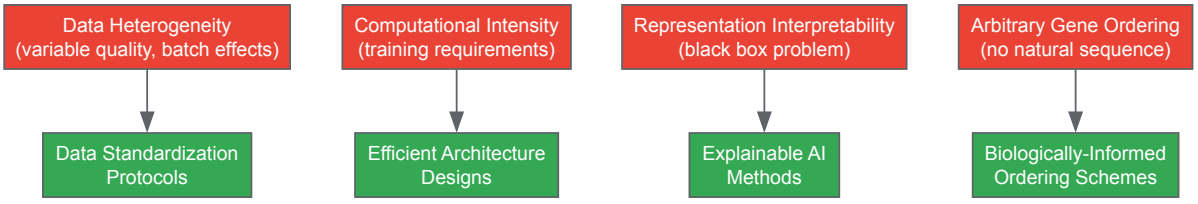
Data and Computational Constraints

- **Data heterogeneity**: Integrating datasets with different technologies, protocols, and quality levels introduces technical artifacts that can confound biological signals [33]
- **Computational intensity**: Training large transformer models on millions of cells requires substantial computational resources, limiting accessibility [33]
- **Batch effects**: Technical variations between datasets can persist in embeddings despite pretraining, requiring specialized correction approaches [33]

Biological Interpretation Barriers

- **Representation interpretability**: Understanding what specific biological knowledge is encoded in model representations remains challenging [33]
- **Gene ordering arbitrariness**: The lack of natural gene sequencing requires artificial ordering schemes whose impact on learned representations is not fully understood [33]
- **Validation complexity**: Thoroughly validating that models capture true biological relationships rather than technical artifacts requires extensive biological expertise [1]

The following diagram illustrates the relationship between these challenges and potential solution directions:



Emerging Innovations

The field of scFMs pretrained with masked gene prediction is rapidly evolving, with several promising research directions:

- **Multimodal integration**: Combining transcriptomic data with epigenetic, proteomic, and spatial information to create more comprehensive cellular representations [33]
- **Transfer learning architectures**: Developing more efficient fine-tuning approaches that require minimal labeled data for specialized applications [33]
- **Interpretability enhancements**: Creating methods to extract biologically meaningful insights from model attention patterns and latent representations [33] [1]
- **Generative capabilities**: Extending models beyond representation learning to in silico simulation of cellular responses to perturbations [33]

Masked gene prediction has established itself as a core pretraining paradigm for single-cell foundation models, enabling learning of transferable biological representations from vast unlabeled datasets. This technical overview has detailed the conceptual foundations, implementation

methodologies, and evaluation frameworks essential for researchers deploying these approaches. As the field matures, addressing current limitations around interpretability, computational demands, and biological validation will be crucial for realizing the full potential of scFMs in both basic research and therapeutic development.

The integration of masked prediction with increasingly diverse multimodal data and more biologically-informed architectures promises to yield even more powerful models capable of unraveling the complex regulatory logic underlying cellular function and dysfunction. For drug development professionals and researchers, these advances offer exciting opportunities to accelerate target discovery, patient stratification, and therapeutic optimization through deeper computational understanding of cellular biology.

Single-cell foundation models (scFMs) represent a transformative approach in computational biology, adapting the "pre-train then fine-tune" paradigm from natural language processing to single-cell omics data. These large-scale deep learning models are pretrained on vast datasets comprising tens of millions of single-cell transcriptomes, enabling them to learn fundamental biological principles that generalize across diverse downstream tasks [6]. The emergence of scFMs addresses critical challenges in single-cell genomics, where the exponential growth of data has created an urgent need for unified frameworks capable of integrating and comprehensively analyzing rapidly expanding biological repositories [6].

These models typically employ transformer architectures to process single-cell data by drawing an analogy to language: individual cells are treated as "sentences" while genes or genomic features become "words" or "tokens" [6]. Through self-supervised pretraining on massive corpora of single-cell data, scFMs develop rich internal representations that capture complex gene-gene relationships and cellular states. This foundational knowledge can then be efficiently adapted to specialized applications with relatively few additional labeled examples, making scFMs particularly valuable for biological discovery where labeled data is often scarce [6] [1].

Within the ecosystem of single-cell analysis, two applications have emerged as critical benchmarks for scFM performance: cell type annotation, which involves classifying cells into known biological categories, and batch integration, which aligns datasets from different experimental conditions to remove technical artifacts while preserving biological variation [1] [34]. These complementary applications represent fundamental prerequisites for constructing unified cell atlases, comparing healthy and diseased tissues, and identifying novel cell states – each essential for advancing both basic biology and therapeutic development [6] [1].

Core Methodological Framework of scFMs

Architectural Foundations and Tokenization Strategies

scFMs predominantly build upon transformer architectures, leveraging attention mechanisms to model complex dependencies between genes within individual cells. Most implementations adopt either encoder-based (BERT-like) or decoder-based (GPT-like) configurations, with each offering distinct advantages for different biological tasks [6]. The encoder-based models utilize bidirectional attention, processing all genes in a cell simultaneously to build comprehensive representations, while decoder-based models employ masked self-attention mechanisms that iteratively predict masked genes conditioned on known expression patterns [6].

A fundamental challenge in applying transformer architectures to single-cell data is the non-sequential nature of gene expression information. Unlike words in a sentence, genes lack inherent ordering, requiring scFMs to implement various tokenization strategies to structure the input. Common approaches include:

- **Expression-based ranking**: Genes are ordered by their expression levels within each cell, creating a deterministic sequence from highest to lowest expressed genes [6]
- **Value binning**: Expression values are partitioned into discrete bins, with each bin representing a distinct token [6]
- **Genomic positioning**: Some newer approaches like scMamba organize genes according to their genomic coordinates, preserving spatial relationships in the genome [35]
- **Patch-based tokenization**: Advanced models process genomic regions as cohesive patches, treating these aggregated features as input tokens [35]

Following tokenization, genes are converted to embedding vectors that typically combine a gene identifier embedding with information about its expression value in the given cell. Positional encoding schemes are then applied to represent the relative order or rank of each gene, enabling the transformer architecture to process the structured input [6].

The power of scFMs stems primarily from their pretraining phase, where models learn generalizable biological principles from massive, diverse collections of single-cell data. Key public data repositories used for scFM pretraining include:

- **CZ CELLxGENE**: Provides unified access to over 100 million annotated single-cells [6]
- **Human Cell Atlas**: Offers broad coverage of cell types and states across multiple organs [6]
- **PanglaoDB and Ensemble Cell Atlas**: Curated compendia aggregating data from multiple sources and studies [6]

During pretraining, scFMs typically employ self-supervised objectives similar to those used in natural language processing. The most common approach involves masked gene prediction, where a subset of genes in each cell is masked, and the model must predict their values based on the

remaining context [6]. Through this process, the model learns the complex covariance structure of gene expression and develops representations that capture fundamental biological relationships.

More advanced pretraining strategies incorporate contrastive learning objectives, particularly for multimodal integration. For instance, scMamba employs cosine similarity regularization to align representations across different omics modalities, enabling more effective integration of complementary data types [35].

Cell Type Annotation with scFMs

Methodological Approaches

Cell type annotation represents one of the most immediate and valuable applications of scFMs, transforming raw gene expression data into biologically meaningful categorizations. scFMs approach this task through several distinct methodologies:

Embedding-based annotation leverages the latent representations learned during pretraining. Cells are projected into an embedding space where distances reflect biological similarity, enabling classification through reference mapping or clustering. The zero-shot capabilities of scFMs allow these embeddings to capture meaningful biological structure even without task-specific fine-tuning, as demonstrated by benchmark studies showing that scFM embeddings preserve relationships consistent with established biological knowledge [1].

Fine-tuned classification adapts the pretrained models to specific annotation tasks through additional training on labeled reference datasets. This approach typically modifies the model's final layers to predict specific cell type categories, leveraging the transfer learning capabilities of foundation models. Studies have shown that fine-tuned scFMs can achieve impressive performance, with models like scGPT demonstrating accuracy of 73.4% in comparative evaluations [30].

Search-based annotation implements content-based retrieval systems analogous to reverse image search. Tools like Cell Annotation Service (CAS) use scFMs to generate compact "signatures" for both query cells and reference databases, enabling rapid identification of similar cells and transfer of annotations [36]. This approach benefits from continuously expanding reference atlases, with current systems incorporating approximately 87 million cells from nearly 1,400 published studies [36].

Experimental Protocols and Benchmarking

Comprehensive benchmarking studies have evaluated the performance of scFMs against traditional cell annotation methods, employing metrics designed to assess both accuracy and biological plausibility:

Protocol for zero-shot embedding evaluation:

- Extract cell embeddings from pretrained scFMs without additional fine-tuning
- Project embeddings into lower-dimensional space using UMAP or t-SNE
- Apply clustering algorithms (e.g., Leiden clustering) to identify cell groups
- Evaluate clustering quality using metrics such as Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI)
- Assess biological consistency using ontology-informed metrics like Lowest Common Ancestor Distance (LCAD) [1]

Protocol for fine-tuning evaluation:

- Initialize model with pretrained weights
- Replace classification head with task-specific layers
- Fine-tune on annotated reference datasets using cross-validation
- Evaluate on held-out test sets using accuracy, F1-score, and other classification metrics
- Compare against baseline methods including traditional machine learning and manual annotation [1] [30]

Table 1: Performance Comparison of scFMs in Cell Type Annotation Tasks

Model	Annotation Accuracy	ARI	NMI	Specialization
scGPT	73.4%	0.65	0.72	General cell types
scBERT	68.2%	0.61	0.69	Immune cells
Geneformer	70.1%	0.63	0.71	Developmental trajectories
scMamba	75.6%	0.68	0.74	Multi-omics integration

Benchmarking results indicate that while scFMs generally outperform traditional methods, no single model dominates across all scenarios. Performance varies based on factors including dataset size, cell type complexity, and computational resources, highlighting the importance of context-specific model selection [1].

Batch Integration with scFMs

Technical Approaches and Challenges

Batch integration addresses a fundamental challenge in single-cell genomics: the presence of non-biological technical variation between datasets generated under different conditions, using different protocols, or at different times. scFMs approach this problem through several technical paradigms:

Latent space alignment leverages the unified representation space learned by scFMs during pretraining. By projecting cells from different batches into a shared embedding space, technical variations are naturally minimized while biological signals are preserved. Models like scMamba implement contrastive learning objectives with cosine similarity regularization to explicitly optimize for batch-invariant representations [35].

Domain adaptation techniques modify the pretraining objectives to explicitly account for batch effects. These approaches may incorporate batch-specific tokens or adversarial training strategies that learn to disentangle biological signals from technical artifacts [6]. Advanced implementations can simultaneously handle multiple integration challenges, including cross-species, cross-tissue, and cross-technology alignment [1].

Deep metric learning approaches, exemplified by methods like scDML, use triplet loss functions to pull cells of the same type together in embedding space while pushing apart cells of different types, regardless of their batch origins [34]. These methods typically operate on initial high-resolution clusters, preserving rare cell populations that might be lost in traditional integration pipelines.

A significant challenge in batch integration is avoiding the removal of meaningful biological variation while eliminating technical artifacts. Recent studies have demonstrated that conventional integration methods can inadvertently remove subtle but biologically important signals [37]. scFMs address this challenge through their comprehensive understanding of cellular biology learned during large-scale pretraining, enabling more nuanced discrimination between technical and biological variation.

Evaluation Frameworks and Metrics

Rigorous evaluation of batch integration methods employs multiple complementary metrics assessing both technical artifact removal and biological signal preservation:

Protocol for batch integration evaluation:

- Apply integration method to multi-batch dataset
- Project integrated data into low-dimensional space (UMAP, t-SNE)
- Apply clustering algorithms to identify cell groups
- Quantify batch mixing using batch-aware metrics (iLISI, ASW_batch, BatchKL)
- Assess biological preservation using cell type-aware metrics (ARI, NMI, ASW_celltype)
- Evaluate rare cell type preservation through focused analysis of small clusters [1] [34]

Advanced evaluation techniques:

- **Biological validation**: Use orthogonal assays or known biological relationships to validate preserved signals [37]
- **Rare cell type analysis**: Quantify the preservation percentage of known rare populations after integration [34]
- **Trajectory preservation**: Evaluate whether continuous biological processes remain coherent after integration [35]

Table 2: Performance Comparison of Integration Methods Across Multiple Datasets

Method	iLISI (Batch Mixing)	ARI (Cell Type)	Rare Cell Preservation	Scalability
scDML	0.82	0.89	92%	High
Harmony	0.78	0.79	85%	High
scVI	0.75	0.81	83%	Medium
scMamba	0.85	0.87	90%	High
Seurat	0.72	0.76	78%	Medium

Evaluation studies consistently show that scFM-based integration methods outperform traditional approaches, particularly in preserving rare cell types and maintaining biological variation. For instance, scDML demonstrates superior performance in both batch mixing and cell type preservation across diverse tissue types and experimental conditions [34].

Integrated Workflow and Visualization

End-to-End Experimental Protocol

Implementing scFMs for cell annotation and batch integration typically follows a structured workflow that leverages the strengths of foundation models while incorporating domain-specific validation:

Data preprocessing and quality control:

- Perform standard QC filtering (minimum genes per cell, minimum cells per gene)
- Identify and remove doublets using Scrublet or similar tools [38]
- Normalize using count depth scaling (e.g., median normalization) with log1p transformation
- Select highly variable genes (2000-5000 genes) while considering batch-specific effects [38]

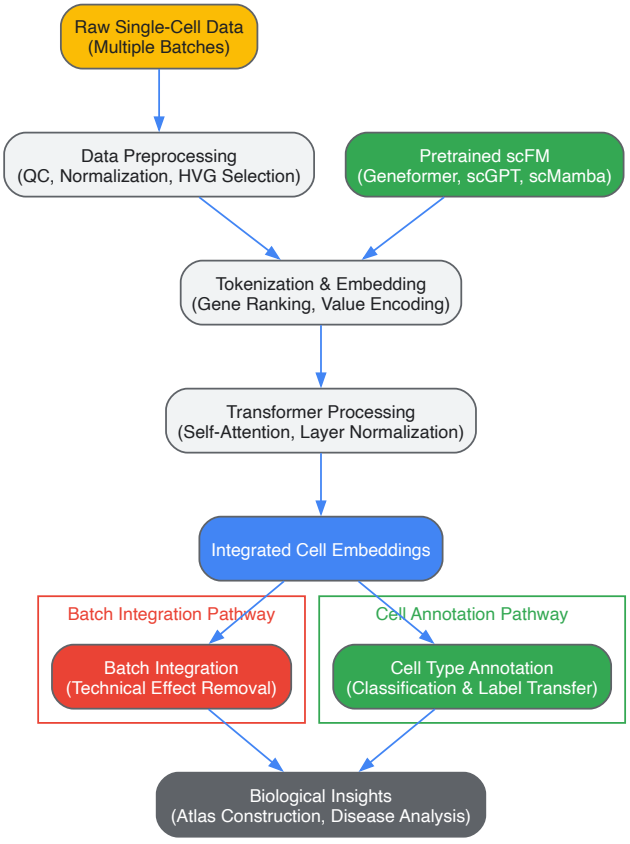
Model selection and application:

- Choose appropriate scFM based on data characteristics and task requirements
- For batch integration: Generate joint embeddings using pretrained or fine-tuned models
- For cell annotation: Either extract embeddings for reference-based annotation or fine-tune for direct classification
- Project results into low-dimensional space for visualization and quality assessment

Validation and iteration:

- Assess integration quality using multiple complementary metrics
- Validate cell annotations using marker gene expression and biological consistency checks
- Perform differential expression analysis to confirm annotation specificity
- Iterate on model parameters or annotation labels as needed

Visual Workflow Representation



scFM Application Workflow for Annotation and Integration

The diagram above illustrates the integrated workflow for applying scFMs to both cell annotation and batch integration tasks. The process begins with raw single-cell data from multiple batches, which undergoes standardized preprocessing before being tokenized and processed through the transformer architecture of a pretrained scFM. The resulting cell embeddings simultaneously support both batch integration (removing technical artifacts while preserving biological variation) and cell annotation (enabling classification and label transfer), ultimately generating biological insights through atlas construction and comparative analysis.

Implementing scFM-based approaches for cell annotation and batch integration requires both computational frameworks and biological reference data. The following table summarizes key resources mentioned in recent literature:

Table 3: Essential Research Resources for scFM Applications

Resource	Type	Function	Application Context
CZ CELLxGENE	Data Repository	Provides unified access to >100 million annotated single-cells	Pretraining data source, reference for annotation [6]
Cell Annotation Service (CAS)	Tool	Machine learning-based search engine for rapid cell annotation	Label transfer for new datasets [36]
scGPT	Software Framework	Decoder-based scFM for various single-cell tasks	Cell type annotation, perturbation prediction [6] [30]
scMamba	Software Framework	scFM with patch-based tokenization for multi-omics	Multi-omics integration, batch correction [35]
Harmony	Algorithm	Iterative PCA correction for dataset integration	Baseline comparison for batch integration [1] [35]
Scanpy	Software Library	Python-based single-cell analysis toolkit	Data preprocessing, visualization, and analysis [38]
CellANOVA	Statistical Method	Recovers biological signals lost during integration	Validation of biological preservation [37]
Human Cell Atlas	Data Resource	Reference atlas of cell types across human tissues	Annotation reference, model pretraining [6]

These resources collectively enable researchers to implement comprehensive workflows for single-cell analysis, from initial data processing through advanced integrative analysis using foundation models.

Single-cell foundation models have established themselves as powerful tools for two of the most critical tasks in single-cell analysis: cell type annotation and batch integration. Through their pretraining on massive, diverse datasets, scFMs develop a fundamental understanding of cellular biology that enables robust performance across diverse biological contexts and technical conditions.

The benchmarking studies summarized in this technical guide demonstrate that scFMs consistently outperform traditional methods, particularly in challenging scenarios involving rare cell types, cross-tissue comparisons, and complex biological systems [1]. Their ability to leverage large-scale pretraining makes them especially valuable as single-cell datasets continue to grow in size and complexity.

Future developments in scFMs will likely focus on several key areas: (1) enhanced interpretability through techniques like transcoder-based circuit analysis, which extracts biologically plausible pathways from model decisions [2]; (2) improved multimodal integration capabilities that more effectively leverage complementary omics data types [35]; and (3) more efficient training and inference methods that reduce computational barriers to adoption.

As these models continue to evolve, they will play an increasingly central role in both basic biological research and therapeutic development, enabling more comprehensive cell atlas construction, more accurate disease characterization, and ultimately more targeted therapeutic interventions. The integration of scFMs into standardized analytical workflows represents a significant advancement in our ability to extract meaningful biological insights from the complex landscape of single-cell data.

Single-cell foundation models (scFMs) represent a paradigm shift in computational biology, leveraging large-scale pretraining on millions of single-cell transcriptomes to create universal biological representations. These models, built primarily on transformer architectures, have demonstrated remarkable capabilities in capturing complex gene-gene relationships and cellular states [33]. The application of scFMs to perturbation prediction and drug response modeling marks a significant advancement in personalized medicine and therapeutic development. By learning the fundamental "language" of biology—where cells are treated as sentences and genes as words—these models can extrapolate cellular behaviors under novel conditions, including previously unseen drug treatments and cell types [33] [39].

The foundational architecture of scFMs enables this capability through several key mechanisms. First, their pretraining on diverse cellular contexts encompassing multiple tissues, species, and disease states provides a comprehensive representation of biological space. Second, the self-attention mechanism inherent in transformer architectures allows scFMs to model complex, non-linear relationships between genes and pathways. Third, through techniques like transfer learning and fine-tuning, these models can adapt their general biological knowledge to specific predictive tasks with limited additional data [33] [40]. This combination of breadth and adaptability makes scFMs uniquely positioned to address the formidable challenges in drug response prediction, particularly the need to generalize to novel chemical compounds and cellular contexts.

Technical Foundations of Perturbation Prediction

Architectural Framework

The application of scFMs to perturbation prediction requires specialized architectural adaptations to handle the unique challenges of the domain. Most scFMs utilize transformer-based architectures, which can be broadly categorized into encoder-based models (e.g., scBERT) and decoder-based models (e.g., scGPT) [33]. These models process single-cell data through a tokenization process where individual genes or genomic features become input tokens, analogous to words in a sentence. Critical to their success is how these models handle the non-sequential nature of genomic data—some approaches rank genes by expression levels within each cell, while others partition genes into expression bins or use normalized counts directly [33].

For perturbation prediction specifically, researchers have developed innovative fine-tuning approaches that preserve the rich biological knowledge encoded during pretraining while adapting to new tasks. The single-cell Drug-Conditional Adapter (scDCA) framework exemplifies this approach, introducing parameter-efficient fine-tuning that trains less than 1% of the original model parameters [40]. This method incorporates a drug-conditional adapter layer that injects molecular information into the model while keeping the original scFM weights frozen, effectively bridging the gap between cellular representations and chemical structures without catastrophic forgetting of pretrained knowledge [40].

Key Methodological Challenges

Several significant technical challenges arise when applying scFMs to perturbation prediction. The high dimensionality and sparsity of single-cell data require specialized handling, as does the integration of multimodal information—particularly chemical structures of drugs, which represent a completely different modality from the gene expression data on which scFMs are pretrained [40]. Additionally, batch effects across experiments and platforms introduce technical noise that can obscure biological signals, necessitating robust integration techniques [33] [1].

Perhaps the most formidable challenge is the limited availability of perturbation data, which creates a few-shot learning scenario. While scFMs are pretrained on millions of cells, experimental data for specific drug perturbations may encompass only hundreds of examples [40]. This data scarcity is further compounded by the need to predict responses for unseen cell types and novel chemical compounds, requiring sophisticated generalization capabilities beyond standard supervised learning approaches [39] [40].

Experimental Design and Benchmarking Frameworks

Standardized Evaluation Protocols

Robust evaluation is critical for assessing scFM performance in perturbation prediction. Two primary evaluation scenarios have emerged in the literature: pooled-data evaluation and cross-data evaluation [27]. In pooled-data evaluation, models are trained and tested on aggregated data from multiple studies, testing the model's ability to integrate diverse data sources. In cross-data evaluation, models are tested on datasets from individual studies not seen during training, providing a more challenging assessment of generalizability [27].

The CRISP framework introduces a specialized evaluation protocol for drug perturbation response prediction that incorporates increasingly challenging scenarios, from unseen cell types to cross-platform predictions [39]. This approach employs transfer learning strategies with foundation models to enable effective information transfer from control to perturbed states even with limited empirical data. Evaluation typically focuses on the model's ability to predict transcriptional responses to novel drugs and generalize to unseen cell lines in a zero-shot manner [40].

Benchmarking Results and Performance Metrics

Comprehensive benchmarking studies reveal varied performance across scFMs for drug response prediction. The table below summarizes key performance metrics from recent large-scale evaluations:

Table 1: Performance Comparison of scFMs in Drug Response Prediction

Model	Evaluation Scenario	Key Performance Metrics	Notable Strengths
scFoundation	Pooled-data evaluation	Mean F1 score: 0.971 (layer-freezing), 0.947 (fine-tuning) [27]	Excels in integrated data analysis
UCE	Cross-data evaluation (fine-tuned)	Mean F1 score: 0.774 (tumor tissue) [27]	Strong fine-tuning capability
scGPT	Cross-data evaluation (zero-shot)	Mean F1 score: 0.858 [27]	Superior zero-shot generalization
CRISP	Unseen cell type prediction	Demonstrated successful transfer learning [39]	Effective for cross-cell-type inference

A separate benchmark evaluating six scFMs against traditional methods revealed that no single model consistently outperforms others across all tasks, emphasizing the need for task-specific model selection [4] [1]. Factors such as dataset size, task complexity, and computational resources

significantly influence optimal model choice. The introduction of biology-informed metrics like scGraph-OntoRWR, which measures consistency of cell type relationships with prior biological knowledge, provides additional dimensions for model evaluation beyond traditional performance metrics [1].

Signaling Pathways and Biological Mechanisms

Key Pathways in Drug Response

scFMs have enabled the identification of critical signaling pathways involved in drug response mechanisms. For example, the application of CRISP to sorafenib in chronic myeloid leukemia (CML) revealed inhibition of the CXCR4 pathway as a key therapeutic mechanism, a finding supported by independent studies and clinical trials [39]. This demonstrates how scFMs can uncover biologically plausible mechanisms that align with established knowledge while potentially revealing novel insights.

The attention mechanisms within transformer-based scFMs provide a unique window into gene-gene interactions and pathway relationships. By analyzing attention weights, researchers can identify which genes and relationships the model deems important for specific predictions, creating opportunities for hypothesis generation about underlying biological mechanisms [33] [1]. This capability is particularly valuable for understanding complex, multi-factorial drug responses that involve coordinated changes across multiple pathways.

Visualizing Key Relationships

The following diagram illustrates the core workflow for perturbation prediction using single-cell foundation models, highlighting the integration of single-cell data with drug information:

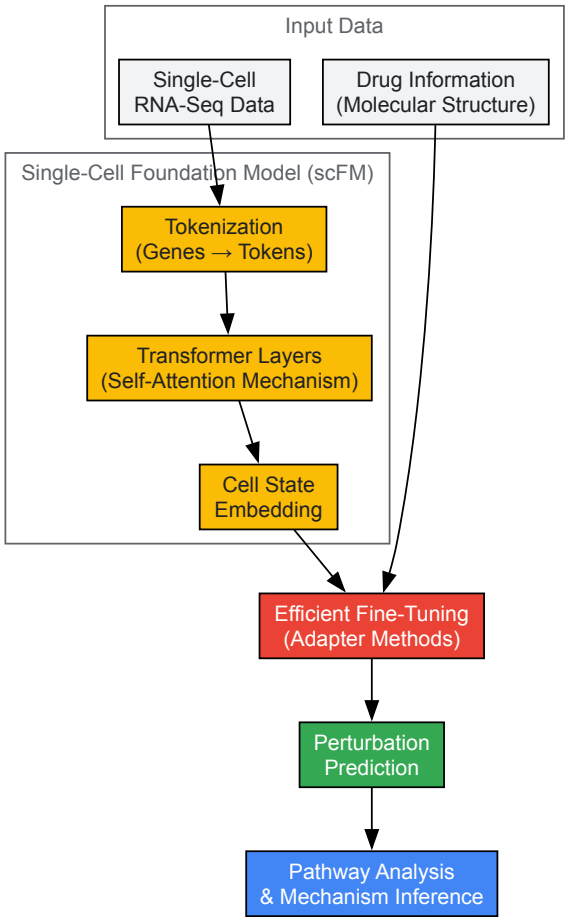


Diagram 1: scFM Perturbation Prediction Workflow

Research Reagent Solutions and Experimental Materials

The effective implementation of scFMs for perturbation prediction requires specific computational resources and datasets. The following table details essential components of the research toolkit:

Table 2: Essential Research Reagents and Computational Resources

Resource Category	Specific Examples	Function/Role	Implementation Notes
Foundation Models	scGPT, Geneformer, scFoundation, UCE, scBERT [27]	Provide pretrained biological representations	Selection depends on task: scGPT for multi-omics, scFoundation for drug response [27]
Computational Frameworks	CRISP [39], scDrugMap [27], scDCA [40]	Specialized architectures for perturbation prediction	scDrugMap offers both command-line tool and web server [27]
Data Resources	CZ CELLxGENE [33], GEO/SRA [33], PanglaoDB [33]	Provide training and benchmarking data	CELLxGENE contains >100 million standardized cells [33]
Fine-Tuning Methods	Low-Rank Adaptation (LoRA) [27], Drug-Conditional Adapters [40]	Enable efficient model adaptation	LoRA trains <1% of parameters [40]
Evaluation Datasets	Primary collection: 326,751 cells from 36 datasets [27]	Benchmark model performance	Span 14 cancer types, 3 therapy types [27]

Implementation Protocols and Methodologies

Step-by-Step Experimental Guide

Implementing scFMs for perturbation prediction involves a systematic process beginning with data preparation and culminating in model evaluation. The following protocol outlines key steps:

- **Data Acquisition and Preprocessing**: Curate single-cell datasets from resources like CELLxGENE or GEO. Implement rigorous quality control including cell filtering, normalization, and batch effect correction. For drug response prediction, ensure proper annotation of perturbation conditions and responses [27].
- **Model Selection and Setup**: Choose an appropriate scFM based on task requirements. For multi-omics integration, scGPT is recommended; for specialized drug response prediction, scFoundation may be preferable [27]. Initialize the model with pretrained weights.
- **Tokenization and Input Representation**: Convert gene expression matrices into token sequences. Common approaches include ranking genes by expression levels or binning expression values. Incorporate positional encodings to provide sequence context [33].
- **Efficient Fine-Tuning**: Implement parameter-efficient fine-tuning using adapter-based methods like LoRA or drug-conditional adapters. For scDCA, this involves training adapter layers that condition on drug molecular structures while keeping base model parameters frozen [40].
- **Validation and Interpretation**: Evaluate model performance using appropriate metrics (F1 score, accuracy, etc.). Perform biological validation through attention analysis and pathway enrichment to ensure predictions align with known biology [1].

Advanced Technical Considerations

For researchers implementing these methods, several advanced considerations can enhance results. First, consider the roughness index (ROGI) as a proxy for dataset complexity to guide model selection [1]. Second, incorporate biology-informed evaluation metrics like scGraph-OntoRWR to assess whether model-predicted cell relationships align with ontological knowledge [1]. Third, for optimal performance in cross-data evaluation scenarios, employ ensemble approaches that leverage multiple scFMs tailored to different aspects of the prediction task.

When generalizing to unseen cell types, the CRISP framework demonstrates that transfer learning strategies specifically designed for foundation models significantly outperform generic approaches [39]. Similarly, for zero-shot prediction to novel cell lines, the scDCA method shows that drug-conditional adapters enable generalization by separating cellular context from drug mechanism [40].

The application of scFMs to perturbation prediction and drug response modeling represents a rapidly advancing frontier with significant potential for transformative impact on therapeutic development. Current research indicates several promising directions for future work, including the development of multi-modal foundation models that integrate single-cell data with protein structures, clinical information, and chemical properties [40]. Additionally, methods for improved interpretability, such as enhanced attention mechanism analysis and biologically constrained model architectures, will be crucial for building trust and facilitating biological discovery.

As benchmark studies have consistently shown, the field is moving beyond simple performance comparisons to more nuanced evaluations of biological relevance and clinical utility [4] [1]. The introduction of frameworks like scDrugMap [27] and methodologies like scDCA [40] provide robust platforms for continued innovation. While challenges remain—particularly in data scarcity, model interpretability, and computational resource requirements—the rapid progress in single-cell foundation models suggests a future where predictive in silico drug screening becomes an integral component of therapeutic development, accelerating the journey from basic research to clinical application.

Single-cell RNA sequencing (scRNA-seq) has revolutionized our understanding of cellular heterogeneity but requires cell dissociation, thereby losing critical information about the native cellular microenvironment [41]. This spatial context is fundamental to biological processes, encompassing cell-cell communication, spatial gradients, and the emergent properties of tissue niches. The emergence of image-based spatial transcriptomics technologies now enables *in situ* profiling of gene expression, revealing spatial components of cellular variation [41]. Concurrently, **single-cell foundation models (scFMs)** have arisen as powerful tools trained on massive datasets to learn universal patterns that can be adapted to diverse downstream tasks [33] [1]. However, most existing scFMs are trained exclusively on dissociated single-cell data, limiting their ability to recover the complexity of spatial microenvironments [41]. **Nicheformer** represents a pivotal advancement—a transformer-based foundation model explicitly designed to learn cell representations that capture spatial context by being trained on both dissociated and spatially resolved transcriptomics data [41] [42]. This in-depth technical guide explores the core architecture, functionalities, and applications of Nicheformer, framing it within the broader context of scFM research and its implications for drug discovery.

Core Architecture and Technical Innovation of Nicheformer

Model Design and Training Strategy

Nicheformer is built on a transformer architecture, which has become the backbone of modern foundation models due to its attention mechanism that effectively captures complex, long-range relationships in data [33]. The model's specific configuration consists of **12 transformer encoder layers**, each equipped with **16 attention heads** and a feed-forward network of size **1,024**, culminating in a **512-dimensional embedding** for each cell and totaling **49.3 million parameters** [41]. This architecture was selected after extensive pretraining experiments demonstrated its superior performance compared to smaller configurations [41].

A critical innovation in Nicheformer is its training on **SpatialCorpus-110M**, a curated collection of over 110 million cells that includes both dissociated single-cell data and spatially resolved transcriptomics data [41]. This corpus spans 73 human and mouse tissues and organs, providing unprecedented diversity. The incorporation of spatial data is not merely quantitative but qualitatively essential; models trained solely on dissociated data, even with three times more cells, showed significantly lower performance on spatial tasks, underscoring the indispensability of spatial data for learning microenvironmental context [41].

Table 1: Nicheformer Model Specifications

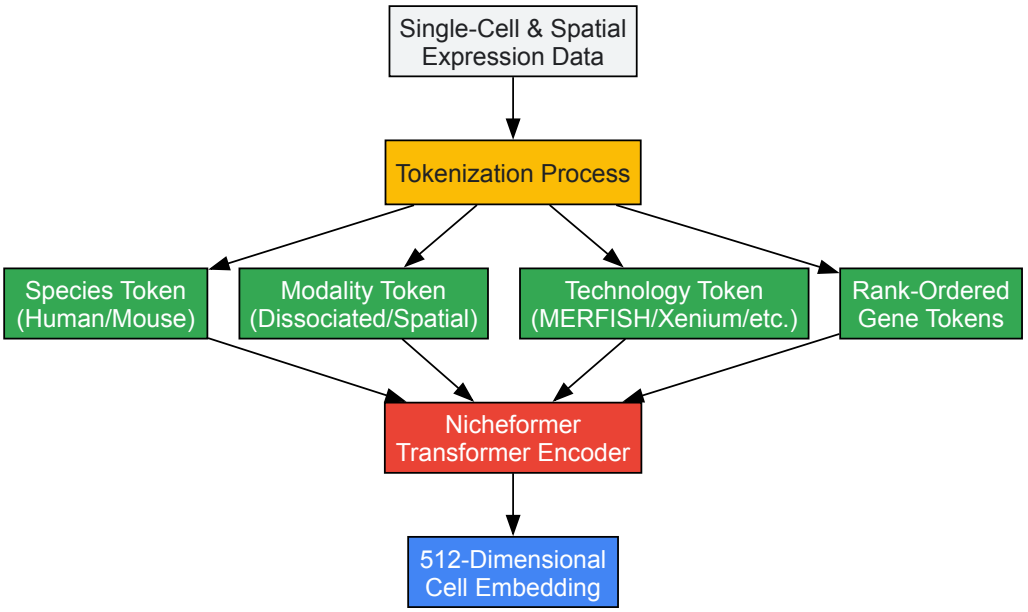
Component	Specification	Biological Significance
Architecture	Transformer Encoder	Models complex gene-gene interactions within a cell
Layers	12	Depth sufficient to capture hierarchical biological relationships
Attention Heads	16	Enables model to focus on different gene subsets simultaneously
Embedding Dimension	512	Balance between information richness and computational efficiency
Parameters	49.3 million	Scale necessary for learning complex biological patterns
Pretraining Corpus	SpatialCorpus-110M	Unifies dissociated and spatial data; enables spatial awareness

Tokenization Strategy for Multimodal Data

Tokenization—the process of converting raw data into discrete model-input units—poses a unique challenge in single-cell biology because gene expression data lacks the inherent sequence of natural language [33]. Nicheformer addresses this by representing each cell as a sequence of **gene tokens ordered by expression level** relative to a technology-specific mean, creating a deterministic sequence for the transformer to process [41]. This rank-based encoding has demonstrated robustness to technical variations and batch effects.

To enable multimodal and cross-species learning, Nicheformer implements several key strategies:

- **Unified Vocabulary**: A shared vocabulary of 20,310 gene tokens was constructed by concatenating orthologous protein-coding genes between humans and mice, plus species-specific genes [41].
- **Contextual Tokens**: Special tokens are incorporated to denote species (human/mouse), modality (dissociated/spatial), and specific spatial technologies (MERFISH, Xenium, CosMx, ISS), enabling the model to learn and adjust for their distinct characteristics [41].
- **Technology-Specific Normalization**: To address technology-dependent biases—where spatial data often yields higher gene counts—Nicheformer computes separate nonzero mean expression vectors for each assay type rather than using a global mean [41].



Benchmarking and Performance Evaluation

Novel Spatial Tasks and Evaluation Framework

The incorporation of spatial data enables Nicheformer to address a fundamentally new class of downstream tasks that previous scFMs trained only on dissociated data cannot perform effectively [41]. These spatially aware tasks represent biologically meaningful and nontrivial problems that move beyond standard cell-type annotation or batch integration:

- **Spatial Label Prediction**: Predicting human-annotated tissue niches or regions based on cellular gene expression profiles, effectively learning the relationship between transcriptome and spatial context [41].
- **Spatial Composition Prediction**: Defining a distance-based spatially homogeneous niche around each cell and predicting local cell-type density or composition, capturing microenvironmental organization principles [41].
- **Spatial Context Transfer**: Applying spatial context learned from spatial transcriptomics to dissociated scRNA-seq data, thereby enriching conventional single-cell data with spatial information [41].

In rigorous benchmarking, Nicheformer systematically outperformed existing foundation models (including Geneformer, scGPT, and UCE) and traditional embedding methods (such as scVI and PCA) on these spatial tasks [41]. This performance advantage persists in both fine-tuning scenarios and linear probing, where only a simple linear layer is trained on top of frozen Nicheformer embeddings [41].

Comparative Analysis with Other scFMs and Traditional Methods

Recent comprehensive benchmarking studies of scFMs provide context for evaluating Nicheformer's advancements. These studies reveal that no single scFM consistently outperforms all others across every task, emphasizing that model selection must be tailored to specific dataset characteristics and task requirements [4] [1]. While scFMs generally demonstrate robustness and versatility, simpler machine learning models can sometimes be more efficient for specific datasets, particularly under resource constraints [1].

Table 2: Performance Comparison of Single-Cell Foundation Models

Model	Training Data	Spatial Awareness	Key Strengths	Limitations
Nicheformer	110M cells (dissociated + spatial)	Native	Excels in spatial tasks; cross-species transfer	Computational intensity
Geneformer	30M cells (dissociated)	Limited	Effective for gene network inference	No inherent spatial context
scGPT	10M+ cells (dissociated)	Limited	Strong generative capabilities	No inherent spatial context
scBERT	Millions of cells (dissociated)	None	Optimized for cell-type annotation	Limited to classification tasks
UCE	Massive-scale dissociated	None	Scalability to very large datasets	No spatial context

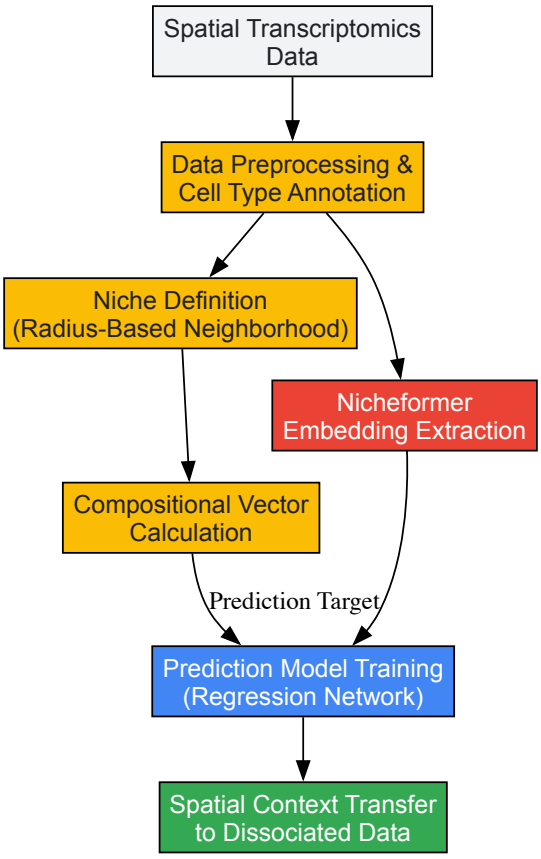
Notably, benchmarking analyses have introduced novel biological relevance metrics, such as **scGraph-OntoRWR**, which measures how well a model's captured cell-type relationships align with established biological knowledge from cell ontologies [1]. The **Lowest Common Ancestor Distance (LCAD)** metric further assesses the severity of cell-type misclassification by measuring ontological proximity between predicted and actual cell types [1]. Nicheformer's design principles suggest inherent advantages on such biologically grounded metrics, though specific results were not provided in the search results.

Experimental Protocols and Methodologies

Workflow for Spatial Composition Prediction

A core application of Nicheformer is predicting the spatial composition of cellular microenvironments. The following detailed protocol outlines how to implement this analysis:

- **Data Preprocessing**: Begin with a spatially resolved transcriptomics dataset (e.g., from MERFISH or Xenium). Annotate cell types using established markers or reference-based annotation tools. Normalize expression counts using technology-specific parameters as implemented in Nicheformer's preprocessing pipeline [41].
- **Niche Definition**: For each cell (the "anchor cell"), define a local neighborhood or niche. This is typically achieved by:
  - Setting a fixed radius (e.g., 50 micrometers) around the anchor cell's spatial coordinates.
  - Identifying all cells whose centroids fall within this radius.
  - Calculating the compositional vector for this niche, representing the proportions of each cell type present [41].
- **Embedding Extraction**: Pass the gene expression profile of the anchor cell through the pretrained Nicheformer model (with frozen weights) to obtain its 512-dimensional cell embedding [41].
- **Model Training for Prediction**:
  - **Architecture**: Use a simple feed-forward neural network with the Nicheformer embedding as input and the niche compositional vector as the prediction target.
  - **Training**: Train this network on a subset of the data where spatial coordinates are known, using mean squared error loss between predicted and actual composition vectors.
  - **Validation**: Validate performance on held-out spatial regions to ensure generalizability [41].
- **Spatial Context Transfer**: To transfer spatial context to dissociated scRNA-seq data, simply pass the dissociated cell expression profiles through the trained pipeline from Step 4. The model will predict the spatial microenvironment each dissociated cell would likely occupy based on its transcriptome [41].



Integration with Cell-Cell Communication Analysis

While Nicheformer focuses on spatial representation learning, its outputs can be powerfully integrated with specialized cell-cell communication tools like **NicheNet** to generate comprehensive hypotheses about intercellular signaling [43] [44]. NicheNet differs from many communication tools by incorporating not just ligand-receptor expression but also downstream transcriptional responses, using a prior knowledge model that integrates ligand-receptor interactions, signaling pathways, and gene regulatory networks [43] [44] [45].

A typical integrated analysis workflow:

- Use Nicheformer to identify spatially co-localized cell populations or predict spatial context for dissociated cells.
- Based on these spatial relationships, define putative sender and receiver cell populations for communication analysis.
- Apply NicheNet to prioritize ligands from sender cells that may explain expression changes in receiver cells [44].
- Validate predictions using NicheNet's signaling path inference to identify potential mediators connecting ligands to target genes [43].

This integrated approach leverages the respective strengths of both platforms: Nicheformer's spatial awareness and NicheNet's specialized knowledge of signaling pathways.

The Scientist's Toolkit: Essential Research Reagents

Implementing Nicheformer and related spatial analyses requires both computational tools and data resources. The following table details key components of the research toolkit.

Table 3: Essential Research Reagents and Resources

Resource	Type	Function/Purpose	Access
Nicheformer Codebase	Software	Primary model implementation; fine-tuning and inference	GitHub: theislab/nicheformer [42]
Pretrained Weights	Model Parameters	Transfer learning; avoids costly pretraining	Mendeley Data [42]
SpatialCorpus-110M	Data Resource	Training data; reference for cross-dataset integration	Upon request from authors [41]
NicheNet R Package	Software	Cell-cell communication inference from expression data	GitHub: saeyslab/nichenetr [43]
CZ CELLxGENE	Data Resource	Curated single-cell datasets for model validation	cellxgene.cziscience.com [1]
Seurat / Scanpy	Software	Standard single-cell analysis preprocessing	CRAN / PyPI

Resource	Type	Function/Purpose	Access
Spatial Data	Data Resource	Validation datasets (MERFISH, Xenium, CosMx)	Vendor portals / original publications

Implications for Drug Discovery and Development

The integration of spatial biology with foundation models holds particular promise for pharmaceutical research, where understanding cellular microenvironment context is critical for target identification and validation [26] [46]. Single-cell technologies already contribute significantly to drug discovery by revealing cellular heterogeneity in disease tissues, identifying novel therapeutic targets, and predicting drug responsiveness [26] [46]. Nicheformer enhances these applications by adding the spatial dimension:

- **Target Identification in Tumor Microenvironments**: Spatial context enables identification of targets specific to disease-associated cell populations within their functional niches, such as immune cells in tumor regions with specific spatial configurations [41] [46].
- **Drug Sensitivity Prediction**: Models can incorporate spatial neighborhood information as a feature for predicting cellular responses to therapeutic perturbations, potentially increasing accuracy over models using dissociated data alone [1].
- **Toxicology and Safety Assessment**: Understanding how drugs affect spatial organization of tissues provides insights into mechanism-based toxicity, moving beyond simple biomarker detection [26].

As the field progresses, the combination of single-cell technologies, spatial resolution, and artificial intelligence is expected to further optimize therapeutic strategies and improve clinical outcomes, particularly in oncology and other complex diseases [46].

Nicheformer represents a significant evolution in single-cell foundation models by fundamentally addressing the critical dimension of spatial context. Its ability to learn joint representations from both dissociated and spatial transcriptomics data enables a new class of spatially aware downstream tasks that were previously inaccessible to computational biology [41]. When integrated with specialized tools like NicheNet for cell-cell communication inference, it provides a powerful framework for generating biologically testable hypotheses about microenvironmental regulation [43] [44].

While current benchmarking indicates that no single scFM is universally superior across all tasks [4] [1], Nicheformer establishes a new state-of-the-art for applications where spatial context is biologically decisive. Future developments will likely focus on enhancing model interpretability, reducing computational demands, and incorporating additional multimodal data streams. As these models mature, they will increasingly serve as pivotal tools in bridging high-resolution molecular profiling with tissue-level pathophysiology, ultimately accelerating the translation of basic biological insights into therapeutic innovations.

Navigating scFM Challenges: Zero-Shot Limits, Data Issues, and Optimization Frameworks

The emergence of single-cell foundation models (scFMs) has generated considerable excitement in computational biology. Trained on millions of single-cell RNA sequencing profiles using self-supervised objectives like masked gene prediction, these models promise to learn universal biological principles and generate powerful cell embeddings transferable to diverse downstream tasks without additional training—a capability known as **zero-shot application** [1] [47]. This potential is particularly valuable for exploratory biological discovery where labeled data for fine-tuning is unavailable [47].

However, recent rigorous benchmarking studies reveal a concerning trend: in zero-shot settings, these sophisticated models frequently **underperform simpler, established methods** across critical tasks including cell type annotation, batch integration, and perturbation prediction [47] [48] [49]. This performance gap challenges assumptions about the foundational biological knowledge captured during pretraining and highlights the need for careful model evaluation and selection in research applications.

Comprehensive Benchmarking Reveals Consistent Performance Gaps

Experimental Designs for Evaluating Zero-Shot Performance

Robust evaluation of scFMs requires standardized benchmarks that assess model capabilities under realistic conditions. Key experimental designs include:

- **Cell Type Clustering**: Models generate cell embeddings without fine-tuning, and clustering algorithms group cells by type. Performance is measured by how well embeddings separate known cell types while resisting confounding by technical batch effects [47] [49].
- **Batch Integration**: Embeddings are evaluated on their ability to mix cells from different experimental batches while preserving biological variation. Both qualitative visualization and quantitative metrics assess integration quality [1] [47].
- **Perturbation Effect Prediction**: Model embeddings are used to predict transcriptional responses to genetic or chemical perturbations without task-specific training [48].

These evaluations typically compare scFMs against traditional baselines including Highly Variable Genes (HVG) selection, anchor-based methods (Seurat), clustering-based integration (Harmony), and generative models (scVI) [1] [47].

Quantitative Performance Comparisons Across Critical Tasks

Table 1: Performance Comparison of Methods on Cell Type Clustering (AvgBIO Score)

Method	PBMC (12k)	Tabula Sapiens	Pancreas	Immune Dataset
HVG	0.75	0.72	0.68	0.71
Harmony	0.78	0.75	0.72	0.74
scVI	0.82	0.79	0.76	0.78
scGPT	0.74	0.65	0.61	0.63
Geneformer	0.58	0.52	0.49	0.51

Data adapted from Genome Biology benchmark studies [47]

Table 2: Batch Integration Performance (Batch Mixing Score)

Method	Pancreas	PBMC	Tabula Sapiens	Immune Dataset
HVG	0.82	0.85	0.79	0.81
Harmony	0.78	0.81	0.72	0.76
scVI	0.81	0.83	0.77	0.74
scGPT	0.69	0.73	0.68	0.71
Geneformer	0.52	0.55	0.51	0.53

Data adapted from Genome Biology benchmark studies [47]

The data reveal that **simpler methods consistently outperform scFMs** in zero-shot settings. In some cases, even basic HVG selection surpasses foundation models. Geneformer particularly struggles, often performing worse than all other methods [47] [49].

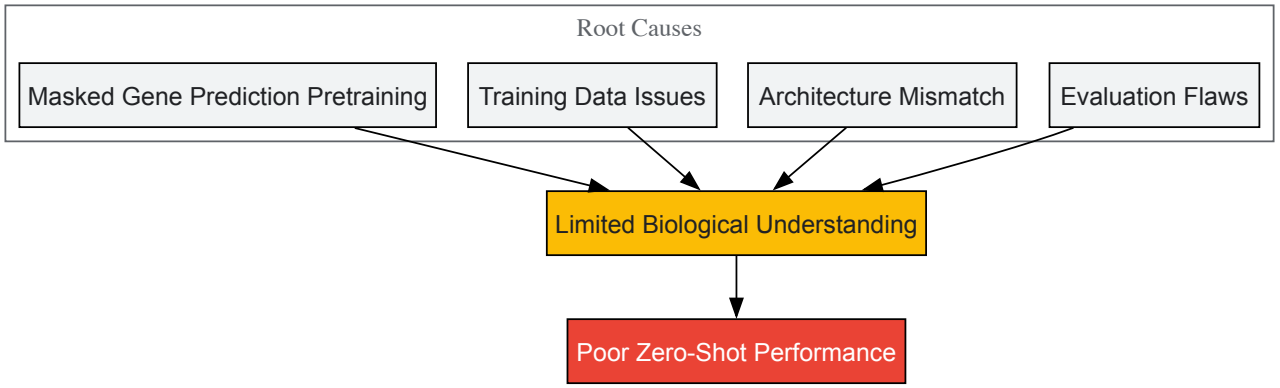
For perturbation prediction, the PertEval-scFM benchmark found scFM embeddings offered **limited improvement over simple baseline models**, particularly under distribution shift where models face data different from their training corpus [48].

Understanding the Limitations: Why scFMs Struggle Zero-Shot

Core Challenges in Current Training Paradigms

Several interconnected factors explain the zero-shot performance gap:

- **Pretraining Objective Misalignment**: Most scFMs employ masked language modeling where they predict randomly masked gene expressions. However, evidence suggests models may not deeply learn gene relationships, instead relying on superficial patterns. For example, scGPT often predicts median expression values regardless of context, indicating limited understanding of gene interactions [49].
- **Architectural Limitations**: Transformers adapted from natural language processing may not optimally capture gene-gene relationships, as genes lack the sequential dependencies of words in sentences [1].
- **Data Quality and Diversity Issues**: While trained on large datasets, the sparsity, noise, and technical variability in single-cell data may hinder learning of robust biological representations that generalize zero-shot [1] [48].
- **Evaluation Artifacts**: Previous emphasis on fine-tuned performance created overly optimistic assessments. Fine-tuning can enable models to exploit dataset-specific patterns without demonstrating true biological understanding [47] [49].



Root Cause Analysis: Why scFMs Struggle with Zero-Shot Tasks

Methodological Guidance for Researchers

Experimental Protocols for Rigorous scFM Evaluation

To properly assess scFM performance, researchers should implement these standardized protocols:

- **Zero-Shot Embedding Extraction**:
  - Download pretrained model weights without fine-tuning
  - Pass raw count matrices through model to extract cell embeddings
  - Prohibit any parameter updates or gradient-based adaptation
  - Use embeddings directly in downstream tasks [47]
- **Comprehensive Baseline Comparison**:
  - Implement HVG selection (2000 most variable genes)
  - Apply established integration methods (Harmony, Seurat, scVI)
  - Evaluate using multiple metrics (AvgBIO, ASW, batch mixing scores)
  - Assess statistical significance of performance differences [1] [47]
- **Biological Ground-Truth Validation**:
  - Utilize cell ontology-informed metrics (scGraph-OntoRWR)
  - Measure consistency with prior knowledge graphs
  - Evaluate clinical relevance through drug response prediction
  - Assess performance on novel cell types not in training data [1]

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Reagents for scFM Benchmarking Studies

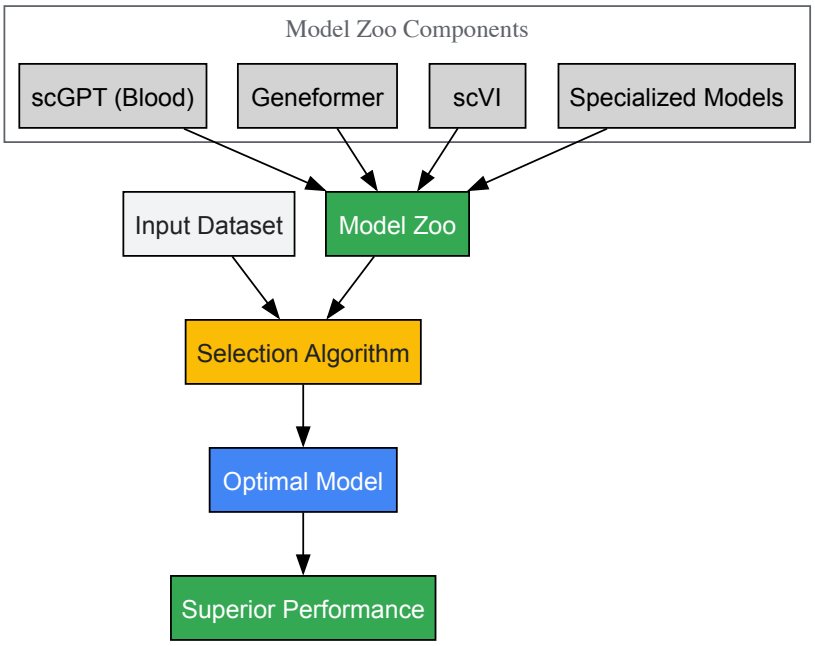
Reagent/Resource	Type	Function in Evaluation	Example Sources
Reference Datasets	Data	Provide ground truth for cell identity	CellxGene, AIDA v2 [1]
Benchmarking Frameworks	Software	Standardize model comparison	PertEval-scFM [48]
Ontology Metrics	Algorithm	Assess biological consistency	scGraph-OntoRWR, LCAD [1]
Traditional Baselines	Method	Establish performance floor	HVG, Harmony, scVI [47]
Visualization Tools	Software	Qualitatively assess embeddings	UMAP, t-SNE plots [47]

Emerging Solutions and Future Directions

Promising Approaches to Bridge the Performance Gap

While current scFMs demonstrate limitations, several promising directions may improve zero-shot capabilities:

- **Biology-Informed Model Architectures**: Moving beyond direct Transformer adaptations to designs specifically crafted for gene interaction networks could better capture biological relationships [1].
- **Enhanced Pretraining Objectives**: Combining masked modeling with explicit biological constraints, such as incorporating gene pathway information during pretraining, may foster more meaningful representation learning [1].
- **Model Zoos and Specialized Ensembles**: As observed in time-series forecasting, creating collections of specialized models with complementary strengths enables dynamic model selection based on task characteristics [50].
- **Novel Evaluation Paradigms**: Developing more sophisticated metrics like roughness index (ROGI) that predict model suitability based on dataset characteristics can guide better model selection [1].



Model Zoo Approach for Optimal scFM Selection

The consistent zero-shot performance gap between sophisticated single-cell foundation models and simpler traditional methods presents both a challenge and opportunity for the field. Rather than dismissing scFMs entirely, researchers should recognize that current limitations stem from identifiable factors including pretraining objectives, architectural choices, and evaluation practices.

For practitioners, this evidence suggests a **cautious approach to adopting scFMs** in discovery settings where zero-shot capability is essential. Established methods like Harmony, scVI, and even HVG selection provide robust baselines that frequently outperform foundation models without their computational costs [47] [49].

Future progress will likely require rethinking foundation model design beyond simply scaling up existing approaches, instead developing architectures and training objectives specifically tailored to biological reasoning. As benchmark methodologies mature—incorporating biologically-grounded metrics and realistic task formulations—the field will be better positioned to develop models that truly capture foundational biological principles transferable to novel discovery contexts.

Addressing Technical Noise and Batch Effect Correction

Technical noise and batch effects present formidable challenges in single-cell RNA sequencing (scRNA-seq) analysis, where unwanted variations from sequencing technologies, laboratory conditions, and experimental protocols can obscure biological signals of interest. The emergence of single-cell foundation models (scFMs) has revolutionized this landscape by offering powerful new approaches for data integration and biological discovery. This technical guide examines batch effect correction methodologies within the broader context of scFM research, providing drug development professionals and researchers with comprehensive frameworks for addressing these persistent analytical challenges. As the field progresses toward unified analysis of massive single-cell datasets, effective batch correction remains paramount for accurate biological interpretation and translation of findings to clinical applications.

The Critical Challenge of Batch Effects in Single-Cell Biology

Batch effects constitute technical or biologically irrelevant variations introduced when samples are processed in different experiments, times, or sequencing platforms [51]. In scRNA-seq data, these effects manifest as systematic differences that can confound true biological variation, potentially

leading to false interpretations in downstream analyses. The primary goal of batch correction is to remove these unwanted technical variations while preserving biologically relevant signals [52] [53].

The single-cell sequencing process introduces specific challenges that complicate batch effect correction. scRNA-seq data characteristically exhibits high dimensionality, sparsity, and low signal-to-noise ratio [1]. A significant phenomenon is "dropout"—events where expressed genes fail to be detected due to the stochastic nature of gene expression or technical failures in RNA capture or amplification [17]. These characteristics distinguish single-cell data from bulk RNA-seq and necessitate specialized computational approaches for effective normalization and batch correction.

Comprehensive Evaluation of Traditional Batch Correction Methods

Method Categories and Underlying Principles

Traditional batch correction methods for scRNA-seq data can be broadly categorized into several algorithmic approaches:

- **Linear model-based methods**: Tools like ComBat and limma, borrowed from bulk RNA-seq analysis, model the linear relationship between batch and gene expression based on Gaussian-distribution assumptions [17] [53]. These approaches assume transcriptomics differences between batches primarily attribute to technical factors that can be modeled and regressed out.
- **Mutual Nearest Neighbors (MNN)-based methods**: Algorithms such as fastMNN, Seurat, and Scanorama identify shared cell types across batches by finding mutual nearest neighbors in reduced-dimensional spaces [17] [53]. Expression differences between cells from the same cell type but different batches are then used to estimate and correct the batch effect.
- **Matrix factorization approaches**: Methods like LIGER use integrative non-negative matrix factorization (NMF) to jointly define cell types from multiple datasets by calculating shared and dataset-specific metagenes [17] [53]. Unlike other methods, LIGER explicitly accounts for the possibility that some differences between datasets may be biological rather than technical in origin.
- **Clustering-based integration**: Harmony formulates an objective function to balance cell-type clustering and degree of dataset mixing in principal component analysis (PCA) space, iteratively removing batch effects while preserving biological structure [17] [53].
- **Deep learning approaches**: Methods like scVI use variational autoencoders to reduce high-dimensional gene-expression matrices to lower-dimensional representations that can be interpreted for relevant biology [17] [53]. More recent innovations like BDACL (Biological-noise Decoupling Autoencoder and Central-cross Loss) reconstruct raw data using autoencoders and employ hierarchical clustering trees to mitigate batch effects without losing rare cell types [51].

Performance Benchmarking Across Scenarios

Table 1: Performance of Selected Batch Correction Methods Across Different Technical Scenarios

Method	Computational Efficiency	Handling of Large Datasets	Preservation of Rare Cell Types	Recommended Scenarios
Harmony	High	Excellent	Moderate	First choice for most scenarios, especially with large datasets [17]
LIGER	Moderate	Good	Good	When biological differences between batches are expected [17]
Seurat 3	Moderate	Good	Good	Datasets with complex cell type hierarchies [17]
fastMNN	Moderate	Moderate	Moderate	Two-batch integrations with overlapping cell types [53]
Scanorama	Moderate	Good	Good	Multiple batches with similar cell type compositions [17]
ComBat	High	Moderate	Poor	When batch information is known and effects are presumed linear [53]
BDACL	Low	Moderate	Excellent	When rare cell type preservation is critical [51]

Large-scale benchmarking studies have evaluated these methods across diverse technical and biological scenarios. One comprehensive assessment evaluated 28 scRNA-seq noise reduction procedures in 55 different scenarios accounting for factors including relative magnitude of batch effects, cell population imbalance, complexity of cell group structures, proportion and similarity of non-overlapping cell populations, dropout rates, and variable library sizes [52] [53].

These evaluations revealed that method performance significantly depends on specific data characteristics. For instance, Harmony demonstrates particularly fast runtime, making it suitable as a first-choice method for large-scale datasets [17]. LIGER and Seurat 3 also show robust performance across multiple scenarios and serve as viable alternatives [17]. However, most traditional methods face challenges in scenarios where batch effects are

subtle yet biologically confounding, often leading to either under-correction or over-correction where biological variation is inadvertently removed [52].

Table 2: Quantitative Performance Metrics for Batch Correction Methods

Method	Batch Mixing (ASW_batch)	Cell Structure Preservation (ASW_group)	Gene Structure Preservation (TPR)	Runtime (Relative)
Harmony	0.72	0.68	N/A	1x
Scanorama	0.65	0.63	0.81	2.5x
Seurat 3	0.69	0.66	0.78	3.1x
fastMNN	0.63	0.61	0.52	2.8x
scVI	0.59	0.58	0.48	5.7x
LIGER	0.67	0.65	N/A	4.2x

Note: Metrics adapted from comprehensive benchmarking studies [52] [17] [53]. Scores represent average performance across multiple scenarios. TPR (True Positive Rate) indicates the percentage of true marker genes preserved between cell types. Harmony and LIGER output low-dimensional embeddings without corrected gene-expression matrices, hence TPR scores are not applicable (N/A).

Single-Cell Foundation Models: A Paradigm Shift

Conceptual Framework and Architecture

Single-cell foundation models represent a transformative approach in computational biology, leveraging large-scale deep learning architectures pretrained on vast single-cell datasets to address multiple downstream tasks including batch correction [1] [33]. These models adapt transformer architectures—originally developed for natural language processing—to single-cell data by treating individual cells as "sentences" and genes or genomic features as "words" or "tokens" [33].

The fundamental innovation of scFMs lies in their self-supervised pretraining on massive, diverse single-cell corpora, enabling them to learn universal biological patterns that can be transferred to specific analytical tasks with minimal fine-tuning [1]. This approach contrasts with traditional methods designed specifically for batch correction, as scFMs learn general representations of cellular biology that inherently distinguish technical artifacts from biological signals.

Prominent Single-Cell Foundation Models

Several scFMs have emerged with distinct architectural implementations and training strategies:

- **Geneformer**: A transformer model pretrained on millions of single-cell transcriptomes that learns generalizable representations of network dynamics [1].
- **scGPT**: Uses a generative pretrained transformer architecture with a unidirectional masked self-attention mechanism that iteratively predicts masked genes conditioned on known genes [33] [54].
- **scBERT**: Employs a bidirectional encoder representations from transformers (BERT)-like architecture with bidirectional attention where the model learns from the context of all genes in a cell simultaneously [33].
- **UCE, scFoundation, LangCell, and scCello**: Additional scFMs with varying architectural innovations and training methodologies [1].

These models typically generate two types of embeddings: gene embeddings that capture functional relationships between genes, and cell embeddings that represent cellular states and identities [1]. The batch correction capability emerges as a byproduct of these learned representations, which ideally capture biological similarity while disregarding technical variations.

Benchmarking scFMs Against Traditional Methods

Comprehensive benchmarking studies have evaluated scFMs against established batch correction methods under realistic conditions. One recent study benchmarked six scFMs against traditional baselines across five datasets with diverse biological conditions, employing 12 evaluation metrics that included unsupervised, supervised, and novel knowledge-based approaches [1].

The findings reveal that while scFMs are robust and versatile tools for diverse applications, simpler machine learning models can sometimes outperform them for specific tasks, particularly under resource constraints [1]. Notably, no single scFM consistently outperforms others across all tasks, emphasizing the need for tailored model selection based on factors such as dataset size, task complexity, biological interpretability, and computational resources [1].

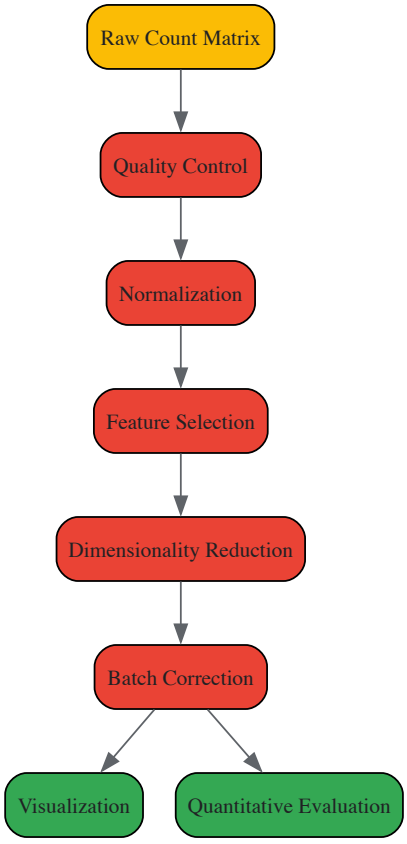
A key advantage of scFMs is their ability to capture biological relationships that align with prior knowledge. Novel evaluation metrics like scGraph-OntoRWR measure the consistency of cell type relationships captured by scFMs with established biological ontologies, while the Lowest Common Ancestor Distance (LCAD) metric assesses the ontological proximity between misclassified cell types [1]. These biologically-grounded evaluation approaches demonstrate that pretrained scFM embeddings indeed capture meaningful biological insights beyond technical artifacts.

Experimental Protocols for Batch Effect Correction

Standardized Workflow for Traditional Methods

A robust batch correction protocol involves multiple sequential steps:

- **Data Preprocessing**: Begin with quality control to remove low-quality cells and genes, followed by basic normalization such as library size adjustment using counts per million (CPM) or trimmed mean of M-values (TMM) normalization [55].
- **Feature Selection**: Identify highly variable genes (HVGs) that exhibit high cell-to-cell variation, as these likely contain biological signals rather than technical noise.
- **Dimensionality Reduction**: Apply principal component analysis (PCA) to capture the main axes of variation in the data while reducing computational complexity for subsequent steps.
- **Batch Correction Method Application**: Implement specific algorithms such as Harmony, LIGER, or Seurat 3 following package-specific guidelines. For Harmony, this involves iteratively clustering cells while maximizing batch diversity within clusters and applying correction factors [17].
- **Visualization and Evaluation**: Project corrected data into two dimensions using UMAP or t-SNE, and calculate quantitative metrics such as kBET (k-nearest neighbor batch-effect test), LISI (local inverse Simpson's index), ASW (average silhouette width), and ARI (adjusted rand index) [17].



scFM-Specific Batch Correction Protocol

When using single-cell foundation models for batch correction:

- **Embedding Extraction**: Load a pretrained scFM and pass your single-cell data through the model in "zero-shot" mode to extract cell embeddings without fine-tuning [1].

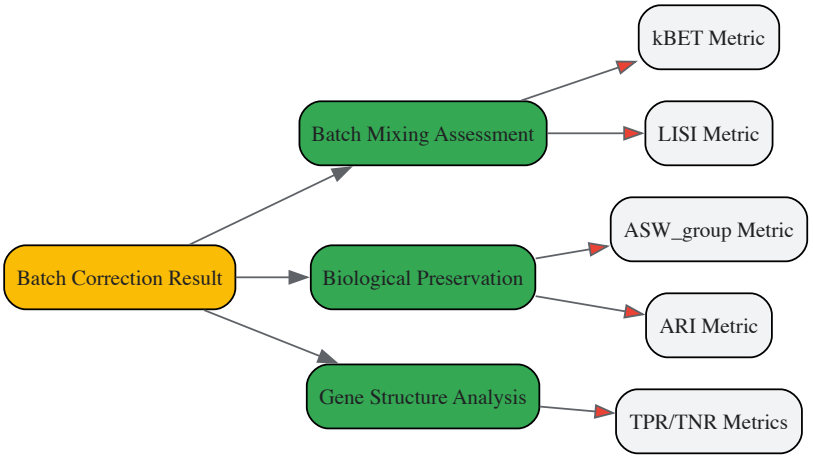
- **Biological Noise Decoupling**: For advanced implementations, employ specialized architectures like the Biological-noise Decoupling Autoencoder (BDA) which separates biological signals from technical noise through reconstruction and clustering [51].
- **Integration with Central-Cross Loss**: Implement the Central-cross Loss (CL) strategy which combines cross-entropy loss for distinguishing cluster labels with Central Loss for encouraging compact cluster formation in the embedding space [51].
- **Hierarchical Cluster Refinement**: Construct similarity matrices and hierarchical clustering trees to delineate relationships within and between batches, gradually merging smaller clusters into larger ones using biological similarity [51].
- **Biological Plausibility Assessment**: Evaluate results using ontology-informed metrics such as scGraph-OntoRWR and LCAD to ensure corrected data aligns with established biological knowledge [1].

Table 3: Essential Computational Tools for Batch Effect Correction

Tool Name	Category	Primary Function	Key Applications
Harmony	Traditional Method	Iterative clustering integration	Rapid integration of large datasets with strong batch effects [17]
Seurat	Traditional Method	Mutual nearest neighbor integration	Multi-dataset integration with complex cellular hierarchies [17] [53]
scVI	Deep Learning	Variational autoencoder for denoising	Integration while modeling count distribution and dropout [53]
scGPT	Foundation Model	Generative pretrained transformer	Multi-task analysis including batch correction and cell type annotation [33] [54]
Geneformer	Foundation Model	Transformer for network dynamics	Context-aware integration leveraging gene network information [1]
BDACL	Advanced Deep Learning	Biological-noise decoupling	Rare cell type preservation during integration [51]

Visualization Framework for Batch Correction Assessment

Effective evaluation of batch correction requires multiple visualization strategies to assess both technical effectiveness and biological preservation.



Future Directions and Clinical Translation

The field of batch effect correction is rapidly evolving, with several promising research directions emerging. There is growing emphasis on developing metrics that assess batch correction for "imperceptible cell-type mixing"—scenarios where batch effects are subtle yet biologically confounding [52]. Additionally, the integration of biological prior knowledge through ontology-informed metrics represents a significant advancement toward more biologically plausible integration [1].

For drug development applications, batch correction methods must preserve clinically relevant cellular states while removing technical artifacts. Foundation models show particular promise in this domain due to their ability to learn from massive collections of clinical samples and capture disease-relevant biological variation [1] [33]. However, challenges remain in model interpretability, computational resource requirements, and validation across diverse patient populations.

The emerging paradigm suggests a hybrid approach where traditional methods serve as efficient baseline tools, while scFMs provide powerful alternatives for complex integration tasks requiring biological nuance. As the field progresses, the development of more specialized foundation models trained on clinically relevant datasets will likely enhance their utility for drug development applications.

Technical noise and batch effect correction remain critical components in the single-cell analysis pipeline, with implications ranging from basic biological discovery to clinical translation. Traditional computational methods provide well-established, efficient approaches for standard integration tasks, while single-cell foundation models offer a transformative new paradigm that captures deep biological principles. The optimal approach depends on specific data characteristics, analytical goals, and computational resources. As single-cell technologies continue to evolve and generate increasingly complex datasets, the development of more sophisticated batch correction methodologies—particularly within the framework of foundation models—will be essential for unlocking the full potential of single-cell genomics in biomedical research and therapeutic development.

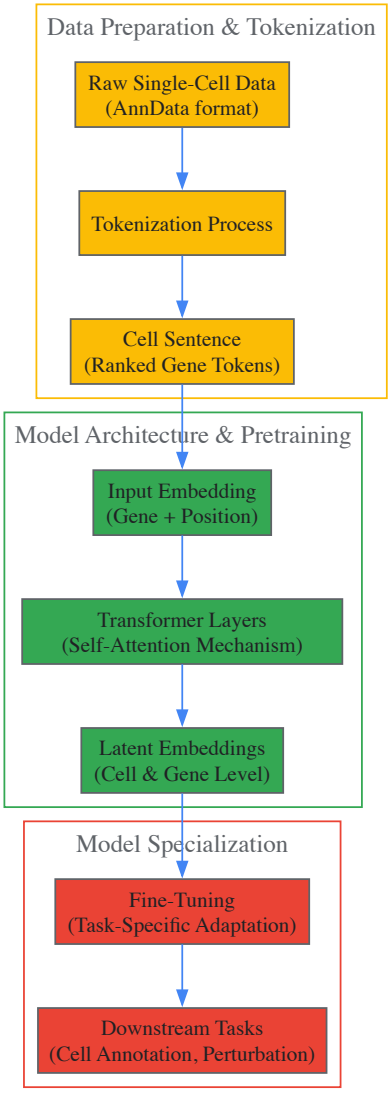
Computational and Resource Demands for Training and Fine-Tuning

Single-cell foundation models (scFMs) are large-scale deep learning models, typically based on transformer architectures, pretrained on vast datasets comprising tens of millions of single-cell transcriptomes [6]. The development and specialization of these models for downstream biological tasks present significant computational challenges that mirror those encountered in the natural language processing domain but with unique biological considerations. The scale of single-cell data continues to grow rapidly, with resources like the Tahoe-100M dataset now comprising over 100 million transcriptomes, creating substantial demands on computational infrastructure for both training and fine-tuning processes [56]. Understanding and managing these resource requirements is essential for researchers aiming to effectively develop and deploy scFMs for applications in cell atlas construction, tumor microenvironment studies, and treatment decision-making [4].

The scFM Training Pipeline: Architecture and Workflows

Core Model Architecture and Data Flow

The development of a single-cell foundation model follows a structured pipeline that transforms raw gene expression data into powerful predictive models. The standard workflow encompasses data preparation, model pretraining, and task-specific fine-tuning, with each stage presenting distinct computational demands. Most successful scFMs utilize transformer architectures, which employ attention mechanisms to learn relationships between genes within cellular profiles [6]. The following diagram illustrates this complete training pipeline:



Tokenization Strategies for Single-Cell Data

A critical preprocessing step for scFMs is tokenization—converting raw gene expression profiles into sequences that transformer models can process. Unlike natural language, gene expression data lacks inherent sequential ordering, requiring researchers to implement various strategies to structure this information:

- **Expression Ranking**: Genes within each cell are ranked by expression levels, creating a deterministic sequence of "top expressed genes" that serves as the cell sentence [57] [6]
- **Value Binning**: Some models partition gene expression values into discrete bins, using these categorical representations as tokens [6]
- **Gene Identifier Embeddings**: Each gene is represented as a token embedding that combines its identifier with expression value information [6]
- **Special Tokens**: Models may incorporate special tokens representing cell metadata, batch information, or experimental conditions to provide additional biological context [6]

Quantitative Resource Demands for scFM Development

Computational Requirements Across Model Scales

The computational resources required for scFM development vary significantly based on model size, dataset scale, and training strategy. The following table summarizes resource demands across different phases of model development:

Development Phase	Compute Requirements (GPU Hours)	Memory Demands	Dataset Scale	Training Time
Full Pretraining	1,000-10,000+ (A100 equivalents)	16-80GB+ GPU Memory	10M-100M+ cells	Days to weeks

Development Phase	Compute Requirements (GPU Hours)	Memory Demands	Dataset Scale	Training Time
Full Fine-Tuning	100-1,000	16-48GB+ GPU Memory	10K-1M cells	Hours to days
Parameter-Efficient Fine-Tuning	10-100	4-16GB GPU Memory	10K-100K cells	Minutes to hours
Inference	<1	2-8GB GPU Memory	Single cells to thousands	Milliseconds to seconds

Table 1: Computational requirements for different phases of scFM development. Values represent estimated ranges based on current practices. [58] [57] [56]

Memory Optimization Techniques and Trade-offs

Multiple optimization techniques can significantly reduce memory demands during training and fine-tuning, each with distinct trade-offs between memory savings, computational overhead, and implementation complexity:

Optimization Technique	Memory Reduction	Runtime Overhead	Implementation Complexity	Best Suited For
Gradient Checkpointing	30-50%	20-30% increase	Low	Large model training
LoRA (Low-Rank Adaptation)	60-80% for fine-tuning	Minimal	Medium	Task adaptation
DeepSpeed ZeRO Stage 2	4x reduction	Moderate	High	Distributed training
DeepSpeed ZeRO Stage 3	8x+ reduction	High	High	Extreme model scaling
FlashAttention	30-70% for attention	10-20% improvement	Medium	Long sequences
Mixed Precision Training	40-60%	10-50% improvement	Low	All training phases

Table 2: Memory optimization techniques for scFM training and fine-tuning with their characteristic trade-offs. [58]

Optimization Strategies for Efficient Fine-Tuning

Parameter-Efficient Fine-Tuning (PEFT) Methods

For most downstream applications, full fine-tuning of scFMs is computationally prohibitive. Parameter-efficient fine-tuning methods dramatically reduce resource requirements by updating only a small subset of model parameters:

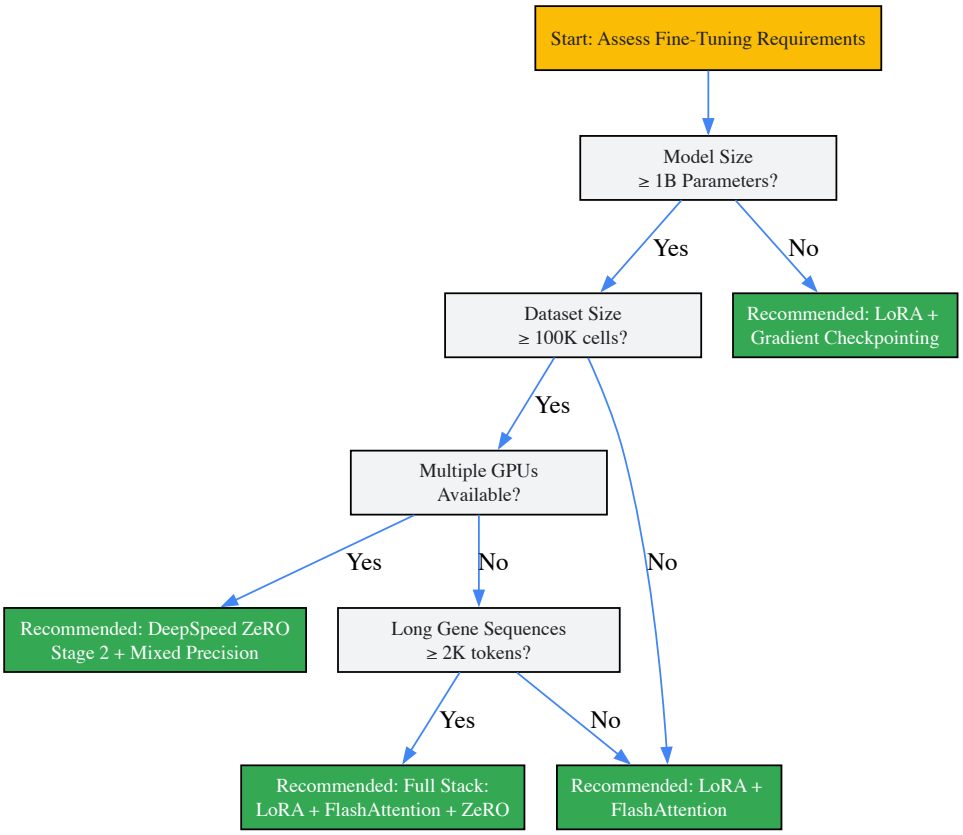
- **Low-Rank Adaptation (LoRA)**: Freezes pre-trained weights and injects trainable rank decomposition matrices into transformer layers, reducing trainable parameters by orders of magnitude (e.g., ~131M instead of 70B parameters for a large model) [58]
- **Prefix Tuning**: Adds trainable embeddings at the model input level while keeping the entire base model frozen [58]
- **Layer-wise Fine-Tuning**: Selectively updates certain layers (typically the last few) while freezing earlier layers that capture general biological patterns [59]

Optimization Combinations for Different Scenarios

Based on empirical studies of optimization techniques, researchers have identified effective combinations for common fine-tuning scenarios:

- **Resource-Constrained Environments**: LoRA + Gradient Checkpointing provides maximum memory savings with manageable runtime impact [58]
- **Balanced Performance**: LoRA + FlashAttention offers a favorable balance of memory efficiency and training speed [58]
- **Large-Scale Fine-Tuning**: DeepSpeed ZeRO Stage 2 + Mixed Precision Training enables fine-tuning of larger models on multiple GPUs [58]

The following diagram illustrates the decision process for selecting appropriate optimization strategies:



Experimental Protocols for Benchmarking scFMs

Standardized Evaluation Framework

Comprehensive benchmarking of scFMs requires a structured evaluation framework encompassing multiple task types and performance metrics. A recent benchmark study evaluated six scFMs against established baselines across realistic biological scenarios [4]:

Gene-Level Tasks:

- **Gene Perturbation Effect Prediction**: Models predict expression changes following genetic or chemical perturbations
- **Gene Network Inference**: Reconstruction of gene regulatory networks from single-cell data

Cell-Level Tasks:

- **Batch Integration**: Removal of technical artifacts across datasets while preserving biological variation
- **Cell Type Annotation**: Automatic classification of cells into known cell types
- **Cancer Cell Identification**: Discrimination of malignant cells within tumor microenvironments
- **Drug Sensitivity Prediction**: Forecasting cellular responses to therapeutic compounds

Evaluation Metrics: The benchmark employed 12 metrics spanning unsupervised, supervised, and knowledge-based approaches, including a novel metric called scGraph-OntoRWR designed to uncover intrinsic knowledge encoded by scFMs [4].

Performance Trends and Scaling Laws

Recent research has revealed that biological language models follow clear scaling laws—performance improves predictably as model size increases [57]. Larger scFMs consistently outperform smaller ones across various biological tasks, from cell type annotation to generating synthetic cells and tissues. For dataset interpretation, consistent gains in semantic similarity scores have been observed when scaling model size in the parameter-efficient regime, with significant improvements in gene overlap percentage for tissue generation as model capacity increases to 27 billion parameters [57].

Essential Research Reagent Solutions

Computational Tools and Infrastructure

Successful scFM development relies on a suite of specialized tools and frameworks that address the unique challenges of single-cell data processing and model training:

Tool/Resource	Function	Key Features	Reference
scDataset	PyTorch IterableDataset for single-cell data	Memory-efficient loading of large .h5ad files; 48× speed-up over alternatives	[56]
AnnData	Standard format for single-cell data	Efficient storage of large, sparse matrices; rich annotation support	[56]
DeepSpeed	Optimization library for training	ZeRO redundancy optimizer; CPU offloading; extreme scaling	[58]
FlashAttention	Optimized attention computation	Linear memory complexity with sequence length; SRAM optimization	[58]
C2S-Scale	LLM for single-cell analysis	Converts cells to sentences; enables natural language interaction	[57]
Tahoe-100M	Large-scale benchmark dataset	100M transcriptomes; 1,100 chemical perturbations; 50 cancer lines	[56]

Table 3: Essential computational tools and resources for scFM development and application.

The pretraining of effective scFMs requires large-scale, diverse single-cell datasets that capture a broad spectrum of biological variation:

- **CZ CELLxGENE**: Provides unified access to annotated single-cell datasets with over 100 million unique cells standardized for analysis [6]
- **Human Cell Atlas**: Multiorgan atlases offering broad coverage of cell types and states across tissues [6]
- **PanglaoDB**: Curated compendium of single-cell data from multiple sources and studies [6]
- **NCBI GEO and SRA**: Public repositories hosting thousands of single-cell sequencing studies [6]

Implementation Guide: Data Loading and Processing

Efficient Data Handling for Large-Scale Training

Working with massive single-cell datasets presents significant input/output challenges that can become training bottlenecks. The scDataset framework provides optimized data loading specifically designed for single-cell omics data stored in AnnData format [56]. Unlike traditional approaches that require loading entire datasets into memory or converting to dense formats, scDataset enables:

- **Direct Disk Reading**: Fast, memory-efficient, and shuffled data loading directly from disk without full dataset loading [56]
- **Block Sampling**: Balanced randomness and I/O efficiency through strategic data access patterns [56]
- **Hardware Scaling**: Efficient training on datasets as large as Tahoe-100M using standard hardware [56]
- **Benchmarked Performance**: 48× speed-up over AnnLoader and substantial improvements over other data loading solutions [56]

Data Preprocessing and Quality Control

Assembling high-quality, nonredundant datasets for pretraining is as important as model architecture in building robust scFMs [6]. Key considerations include:

- **Dataset Selection**: Careful curation of datasets with appropriate biological diversity and technical quality [6]
- **Cell Filtering**: Removal of low-quality cells based on established QC metrics [6]
- **Gene Selection**: Filtering to informative genes while maintaining biological signal [6]
- **Batch Effect Management**: Implementation of strategies to address technical variation across experiments [6]

Future Directions in Efficient scFM Development

The field of single-cell foundation models continues to evolve rapidly, with several promising directions for improving computational efficiency:

- **Model Compression**: Techniques such as quantization, pruning, and knowledge distillation to reduce inference costs
- **Federated Learning**: Approaches that enable model training across distributed datasets without data sharing
- **Multi-Modal Architectures**: Efficient integration of diverse data types (transcriptomics, epigenomics, proteomics, spatial)
- **Automated Hyperparameter Optimization**: Methods for efficiently navigating the complex parameter spaces of scFMs
- **Hardware-Specific Optimizations**: Customized implementations for emerging accelerator technologies

As scFMs mature, balancing computational demands with biological insight will remain a central challenge, requiring continued innovation in both algorithmic approaches and computational infrastructure.

Single-cell foundation models (scFMs) have emerged as transformative tools in computational biology, achieving strong performance on diverse downstream tasks such as cell type annotation, batch integration, and drug response prediction [60] [4]. These models learn universal patterns from massive single-cell transcriptomics datasets through self-supervised pretraining, capturing complex biological relationships within their latent representations [1]. However, their exceptional performance comes with a significant challenge: these models operate as **black boxes**, with limited transparency into how they generate predictions or what biological knowledge they encode [60]. This interpretability gap severely restricts their

utility for biological discovery, as researchers cannot fully understand the basis for model decisions or extract novel biological insights from the learned representations [61].

The fundamental hurdle lies in the fact that while scFMs can detect intricate patterns in high-dimensional single-cell data, their internal workings and decision-making processes remain opaque [60] [61]. This opacity creates a barrier to trust and adoption, particularly in high-stakes domains like **drug development** and **clinical research**, where understanding the rationale behind predictions is as crucial as the predictions themselves [62]. Moreover, without effective interpretability methods, researchers cannot leverage these powerful models for their primary purpose: generating **testable biological hypotheses** about cellular processes, disease mechanisms, and therapeutic targets [63]. This whitepaper examines the core interpretability challenges in scFMs, evaluates current methodological solutions, and provides a technical framework for extracting biologically meaningful insights from these complex models.

The Interpretability Challenge in Single-Cell Foundation Models

Fundamental Limitations of Current Approaches

The interpretability problem in scFMs stems from several interconnected factors. First, the **sheer complexity** of these models, with their deep architectures and millions of parameters, makes it difficult to trace how specific inputs lead to particular outputs [61]. Second, biological sequences and cellular states are not inherently human-interpretable, creating a semantic gap between model representations and biological understanding [60]. Third, traditional interpretability approaches like differential expression analysis provide only **correlational insights** rather than revealing causal relationships captured by the models [60].

Current evaluation paradigms often fail to assess whether scFMs capture biologically meaningful patterns. As noted in recent benchmarking studies, it remains unclear how effectively these models extract unique biological insights beyond what standard methods can discover [4] [1]. This limitation is particularly problematic given that a key promised advantage of scFMs is their ability to uncover novel biology from large-scale data. Without robust interpretability frameworks, verifying whether models learn biologically relevant representations versus exploiting technical artifacts in the data becomes challenging [1].

Performance-Interpretability Tradeoffs in Model Selection

Benchmarking studies reveal that no single scFM consistently outperforms others across all tasks, emphasizing the need for tailored model selection based on factors including biological interpretability requirements [4] [1]. Interestingly, simpler machine learning models sometimes outperform complex foundation models on specific tasks, particularly under resource constraints or when dataset size is limited [1]. This creates a critical tradeoff between predictive performance and interpretability that researchers must navigate based on their specific goals.

Table 1: Benchmarking Performance of Single-Cell Foundation Models Across Biological Tasks

Model	Batch Integration	Cell Type Annotation	Drug Response Prediction	Biological Relevance
scGPT	High	Medium	High (0.858 F1 in zero-shot)	Medium
scFoundation	Medium	High	High (0.971 F1 with fine-tuning)	High
UCE	High	High	Medium (0.774 F1 with fine-tuning)	High
Geneformer	Medium	Medium	Low	Medium
Traditional ML	Low	High	Variable	Highly Interpretable

The table above summarizes performance patterns observed across multiple benchmarking studies [64] [4] [1]. Notably, models excelling in predictive tasks do not necessarily provide greater biological insights, highlighting the need for specialized interpretability approaches regardless of which scFM is selected.

Technical Frameworks for Interpretability

Concept-Based Interpretability with Biological Priors

A promising approach for enhancing scFM interpretability involves **concept-based frameworks** that extract human-understandable concepts from model internals. Claye et al. introduced a novel interpretability framework for single-cell RNA-seq models that moves beyond correlational approaches by incorporating **attribution methods with counterfactual perturbations** [60]. This method identifies genes that directly influence concept activation, providing causal insights into model behavior rather than mere correlations.

The framework employs two complementary interpretation approaches: (1) **expert-driven analysis** facilitated by interactive interfaces that allow domain experts to explore concepts in context, and (2) **ontology-driven methods** with attribution-based biological pathway enrichment that systematically maps concepts to established biological knowledge [60]. When applied to Top-K Sparse Auto-Encoders trained on immune cell

datasets, this approach demonstrated that concepts improve interpretability compared to individual neurons while preserving the richness of latent representations [60].

Multi-View Graph Representation Learning

Another advanced interpretability framework combines deep learning with explainable AI (XAI) through **Multi-view Graph-level Representation Learning (MGRL)** [63]. This approach integrates prior biological network information, such as protein-protein interaction (PPI) networks, with single-cell data to build predictive models that are subsequently interpreted using XAI techniques [63]. The MGRL architecture fuses a deep graph convolutional neural network (DeeperGCN) with a multi-layer perceptron (MLP), enabling the model to capture both local topological information and global expression patterns.

Table 2: Key Components of the MGRL Interpretability Framework

Component	Function	Biological Relevance
PPI Network Integration	Provides spatial context for genes within signaling domains	Models biological pathway structure and interactions
DeeperGCN	Captures local joint topological and gene expression information	Identifies functionally related gene modules
MLP	Extracts gene expression patterns without topological constraints	Discovers expression correlations independent of known interactions
PGExplainer	Identifies predictive PPI edges and genes	Highlights biologically relevant network components

When applied to aging using one of the largest single-cell transcriptomic datasets encompassing over a million immune cells from 981 donors, this DL-XAI framework revealed a ribosomal gene subnetwork whose expression correlates with age independently of cell type [63]. This discovery would not have been possible using standard machine-learning methods, demonstrating how interpretable deep learning can extract novel biological insights from complex data.

Experimental Protocols for Interpretability Assessment

Protocol 1: Concept Extraction and Interpretation

The following protocol outlines the methodology for extracting and interpreting biological concepts from scFMs, based on the approach described by Claye et al. [60]:

- **Concept Extraction**: Train Top-K Sparse Auto-Encoders on the latent representations of scFMs to decompose activations into sparse, interpretable concepts.
- **Gene Attribution**: Apply attribution methods with counterfactual perturbations to identify genes that significantly influence concept activation. This involves:
  - Generating perturbed input sequences with systematic variations in gene expression
  - Measuring the effect on concept activation scores
  - Calculating attribution scores for each gene-concept pair
- **Expert-Driven Interpretation**:
  - Develop an interactive interface visualizing concepts alongside relevant metadata
  - Enable domain experts to explore relationships between concepts and biological conditions
  - Collect qualitative assessments of concept biological relevance
- **Ontology-Driven Interpretation**:
  - Perform attribution-based enrichment analysis using biological pathway databases
  - Map significant genes to Gene Ontology terms and canonical pathways
  - Calculate statistical significance of concept-pathway associations

This protocol successfully identified interpretable immune cell programs in single-cell RNA-seq models, enabling domain experts to validate the biological relevance of extracted concepts [60].

Protocol 2: Biological Knowledge-Guided Benchmarking

For comprehensive assessment of biological interpretability, the following benchmarking protocol evaluates how well scFMs capture established biological knowledge [1]:

- **Gene-Level Task Evaluation**:

    - Extract gene embeddings from scFM input layers
    - Evaluate ability to predict gene functions using Gene Ontology annotations
    - Assess tissue specificity prediction accuracy
    - Compare against baseline methods like FRoGS (Functional Representation of Gene Signatures)

- **Cell-Level Task Evaluation**:
    - Generate zero-shot cell embeddings from pretrained scFMs
    - Evaluate performance on dataset integration and cell type annotation
    - Apply novel ontology-informed metrics including:
        - scGraph-OntoRWR: Measures consistency of cell type relationships with biological knowledge
        - LCAD (Lowest Common Ancestor Distance): Quantifies ontological proximity between misclassified cell types

- **Attention Analysis**:
    - Extract attention weights from transformer-based scFMs
    - Identify genes receiving high attention for specific predictions
    - Correlate attention patterns with known biological pathways

This protocol revealed that pretrained scFM embeddings indeed capture biological insights into the relational structure of genes and cells, which benefits downstream tasks [1].

Visualization and Computational Tools

Accessible Visualization for Single-Cell Data

Effective visualization is crucial for interpreting complex single-cell data and model outputs. Traditional scatter plots of single-cell data (e.g., UMAP, t-SNE) often use color as the sole visual cue, creating accessibility challenges for the substantial proportion of researchers with color vision deficiencies (CVDs) [65]. The **scatterHatch** R package addresses this limitation by creating accessible scatter plots through redundant coding of cell groups using both colors and patterns [65].

The package implements a sophisticated workflow that:

- Classifies points as belonging to dense clusters or sparse distributions
- Plots coarse patterns over dense point clusters
- Individually plots matching patterns over each sparse point
- Provides six default patterns with customizable aesthetics

This approach significantly enhances interpretability for all users, particularly when visualizing data with numerous cell groups where color differentiation becomes challenging [65]. Adoption of such accessible visualization tools should become standard practice for communicating single-cell research findings.

Workflow Diagram: Interpretability Analysis Framework

The following diagram illustrates the integrated workflow for extracting biologically meaningful insights from single-cell foundation models:

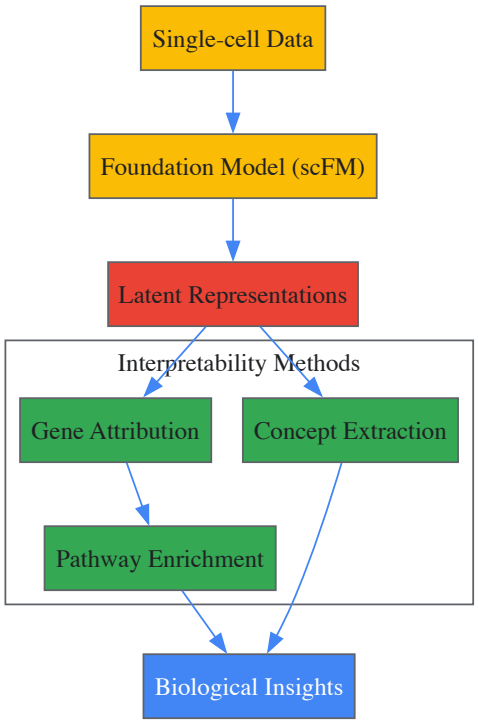


Diagram 1: Interpretability Analysis Framework for Single-Cell Foundation Models

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Computational Tools for scFM Interpretability Research

Tool/Resource	Type	Function	Application Context
scGPT	Foundation Model	Single-cell multi-omics foundation model	Cell type annotation, perturbation prediction, biological concept discovery
Geneformer	Foundation Model	Transformer model pretrained on single-cell data	Gene network analysis, disease mechanism identification
scGraph-OntoRWR	Evaluation Metric	Measures biological consistency of embeddings	Benchmarking scFMs against prior biological knowledge
PGExplainer	Interpretability Tool	Explains graph neural network predictions	Identifying predictive genes and network components in MGRL frameworks
scatterHatch	Visualization Package	Creates accessible scatter plots with patterns	Communicating results for diverse audiences, including CVD users
Top-K Sparse Auto-Encoders	Interpretability Method	Extracts discrete concepts from model activations	Concept-based interpretation of scFM representations
Protein-Protein Interaction Networks	Biological Prior Knowledge	Provides structural context for gene relationships	Integrating biological knowledge into interpretable models

Overcoming interpretability hurdles is essential for realizing the potential of single-cell foundation models to drive biological discovery and therapeutic development. The frameworks and methodologies outlined in this whitepaper provide a pathway for researchers to extract biologically meaningful insights from these complex models. By combining **concept-based interpretation** with **biological knowledge-guided evaluation** and **accessible visualization**, researchers can bridge the gap between model performance and biological understanding.

As the field advances, future developments should focus on creating more **intrinsic interpretability** in model architectures, establishing standardized **evaluation benchmarks** for biological relevance, and developing **interactive tools** that enable domain experts to directly engage with and interpret model behavior. Through these advances, single-cell foundation models can transition from black-box predictors to trustworthy partners in scientific discovery, generating novel biological insights and accelerating progress in biomedicine.

The emergence of single-cell foundation models (scFMs) represents a transformative advancement in computational biology, enabling researchers to extract profound insights from single-cell RNA sequencing data at unprecedented scales. However, this rapid innovation has created a significant

challenge: the field is now characterized by **heterogeneous architectures** and **disparate coding standards** across various models, making consistent application and rigorous benchmarking exceedingly difficult [66]. This lack of standardization hinders reproducibility, obstructs fair performance comparisons, and ultimately slows the translation of these powerful tools to biological discovery and therapeutic development.

The BioLLM (biological large language model) framework addresses this critical bottleneck by providing a **standardized ecosystem** for integrating and evaluating scFMs [66]. By establishing unified interfaces and consistent evaluation protocols, BioLLM enables researchers to bypass technical incompatibilities and focus on scientific inquiry. This technical guide examines how standardization solutions like BioLLM are transforming the scFM landscape, providing researchers with robust methodologies for model assessment, and offering drug development professionals validated approaches for leveraging these tools in critical research applications.

The BioLLM Framework: Architecture and Core Components

Unified Interface Design

BioLLM implements a **cohesive interface** that abstracts away the architectural differences between diverse scFMs, creating a consistent user experience regardless of the underlying model implementation [66]. This design eliminates the need for researchers to learn and navigate the unique coding patterns and data structures required by each individual model, significantly reducing the technical barrier to entry [67]. The framework's modular architecture allows for seamless integration of new models as they emerge, future-proofing the ecosystem against ongoing innovation in the rapidly evolving field of single-cell analysis.

The framework provides **standardized APIs** that encapsulate the complete model lifecycle, from data loading and preprocessing to inference and result interpretation [68]. This consistency enables researchers to switch between different scFMs with minimal code modifications, facilitating direct performance comparisons and ensuring that evaluation results reflect true model capabilities rather than implementation artifacts [66]. The comprehensive documentation accompanying these APIs further enhances usability, allowing both computational biologists and drug development professionals to quickly leverage advanced scFMs without deep technical expertise in each specific model [69].

Supported Models and Integration Approach

BioLLM currently integrates several prominent scFMs, each with distinct architectural characteristics and training methodologies [66]. The framework's evaluation has revealed specialized capabilities across these models, informing context-specific recommendations:

- **scGPT** demonstrates robust performance across diverse task categories, excelling in both zero-shot and fine-tuning scenarios [66] [4]. Its strong generalization capabilities make it particularly valuable for exploratory research where task specificity is low.
- **Geneformer** and **scFoundation** exhibit specialized strengths in gene-level tasks, leveraging effective pretraining strategies that capture fundamental biological relationships [66]. These models are particularly adept at gene network inference and expression prediction tasks.
- **scBERT** shows more limited performance, likely attributable to its smaller model size and restricted training data [66]. This observation highlights the importance of scale and data diversity in building effective biological foundation models.

Table: Single-Cell Foundation Models Integrated in BioLLM

Model	Architecture	Pretraining Data	Specialized Strengths	Notable Limitations
scGPT	Transformer-based	Extensive single-cell datasets	Strong all-around performer; excels in zero-shot learning and fine-tuning	Computationally intensive for large-scale analyses
Geneformer	Transformer-based	Human transcriptomes	Excellent gene-level task performance; effective pretraining strategy	Less versatile for cell-level tasks
scFoundation	Transformer-based	Diverse single-cell atlases	Strong gene-level capabilities; scalable architecture	Requires fine-tuning for optimal performance
scBERT	BERT-based	Limited single-cell data	Efficient for basic annotation tasks	Smaller model size; limited training data constrains performance

Benchmarking Methodology: Standardized Evaluation Protocols

Task Categorization and Evaluation Metrics

BioLLM implements a comprehensive benchmarking approach that assesses model performance across biologically meaningful tasks categorized into **gene-level** and **cell-level** analyses [4]. This hierarchical evaluation strategy ensures that models are tested against realistic biological questions that researchers encounter in both basic science and drug development contexts. The framework employs **twelve distinct metrics** spanning unsupervised, supervised, and knowledge-based paradigms to provide a multidimensional performance assessment [4].

A notable innovation in BioLLM's evaluation toolkit is **scGraph-Ontology Random Walk with Restart (scGraph-OntoRWR)**, a novel metric specifically designed to uncover intrinsic biological knowledge encoded by scFMs beyond what standard performance measures can capture [4]. This knowledge-centric evaluation approach complements traditional accuracy-based metrics, providing insights into how well models capture the fundamental biological relationships that underpin cellular function and disease mechanisms.

Experimental Workflows

The evaluation of scFMs within BioLLM follows structured experimental workflows designed to ensure consistency and reproducibility across different model architectures and task types. The diagram below illustrates the core benchmarking workflow that guides model assessment:

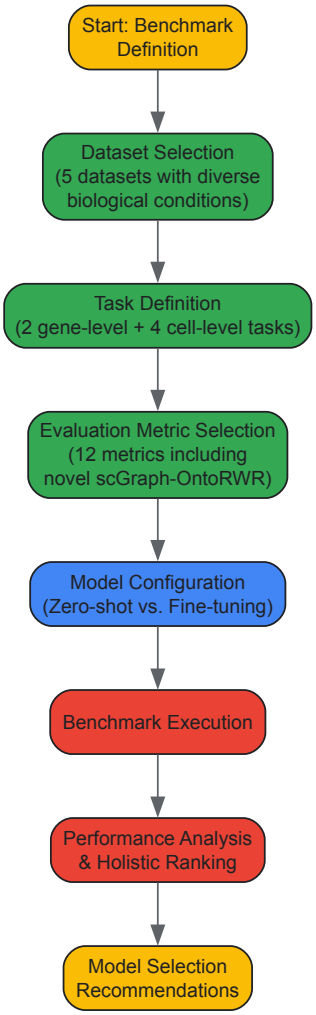


Diagram Title: BioLLM Benchmarking Workflow

For drug development applications, BioLLM implements specialized evaluation protocols that assess model performance on clinically relevant prediction tasks. These include **cancer cell identification** across seven different cancer types and **drug sensitivity prediction** for four therapeutic compounds [4]. This clinically-focused benchmarking ensures that scFMs are evaluated against realistic translational research scenarios, providing drug development professionals with meaningful performance indicators for selecting models most suited to their specific applications.

Key Research Reagents and Computational Tools

The effective implementation and evaluation of scFMs requires both computational resources and biological datasets. The table below details essential components of the scFM research toolkit:

Table: Essential Research Reagents and Computational Tools for scFM Evaluation

Resource Category	Specific Examples	Function/Role in Evaluation	Implementation Notes
Computational Frameworks	BioLLM, PyTorch, TensorFlow	Provides standardized APIs and model integration capabilities	Requires CUDA 11.7+ for GPU acceleration; flash-attn <1.0.5 for optimal performance [67]

Resource Category	Specific Examples	Function/Role in Evaluation	Implementation Notes
Biological Datasets	Diverse single-cell atlases, Cancer cell datasets, Drug response data	Enables realistic benchmarking across biological and clinical contexts	Five datasets with diverse biological conditions; seven cancer types; four drugs for sensitivity prediction [4]
Evaluation Metrics	scGraph-OntoRWR, Standard classification metrics, Unsupervised metrics	Quantifies model performance across multiple dimensions	Twelve total metrics spanning unsupervised, supervised, and knowledge-based paradigms [4]
Benchmarking Tasks	Cell type annotation, Batch integration, Cancer cell ID, Drug sensitivity	Tests model capabilities on biologically meaningful problems	Two gene-level and four cell-level tasks representing realistic research scenarios [4]

Performance Benchmarking Results and Comparative Analysis

Quantitative Performance Assessment

BioLLM's comprehensive evaluation of scFMs has yielded nuanced insights into the relative strengths and limitations of different model architectures across various task types. The systematic benchmarking reveals that **no single scFM consistently outperforms all others** across every task category, emphasizing the importance of context-dependent model selection [4]. This finding underscores the necessity of frameworks like BioLLM that enable researchers to match specific model strengths to their analytical needs.

The quantitative assessment demonstrates that while scFMs generally serve as robust and versatile tools for diverse applications, **simpler machine learning models** can sometimes demonstrate superior efficiency when adapting to specific datasets, particularly under significant computational resource constraints [4]. This observation suggests a pragmatic approach where researchers might opt for traditional methods for well-defined, narrow tasks while reserving scFMs for more complex, exploratory analyses requiring generalization capabilities.

Table: Model Performance Across Task Categories

Model	Cell Type Annotation	Batch Integration	Cancer Cell Identification	Drug Sensitivity Prediction	Gene-Level Tasks	Overall Ranking
scGPT	High	High	High	Medium-High	High	1st
Geneformer	Medium	Medium	Medium	Medium	High	2nd
scFoundation	Medium	Medium-High	Medium	Medium	High	3rd
scBERT	Low-Medium	Low	Low-Medium	Low	Medium	4th

Zero-Shot versus Fine-Tuning Performance

A critical dimension of BioLLM's evaluation is the assessment of model performance in both **zero-shot** (without task-specific training) and **fine-tuning** (with limited task-specific adaptation) scenarios [66]. This distinction has profound practical implications for researchers, as zero-shot capabilities determine a model's utility for exploratory analysis where labeled training data may be scarce, while fine-tuning performance indicates the potential for specialization to specific research questions.

The benchmarking results indicate that **scGPT demonstrates particularly strong zero-shot capabilities**, maintaining robust performance across diverse tasks without requiring task-specific adaptation [66]. This makes it especially valuable for discovery-phase research where the analytical targets may not be well-defined in advance. In contrast, other models show more significant performance improvements with fine-tuning, suggesting they may be better suited for applications where some labeled data is available to guide specialization toward specific analytical objectives.

Implementation Guide: From Framework Setup to Advanced Applications

Technical Installation and Configuration

Implementing BioLLM requires specific technical configurations to ensure optimal performance and compatibility with integrated scFMs. The framework is installed from source, with particular attention to dependencies that have specific hardware and software requirements [67]:

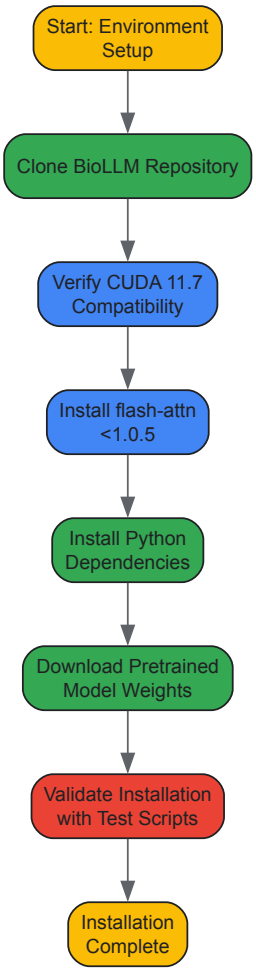


Diagram Title: BioLLM Installation Workflow

A critical dependency management consideration involves **flash-attn**, which requires specific GPU capabilities and CUDA version compatibility [67]. The installation process is most reliable with CUDA 11.7 and flash-attn versions below 1.0.5, as newer versions have reported installation issues that can obstruct framework deployment. Researchers should verify their hardware compatibility before installation and consult the project's GitHub repository for troubleshooting specific dependency conflicts [67].

Application in Drug Development Pipelines

For drug development professionals, BioLLM enables the integration of scFMs into critical research and development workflows, particularly for tasks such as **drug sensitivity prediction**, **tumor microenvironment characterization**, and **treatment response modeling** [4]. The standardized benchmarking provided by BioLLM helps identify the most appropriate models for specific pharmaceutical applications, enhancing the reliability of computational approaches in the drug development pipeline.

The framework's evaluation of scFMs on clinically relevant tasks provides performance baselines that guide model selection for therapeutic development. For example, models demonstrating strong performance in cancer cell identification across multiple cancer types would be prioritized for applications in oncology drug discovery, while those excelling at predicting drug sensitivity would be leveraged for preclinical compound prioritization [4]. This evidence-based approach to model selection increases confidence in computational predictions that inform critical decisions in the drug development process.

Future Directions and Community Adoption

The development of BioLLM represents a significant step toward standardized evaluation of scFMs, but the field continues to evolve rapidly. Future framework enhancements will likely address emerging challenges such as **multimodal data integration**, **temporal modeling capabilities**, and **improved interpretability** for biological insight generation. Community adoption and contribution mechanisms, including GitHub issue tracking and pull request workflows, ensure that the framework remains responsive to evolving research needs [67].

As the single-cell biology field matures, standardization initiatives like BioLLM will play an increasingly critical role in bridging the gap between methodological innovation and biological discovery. By providing consistent evaluation paradigms and reducing technical barriers to implementation, these frameworks accelerate the translation of computational advances to meaningful biological insights and therapeutic

breakthroughs. The continued development and community engagement around BioLLM promises to enhance the reproducibility, reliability, and applicability of scFMs across diverse research contexts.

scFM Benchmarking 2025: A Rigorous Comparison of Leading Models

The emergence of single-cell foundation models (scFMs) represents a paradigm shift in computational biology, promising to unlock deep biological insights from single-cell sequencing data. Models like scGPT and Geneformer, pre-trained on tens of millions of single-cell transcriptomes, propose a new analytical approach where foundational knowledge of cellular biology can be rapidly specialized for diverse downstream applications [47] [70]. The central premise of these models hinges on a critical distinction in their deployment strategy: **zero-shot** application, where pre-trained models are used without modification on new tasks, versus **fine-tuned** application, where models are further trained on task-specific data. This technical guide establishes a rigorous benchmarking framework to evaluate model performance across these distinct deployment paradigms, providing researchers and drug development professionals with methodologies to assess the true capabilities and limitations of scFMs within single-cell research.

Current Landscape of Single-Cell Foundation Models

The architectural philosophy of scFMs is largely inspired by large language models (LLMs) in natural language processing. These models conceptualize single-cell expression profiles as a form of biological language [70]:

- **Tokenizer**: Current scLLMs require tokenization of biological "words," which involves converting genes into vectors for subsequent learning. The tokenizer must combine the gene name and its corresponding expression value, creating a gene token vector for each input cell.
- **Encoder**: The current scLLMs utilize the Transformer architecture to encode gene relationships, stacking multiple transformer blocks to capture interrelated gene patterns.
- **Pretrainer**: scLLMs typically employ the Masked Language Model (MLM) objective, randomly masking certain non-zero gene tokens and predicting the original tokens based on the context provided by non-masked genes.

Several models have gained prominence in the field, including scGPT (pre-trained on over 33 million human cells), Geneformer (trained on 29.9 million transcriptomes), and scBERT (trained on 1.1 million cells from PanglaoDB) [70]. Understanding these foundational architectures is crucial for designing appropriate benchmarking experiments.

Quantitative Performance Comparison: Zero-Shot vs. Fine-Tuned

Rigorous evaluation reveals significant performance disparities between zero-shot and fine-tuned applications of scFMs across critical tasks in single-cell analysis.

Table 1: Performance Comparison of Zero-Shot vs. Fine-Tuned scFMs on Cell Type Clustering

Model	Evaluation Mode	AvgBIO Score	ASW Metric	Performance vs. Baseline Methods
scGPT	Zero-Shot	Low	Variable	Underperforms scVI, Harmony, and HVG on most datasets [47]
Geneformer	Zero-Shot	Low	Poor	Consistently outperformed by simpler baselines [47]
scGPT	Fine-Tuned	High	High	State-of-the-art results on specialized tasks [71]
Geneformer	Fine-Tuned	High	High	Strong performance after task-specific adaptation [70]

Table 2: Performance of scGPT Variants with Different Pre-training Datasets

Model Variant	Pre-training Data	PBMC (12k) Performance	Generalizability Across Tissues
Random Initialization	None	Poor	Limited
scGPT Kidney	814,000 kidney cells	Moderate	Tissue-specific
scGPT Blood	10.3 million blood/bone marrow cells	Good	Strong on blood, moderate on others
scGPT Human	33 million non-cancerous human cells	Good	Broad but sometimes inferior to scGPT Blood [47]

The data reveal a consistent pattern: while zero-shot performance remains limited, strategic fine-tuning enables scFMs to achieve state-of-the-art results. A key study demonstrated that fine-tuned scGPT significantly outperformed Geneformer in cell type annotation, though contrasting findings exist, highlighting the importance of adaptation methodology [70].

Experimental Protocols for Benchmarking

Protocol 1: Zero-Shot Cell Type Clustering Evaluation

Objective: Assess the intrinsic quality of scFM embeddings for separating known cell types without additional training.

Methodology:

- **Embedding Generation**: Pass single-cell data through the pre-trained model (without fine-tuning) to extract cell embeddings.
- **Dimensionality Reduction**: Apply UMAP or t-SNE to embeddings for visualization.
- **Clustering**: Perform Leiden or Louvain clustering on the embeddings.
- **Evaluation Metrics**: Calculate:
  - Average BIO (AvgBIO) score
  - Average silhouette width (ASW)
  - Normalized Mutual Information (NMI)
- **Baseline Comparison**: Compare against established methods (scVI, Harmony) and simple feature selection (Highly Variable Genes).

This protocol revealed that both Geneformer and scGPT underperformed relative to selecting highly variable genes (HVG) and using more established methods like Harmony and scVI in cell type clustering [47].

Protocol 2: Batch Integration Assessment

Objective: Evaluate the model's ability to remove technical batch effects while preserving biological variation.

Methodology:

- **Dataset Selection**: Utilize benchmark datasets with known batch effects (e.g., Pancreas dataset with five different sources).
- **Embedding Generation**: Extract cell embeddings from scFMs in zero-shot mode.
- **Visualization**: Generate UMAP plots coloring by both batch and cell type.
- **Quantitative Metrics**:
  - Batch mixing scores (e.g., graph connectivity)
  - Principal component regression (PCR) score
  - Cell-type separation metrics (e.g., ASW_celltype)
- **Comparative Analysis**: Assess against specialized integration methods (Harmony, scVI).

This evaluation demonstrated that Geneformer's embedding space often failed to retain information about cell type, with clustering primarily driven by batch effects [47].

Protocol 3: Parameter-Efficient Fine-Tuning (PEFT)

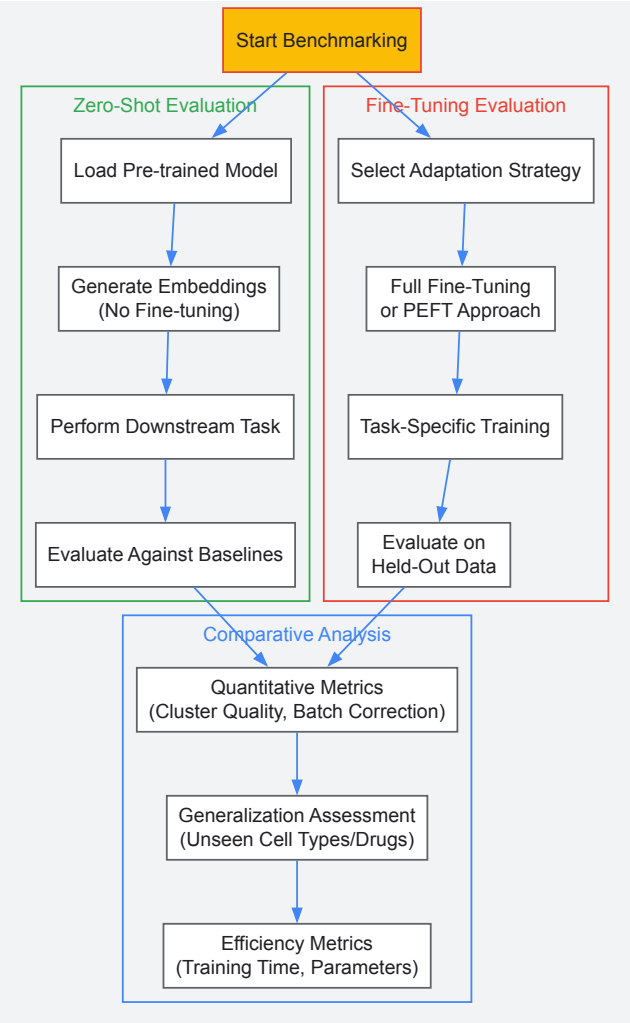
Objective: Evaluate adaptation strategies that preserve pre-trained knowledge while specializing for downstream tasks.

Methodology:

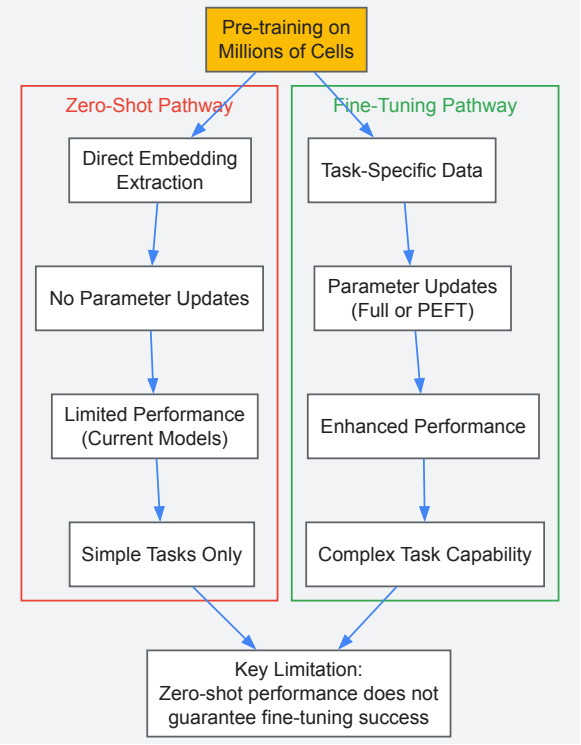
- **Adapter Design**: Introduce small, trainable parameters while keeping original model weights frozen.
- **Training Configuration**:
  - Base model: Frozen scGPT
  - Trainable parameters: <1% of original model [71]
  - Drug-conditional adapter layers for molecular perturbation prediction
- **Evaluation Tasks**:
  - Molecular perturbation response prediction
  - Zero-shot generalization to unseen cell lines
- **Comparative Conditions**: Compare against full fine-tuning and zero-shot baselines.

This approach has demonstrated state-of-the-art results across all settings, with significant improvements in few-shot and zero-shot generalization to new cell lines compared to existing baselines [71].

Visualization of Benchmarking Workflows



Benchmarking Workflow Comparison



Performance Limitation Relationship

The Scientist's Toolkit: Essential Research Reagents

Table 3: Essential Computational Tools for scFM Benchmarking

Tool/Resource	Type	Function in Benchmarking	Key Features
scGPT	Foundation Model	Base model for fine-tuning/zero-shot evaluation	33M cell pre-training, transformer architecture [70]
Geneformer	Foundation Model	Comparative model for benchmarking	6-layer or 12-layer architecture, 29.9M cell training [47]
CELLxGENE	Data Platform	Source of standardized benchmarking datasets	Curated single-cell data, multiple tissues and conditions [47]
Harmony	Integration Method	Baseline for batch correction evaluation	PCA-based integration, preserves biological variation [47]
scVI	Probabilistic Model	Baseline for clustering and integration	Deep generative model, handles technical noise [47]
Parameter-Efficient Fine-Tuning (PEFT)	Adaptation Technique	Efficient model specialization	<1% parameter training, avoids catastrophic forgetting [70]
Drug-Conditional Adapter	Specialized Component	Molecular perturbation prediction	Links scFMs to chemical structures, enables zero-shot prediction [71]

Discussion and Future Directions

The benchmarking evidence reveals a critical conclusion: current single-cell foundation models demonstrate limited capability in zero-shot settings but achieve state-of-the-art performance when properly fine-tuned. This pattern suggests that while pre-training captures broad biological patterns, task-specific adaptation remains essential for optimal performance [47] [70].

The inconsistent zero-shot performance raises important questions about what exactly these models learn during pre-training. Evaluation of scGPT's masked gene expression prediction capability revealed limitations, with the model often predicting median expression values regardless of true expression levels [49]. This fundamental shortcoming may explain why zero-shot embeddings frequently underperform simple baselines like highly variable gene selection.

Future benchmarking efforts should focus on developing more sophisticated evaluation frameworks that:

- Systematically dissect the relationship between pre-training objectives and downstream task performance
- Establish standardized datasets for cross-model comparison
- Develop more nuanced metrics that capture biological plausibility beyond technical clustering scores
- Explore hybrid approaches that combine the strengths of foundation models with task-specific architectures

As the field progresses, rigorous benchmarking methodologies will be essential for distinguishing genuine biological understanding from statistical artifacts in model performance, ultimately guiding the development of more capable and reliable single-cell foundation models for biomedical research and drug discovery.

Single-cell foundation models (scFMs) represent a transformative advancement in computational biology, leveraging large-scale pretraining on massive single-cell RNA-sequencing datasets to learn fundamental biological principles. These models, including scGPT, Geneformer, scBERT, and others, aim to capture the "transcriptional grammar" of cells, enabling researchers to predict cellular responses to perturbations, annotate cell types with unprecedented accuracy, and integrate diverse biological datasets [72]. The promise of these models lies in their potential to generalize across diverse biological contexts and facilitate discovery in settings where labeled data is scarce or unavailable.

This whitepaper provides a comprehensive technical analysis of the current landscape of single-cell foundation models, with a specific focus on their architectural innovations, performance benchmarks across key biological tasks, and practical limitations. Within the broader thesis of scFM research, we examine the critical challenge of balancing model scale with biological insight, exploring whether these complex architectures genuinely outperform established simpler methods or merely represent sophisticated solutions in search of problems. By synthesizing evidence from recent rigorous evaluations, we aim to guide researchers, scientists, and drug development professionals in selecting appropriate models for their specific applications and understanding the current frontiers of this rapidly evolving field.

Model Architectures & Technical Profiles

Architectural Principles and Pretraining Approaches

Single-cell foundation models predominantly adapt transformer architectures, originally developed for natural language processing, to biological data by treating genes as words and cellular gene expression profiles as sentences [73] [72]. This conceptual mapping enables the application of sophisticated language modeling techniques to cellular transcriptomics.

scBERT utilizes a BERT (Bidirectional Encoder Representations from Transformers) architecture adapted for single-cell data. The model creates gene embeddings through gene2vec to encode semantic similarities between genes and incorporates expression embeddings generated through term-frequency analysis to discretize continuous expression variables [72]. These embeddings serve as token inputs to the transformer architecture. scBERT follows a two-stage process: self-supervised pretraining on large amounts of unlabeled scRNA-seq data from sources like PanglaoDB, followed by supervised fine-tuning on task-specific data for applications like cell type annotation [74] [72].

scGPT employs a generative pretrained transformer framework designed for single-cell multi-omics data. The model uses a similar foundation of masked language modeling pretraining but extends its capabilities to integrate multiple modalities of single-cell data [75] [4]. Recent developments include scGPT-spatial, which incorporates spatial transcriptomics data through continual pretraining, enabling the model to capture spatial relationships between cells in addition to transcriptional profiles [75].

Geneformer operates on a transformer architecture pretrained on a massive corpus of single-cell data from various tissues and organisms. The model employs a casual language modeling objective rather than masked language modeling, potentially making it more suitable for generative tasks and temporal modeling of cellular processes [47] [76].

While these models share common transformer foundations, their specialized architectures, pretraining objectives, and data tokenization strategies lead to significantly different performance characteristics across biological tasks, as revealed in recent benchmarking studies.

Model Capabilities and Specializations

Table 1: Core Architectural Characteristics of Major Single-Cell Foundation Models

Model	Architecture	Pretraining Objective	Key Specializations	Primary Applications
scBERT	BERT-based transformer	Masked language modeling	Gene embedding via gene2vec, expression binning	Cell type annotation, novel cell type discovery [74] [72]
scGPT	GPT-based transformer	Generative pretraining	Multi-omics integration, spatial transcriptomics	Perturbation prediction, batch integration, cell classification [75] [47] [4]
Geneformer	Transformer	Causal language modeling	Representation learning for cellular states	Cell type classification, gene network analysis [47] [76]
scFoundation	Not detailed in results	Not detailed in results	Not detailed in results	Not detailed in results

Comparative Performance Benchmarking

Zero-Shot Capability Assessment

The zero-shot performance of foundation models—where pretrained models are applied without task-specific fine-tuning—is critically important for exploratory biological research where labeled data may be unavailable. Recent rigorous evaluations reveal significant limitations in current scFMs in this setting.

A comprehensive assessment of scGPT and Geneformer in zero-shot cell type clustering demonstrated that both models frequently underperform simpler established methods [47]. When evaluated using Average BIO (AvgBio) score and average silhouette width (ASW) metrics across multiple datasets, both proposed foundation models were consistently outperformed by simple selection of highly variable genes (HVG) and more established methods like Harmony and scVI [47]. Surprisingly, HVG selection surpassed both Geneformer and scGPT across all evaluation metrics, raising questions about the actual value contributed by complex transformer architectures in basic clustering tasks [47].

The PertEval-scFM benchmark specifically evaluated zero-shot performance for perturbation effect prediction, testing whether contextualized representations from scFMs enhance prediction of how cells change following genetic perturbations [77]. The results indicated that scFM embeddings offer limited improvement over simple baseline models in the zero-shot setting, particularly under distribution shift where models encounter strong or atypical perturbations not well-represented in training data [77].

Cell Type Annotation and Novel Cell Type Discovery

Cell type annotation represents a fundamental application of scFMs, with several models claiming advanced capabilities in accurately classifying cell types and identifying novel cellular states.

scBERT has demonstrated superior performance in cell type annotation tasks, outperforming methods like Seurat in validation mean accuracy (0.8510 vs. 0.8013) on datasets such as the NeurIPS multi-omics dataset [72]. The model also shows robustness to batch effects, maintaining performance across datasets generated with different technologies [74]. However, scBERT's performance is significantly influenced by cell-type distribution imbalance, with skewed distributions substantially impacting annotation accuracy and novel cell type detection capability [72].

In comprehensive benchmarking across multiple models and tasks, no single scFM consistently outperformed all others across all evaluation metrics [4]. Performance varied significantly based on dataset size, task complexity, and specific biological context, emphasizing the need for researchers to select models based on their specific application requirements rather than assuming universal superiority of any single approach.

Batch Integration and Data Harmonization

Batch integration—removing technical artifacts from multiple data sources while preserving biological signal—represents another critical benchmark for scFMs. Quantitative evaluation with batch integration metrics reveals a mixed performance landscape.

Geneformer consistently underperforms relative to scGPT, Harmony, scVI, and HVG across most datasets in batch correction tasks [47]. Visualization of embeddings from the Pancreas benchmark dataset showed that while Geneformer and scGPT can integrate different experiments conducted with the same experimental technique, they generally fail to correct for batch effects between different techniques [47]. While scGPT's cell embedding space offers some separation between cell types, the primary structure in dimensionality reduction remains driven by batch effects rather than biological signal [47].

Notably, scGPT demonstrates stronger performance on complex datasets where both technical and biological batch effects are present (Tabula Sapiens and Immune datasets), potentially because these datasets were included in its pretraining corpus [47]. This highlights a challenge in evaluating these models: the difficulty in disentangling genuine learning from potential data leakage during pretraining.

Perturbation Effect Prediction

Predicting cellular responses to genetic or chemical perturbations represents a crucial application with significant implications for drug development and disease modeling. The PertEval-scFM benchmark systematically evaluates this capability across multiple scFMs.

Results indicate that current-generation scFMs provide limited improvement over simple baseline models for perturbation effect prediction, especially in zero-shot settings where models cannot be fine-tuned on task-specific data [77]. The benchmark highlights that these models struggle particularly with strong or atypical perturbations that represent distribution shifts from their training data [77]. This limitation significantly impacts real-world applications where predicting responses to novel therapeutic interventions often requires extrapolation beyond established training distributions.

Independent research corroborates these findings, with one study concluding that "deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines" [4]. This suggests that while scFMs represent architecturally sophisticated approaches, their practical utility for critical tasks like drug sensitivity prediction remains limited compared to simpler, more interpretable methods.

Table 2: Quantitative Performance Comparison Across Biological Tasks

Model	Cell Type Annotation (Accuracy)	Batch Integration (Score)	Perturbation Prediction	Zero-Shot Clustering
scBERT	0.8510 (NeurIPS dataset) [72]	Not comprehensively evaluated	Not evaluated	Not primary application
scGPT	Variable across datasets [47]	Moderate (better on complex batches) [47]	Limited improvement over baselines [77]	Underperforms HVG and scVI [47]
Geneformer	Variable across datasets [47]	Consistently underperforms [47]	Limited improvement over baselines [77]	Underperforms HVG and scVI [47]
HVG (Baseline)	Not applicable	High (top performer) [47]	Simple but effective baseline [77]	Outperforms complex scFMs [47]

Experimental Protocols for Model Evaluation

Standardized Benchmarking Frameworks

Rigorous evaluation of scFMs requires standardized frameworks that control for potential confounding factors and ensure fair comparisons across models. The PertEval-scFM framework provides a standardized approach specifically designed for evaluating perturbation effect prediction [77]. The benchmark tests whether zero-shot embeddings produced by scFMs contain meaningful information for predicting perturbation effects by providing a pair of cells—one perturbed and one unperturbed—to a simple model that uses representations from the scFMs to predict how the cell changed [77].

For zero-shot capability assessment, researchers have employed evaluation protocols that test models on datasets with varying degrees of similarity to their pretraining corpora [47]. This involves quantifying performance metrics like AvgBio and ASW for clustering tasks, and principal component regression (PCR) scores for batch integration, while carefully tracking dataset overlaps that might artificially inflate performance metrics [47].

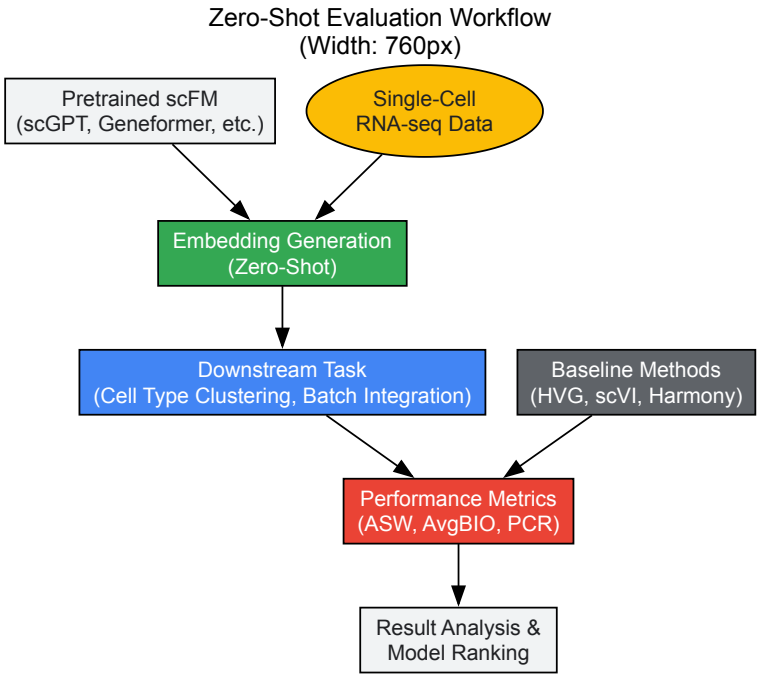
Ablation Studies and Pretraining Impact Assessment

To disentangle the effects of pretraining from architectural choices, researchers have conducted systematic ablation studies. One approach involves comparing multiple variants of the same architecture with different pretraining regimes [47]. For example, evaluations of scGPT have included: a randomly initialized version (no pretraining), scGPT pretrained on 814,000 kidney cells (tissue-specific), scGPT pretrained on 10.3 million blood and bone marrow cells (partially specialized), and scGPT pretrained on 33 million non-cancerous human cells (comprehensive) [47].

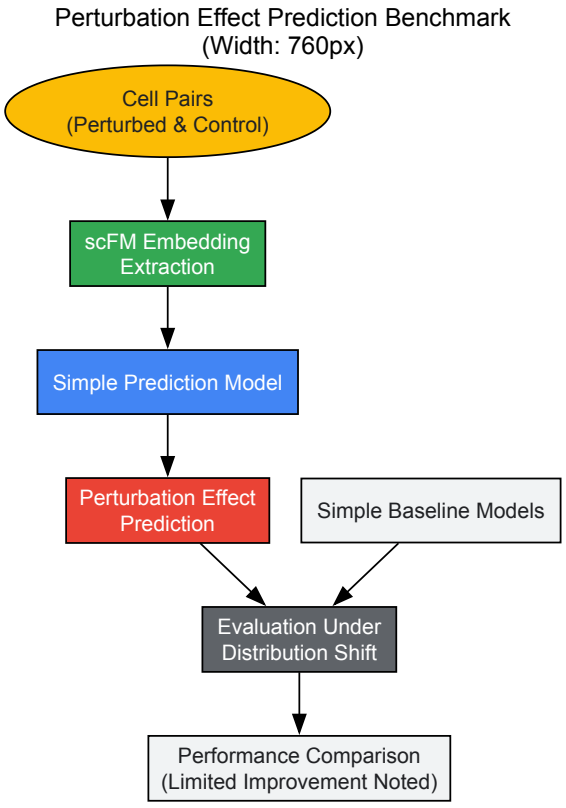
These studies demonstrate that while pretraining generally provides clear improvements over randomly initialized models, the relationship between pretraining dataset size and model performance is not always linear or predictable [47]. In some cases, tissue-specific pretraining on smaller datasets can outperform more comprehensive pretraining on certain tasks, suggesting that dataset relevance may be more important than sheer volume for specific applications.

Visualization of Model Evaluation Workflows

Zero-Shot Evaluation Pipeline



Perturbation Effect Prediction Framework



The Scientist's Toolkit: Essential Research Reagents

Table 3: Essential Research Resources for scFM Evaluation and Application

Resource Name	Type	Function in Research	Access Information
PertEval-scFM	Benchmark Framework	Standardized evaluation of perturbation effect prediction	GitHub: github.com/aaronwtr/PertEval [77]
PanglaoDB	Pretraining Data	Large collection of scRNA-seq data for model pretraining	Publicly available at panglao.se [74] [72]
CELLxGENE	Data Resource	Curated single-cell data for pretraining and evaluation	Publicly available census data [47]
scBERT Codebase	Model Implementation	Reference implementation for scBERT model	GitHub: TencentAILabHealthcare/scBERT [74] [72]
Zheng68k Dataset	Benchmark Data	PBMC dataset for cell-type annotation performance assessment	Available via 10x Genomics [74]
NeurIPS Dataset	Evaluation Data	Multi-omics data from hematopoietic stem cells for validation	Kaggle: Open Problems in Multimodal Single-Cell Data [72]

The comprehensive evaluation of single-cell foundation models presented in this analysis reveals a field in transition, marked by significant architectural achievements but substantial practical limitations. While models like scGPT, Geneformer, and scBERT demonstrate impressive capabilities in specific tasks like cell type annotation, their performance in critical zero-shot settings and perturbation prediction often fails to exceed simpler, established methods [77] [47] [72].

The broader thesis emerging from current scFM research suggests that model complexity alone does not guarantee biological insight. The consistent outperformance of simple highly variable gene selection over sophisticated transformer architectures in clustering tasks [47], coupled with the limited improvement of scFMs over linear baselines in perturbation prediction [77] [4], indicates fundamental challenges in translating architectural sophistication to practical utility.

Future development of single-cell foundation models should prioritize biological plausibility over sheer scale, specialized capabilities over general claims, and rigorous zero-shot evaluation over fine-tuned performance. For researchers and drug development professionals, this analysis suggests a cautious approach to adopting these technologies—leveraging their strengths for specific applications like cell type annotation while maintaining

simpler methods as baselines for critical tasks like perturbation prediction. As the field matures, the integration of biological prior knowledge, improved data tokenization strategies, and more sophisticated pretraining objectives may eventually fulfill the promise of foundation models to transform single-cell biology.

The emergence of single-cell foundation models (scFMs) represents a paradigm shift in computational biology, introducing large-scale, self-supervised models trained on millions of single-cell transcriptomes [10]. These models promise to learn universal biological knowledge during pretraining, endowing them with emergent capabilities for zero-shot learning and efficient adaptation to various downstream tasks [1]. However, as these models grow in complexity and prevalence, the need for rigorous, biologically grounded evaluation metrics becomes increasingly critical. Current benchmarking studies reveal that no single scFM consistently outperforms others across all tasks, emphasizing the need for tailored model selection based on dataset size, task complexity, and biological interpretability [4] [1]. This technical guide provides a comprehensive framework for evaluating scFMs, focusing on three cornerstone assessment domains: cell embedding quality, batch integration efficacy, and biological grounding. We synthesize current benchmarking approaches, introduce novel metrics addressing existing gaps, and provide standardized experimental protocols to ensure reproducible and biologically meaningful model assessment.

Core Performance Metrics for Single-Cell Foundation Models

Cell Embedding Quality Metrics

Cell embeddings form the foundational representation learned by scFMs, and their quality determines performance across downstream applications. Evaluation metrics for embeddings must assess both their structural integrity and ability to preserve biological information.

Table 1: Metrics for Evaluating Cell Embedding Quality

Metric Category	Specific Metric	Technical Definition	Interpretation	Ideal Value
Local Neighborhood Preservation	kNN Accuracy	Proportion of a cell's nearest neighbors in the embedding space that share the same cell type	Measures purity of local cell-type neighborhoods; higher values indicate better separation of cell types	>90% [78]
	kNN Recall	Proportion of a cell's high-dimensional nearest neighbors preserved in the embedding space	Quantifies preservation of original high-dimensional structure; higher values indicate less distortion	Varies; UMAP/TSNE achieve >15% vs <5% for PCA [78]
Cluster Quality	Silhouette Coefficient	Measures compactness and separation of predefined classes in embedding space	Higher values indicate tighter, better-separated clusters; can exceed original high-dimensional space	>0.3 advantage over baselines [78]
	Adjusted Mutual Information (AMI)	Information-theoretic measure between cluster assignments and ground truth labels	Higher values indicate clustering that better recovers true cell types; less sensitive to number of clusters	>0.25 advantage over baselines [78]
Global Structure Preservation	scGraph	Graph-based similarity comparing cell-type relationships in embedding with consensus biological knowledge	Higher scores indicate better preservation of hierarchical biological relationships; flags distorted structures [79]	

Traditional metrics like kNN accuracy and silhouette coefficients effectively measure local neighborhood preservation and cluster separation. However, recent research highlights that these metrics can be gamed by methods that create artificially separated "islands" of cell types while distorting broader biological relationships. The scGraph metric addresses this limitation by evaluating whether embeddings preserve the natural continuum of developmental trajectories and functional relationships between cell types [79].

Batch Integration Metrics

Batch effects constitute systematic technical variations confounding biological signals, and their removal through integration is crucial for joint analysis across datasets. Evaluation of batch integration must balance two competing objectives: removing technical artifacts while preserving meaningful biological variation.

Table 2: Metrics for Evaluating Batch Integration Performance

Metric	Technical Definition	What it Measures	Limitations
kBET (k-nearest neighbor Batch	Tests if local batch label distribution matches global	Batch mixing at local level; lower rejection rates indicate better integration	Sensitive to parameter k; requires cell identity labels [17]

Metric	Technical Definition	What it Measures	Limitations
Effect Test)	distribution using χ^2-test		[80]
LISI (Local Inverse Simpson's Index)	Measures diversity of batches or cell types in local neighborhoods	Effective number of batches or cell types in neighborhood; higher LISI (batch) = better mixing, higher LISI (cell type) = better separation	Computationally intensive; interpretation depends on context [17]
ASW (Average Silhouette Width)	Measures compactness of cell types and separation from other cell types	Cell type preservation after integration; higher values indicate better biological structure preservation	Does not directly measure batch mixing [17]
ARI (Adjusted Rand Index)	Measures similarity between clustering results and ground truth labels	Conservation of cell identity clusters after integration; higher values indicate better biological preservation	Requires known ground truth labels [17]

Benchmarking studies have identified top-performing methods for different integration scenarios. For simple batch correction tasks with consistent cell type compositions across batches, **Harmony** and **Seurat** consistently perform well. For more complex integration tasks involving different protocols or non-identical cell types, deep learning approaches like **scVI**, **scGen**, and **scANVI**, as well as the linear embedding method **Scanorama**, demonstrate superior performance [80].

Biologically-Grounded Evaluation Metrics

Moving beyond technical validation, the most advanced evaluation frameworks incorporate biological knowledge to assess whether embeddings capture meaningful biological relationships.

Table 3: Biology-Informed Metrics for Single-Cell Foundation Models

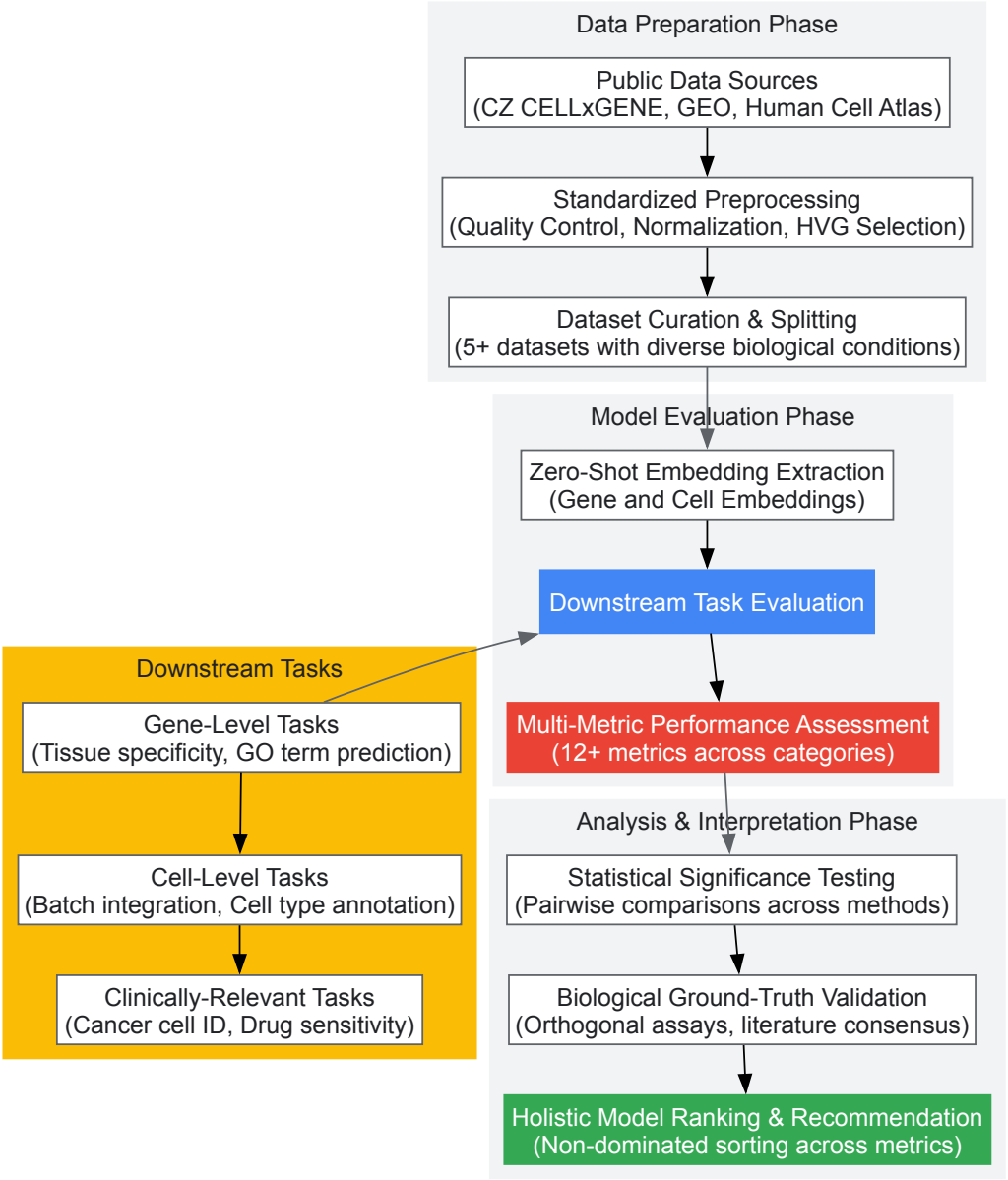
Metric	Basis of Evaluation	Application in scFMs	Advantage over Agnostic Metrics
scGraph-OntoRWR	Consistency of cell-type relationships with ontological knowledge using random walks on ontology graphs	Measures intrinsic biological knowledge encoded in embeddings without need for fine-tuning	Directly evaluates biological relevance rather than just technical separation [1]
LCAD (Lowest Common Ancestor Distance)	Ontological proximity between misclassified cell types in cell type annotation tasks	Assesses severity of annotation errors based on cellular hierarchy	Recognizes that misclassification between closely-related types is less severe than between distant types [1]
Pathway & GO Term Enrichment	Ability of gene embeddings to predict Gene Ontology terms and biological pathways	Evaluates whether functionally related genes cluster in embedding space	Validates that embedding captures functional biological relationships beyond expression patterns [1]
Topic Model Interpretability Metrics	Diversity and consistency of topics identified in embedded topic models (e.g., scE2TM)	Quantifies interpretability of latent factors through 10 specialized metrics	Addresses "interpretation collapse" where models focus only on highly-expressed genes [81]

The integration of biological knowledge into evaluation metrics is particularly crucial for assessing scFMs in zero-shot settings, where the model's inherent biological understanding—without task-specific fine-tuning—determines its utility for discovery-driven research [1].

Experimental Protocols for Metric Evaluation

Standardized Benchmarking Pipeline for scFMs

Implementing a rigorous evaluation framework for scFMs requires standardized protocols across datasets, preprocessing steps, and evaluation metrics. The following workflow provides a comprehensive assessment strategy:



Data Curation and Preprocessing:

- Source diverse datasets from public repositories (CZ CELLxGENE, GEO, SRA) encompassing multiple tissues, species, and experimental conditions [1] [10].
- Apply uniform quality control thresholds: filter cells with mitochondrial content >20% and gene counts outside 200-2500 range for human data.
- Implement standardized normalization (e.g., log(TPM/10^5+1)) and highly variable gene selection (2000-5000 genes) across all datasets.
- Partition data into training/validation/test splits ensuring no data leakage between model development and evaluation phases.

Embedding Extraction and Downstream Task Evaluation:

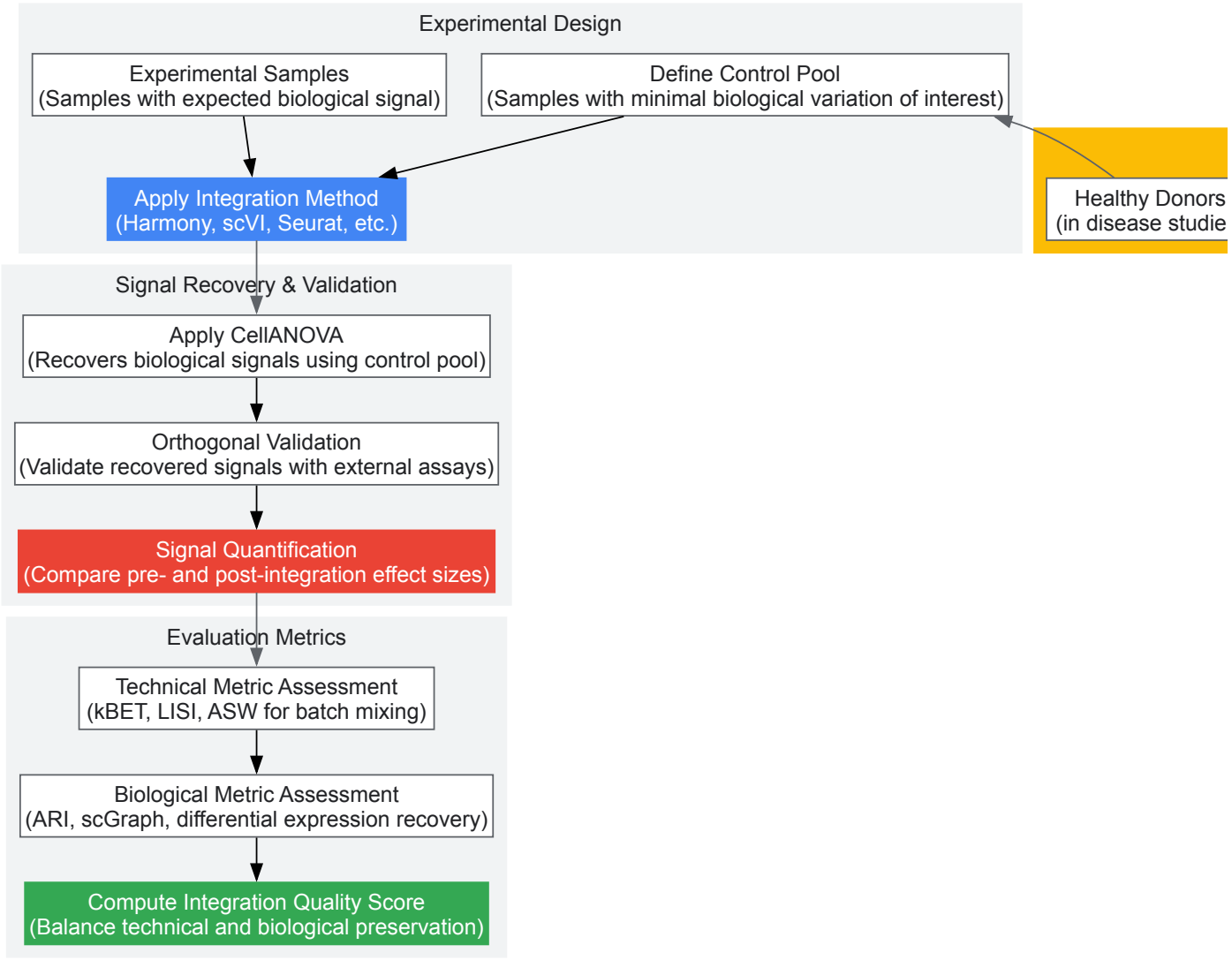
- Extract zero-shot embeddings from pretrained scFMs without fine-tuning to assess inherent model capabilities [1].
- Evaluate on two gene-level tasks (tissue specificity prediction, GO term recovery) and four cell-level tasks (batch integration, cell type annotation, cancer cell identification, drug sensitivity prediction) [4].
- For batch integration assessment, utilize datasets with known batch effects from different technologies (10x, SMART-seq, Drop-seq) and varying biological conditions [17].

Performance Quantification and Statistical Analysis:

- Compute comprehensive metric suites spanning technical and biological dimensions (12+ metrics recommended) [4] [1].
- Employ statistical significance testing (paired t-tests with multiple testing correction) across repeated runs with different random seeds.
- Generate holistic model rankings using non-dominated sorting algorithms that aggregate performance across all metrics and tasks [1].

Controlled Experiment for Batch Integration Assessment

Current integration methods often overcorrect and remove biologically meaningful variation alongside technical artifacts. The following protocol implements a rigorous assessment of integration fidelity:



Experimental Design Considerations:

- Implement case-control designs with clearly defined control pools (e.g., healthy donors in disease studies, baseline timepoints in longitudinal studies) [82] [37].
- For method benchmarking, utilize datasets with orthogonal validation (e.g., CITE-seq protein measurements, fluorescent reporter assays) to establish biological ground truth [37].
- Include datasets with varying complexity: identical cell types across technologies, partially overlapping cell types, and complex nested batch effects [17].

Signal Recovery and Quantification:

- Apply CellANOVA or similar statistical frameworks to recover biological signals erased during integration [82] [37].
- Quantify effect sizes for known biological signals (e.g., disease-associated differential expression, treatment responses) before and after integration.
- Compute the ratio of preserved biological signal to removed batch effect as an overall integration fidelity score.

Validation and Interpretation:

- Validate recovered signals through orthogonal assays (e.g., protein expression, functional assays) when available [37].
- Perform differential expression analysis on integrated data and compare results with ground truth expectations.
- Generate quantitative integration reports detailing the percentage of biological variance preserved and technical variance removed.

Essential Research Reagents and Computational Tools

Table 4: Essential Research Reagents and Computational Tools for scFM Evaluation

Category	Item	Specification/Version	Application Context	Quality Control
Reference Datasets	CZ CELLxGENE Census	v2023.11.01 (50M+ cells)	Pretraining corpus and benchmark standardization	Manual annotation review, cell type consistency checks [10]
	Asian Immune Diversity Atlas (AIDA) v2	Independent dataset from CellxGene	Unbiased validation to prevent data leakage	Standardized preprocessing pipeline [1]
Benchmarking Suites	scIB	Python package (15+ metrics)	Comprehensive integration benchmarking	Cross-metric consistency validation [80]
	scE2TM Interpretability Suite	10 quantitative interpretability metrics	Topic model evaluation for embedded methods	Diversity and consistency threshold checks [81]
Integration Methods	Harmony, scVI, Seurat	Latest stable versions	Baseline comparisons for batch integration	Parameter tuning as per original publications [17] [80]
Biological Knowledge Bases	Gene Ontology, Cell Ontology	Regular updates	Biological grounding of evaluation metrics	Annotation quality filters, evidence code weighting [1]

The evaluation of single-cell foundation models requires a multifaceted approach that balances technical metrics with biological groundedness. As demonstrated through comprehensive benchmarking studies, no single scFM dominates across all tasks, emphasizing the need for task-specific model selection guided by rigorous evaluation frameworks [4] [1]. The field is moving beyond purely technical assessments toward biology-informed metrics that validate whether computational representations capture genuine biological relationships. Methods like scGraph and scGraph-OntoRWR represent important advances in this direction, addressing the limitation of previous metrics that could be gamed by creating artificially separated cell islands without preserving true biological continua [79]. For practitioners, we recommend adopting a comprehensive evaluation strategy that assesses batch integration efficacy, cell embedding quality, and biological relevance using the standardized protocols and metrics outlined in this guide. As scFMs continue to evolve, maintaining this rigorous, biologically-grounded approach to evaluation will be essential for ensuring these powerful tools deliver meaningful insights into cellular function and disease mechanisms.

Single-cell foundation models (scFMs) represent a transformative advance in the analysis of single-cell genomics data. Trained on millions of cells, these models promise to learn universal biological principles that can be adapted to various downstream tasks. However, a critical question remains for researchers and drug development professionals: how does one select the optimal model for a specific biological or clinical question? This whitepaper synthesizes findings from a comprehensive benchmark study to provide a definitive guide for task-specific model selection. We demonstrate that no single scFM consistently outperforms all others across diverse applications. Success depends on a deliberate strategy aligned with the target task's specific requirements, dataset characteristics, and available computational resources. Herein, we provide structured data, detailed protocols, and a practical toolkit to empower scientists to make informed decisions, thereby maximizing the impact of scFMs in biological research and therapeutic development.

The advent of high-throughput single-cell RNA sequencing (scRNA-seq) has generated vast amounts of data, providing an unprecedented granular view of cellular heterogeneity [1]. Single-cell foundation models (scFMs) are large-scale deep learning models, typically based on transformer architectures, pretrained on these extensive and diverse datasets in a self-supervised manner [33]. The premise is that by exposing a model to millions of cells from various tissues and conditions, it can learn fundamental principles of cellular biology, resulting in a foundational tool that can be efficiently adapted—or used in a zero-shot setting—for a wide range of downstream tasks such as cell type annotation, batch integration, and drug sensitivity prediction [4] [33].

Despite their potential, the practical application of scFMs is fraught with a key challenge: the absence of a universally superior model. A recent, extensive benchmark study evaluating six prominent scFMs against established baseline methods confirmed this, concluding that "no single scFM consistently outperforms others across all tasks" [4] [1]. This finding underscores the critical importance of a nuanced, task-oriented approach to model selection. The performance of an scFM is contingent on a complex interplay of factors, including the nature of the task (e.g., gene-level vs. cell-level), the size and quality of the dataset, the complexity of the biological question, and the computational budget. This guide is designed to navigate this complexity, providing a structured framework for identifying the model with the right strengths for the job at hand.

Comprehensive Benchmarking Results

A holistic benchmark, published in *Genome Biology* in 2025, provides a rigorous empirical basis for model selection. The study evaluated six leading scFMs (including Geneformer, scGPT, UCE, scFoundation, LangCell, and scCello) against traditional baseline methods across two gene-level and four cell-level tasks under realistic conditions [4] [1]. Performance was assessed using 12 metrics, incorporating unsupervised, supervised, and novel knowledge-based approaches like scGraph-OntoRWR, which evaluates the biological consistency of learned cell-type relationships [1].

The overarching finding is that while scFMs are robust and versatile, simpler machine learning models can be more efficient and effective for specific datasets, particularly under resource constraints [4]. The following tables synthesize the key quantitative findings from this benchmark, offering a clear comparison of model performance across different tasks.

Table 1: Overall Model Ranking Across Diverse Tasks (General Performance)

Model	Overall Rank	Strengths	Noted Weaknesses
scGPT	1	Versatile; strong in batch integration & clinical task adaptation	Computational intensity for very large datasets
Geneformer	2	Effective gene-level representation; good generalizability	Can be outperformed on specific cell-level annotations
scFoundation	3	Robust large-scale pretraining	Less efficient adaptation to small, specific datasets
UCE	4	Good integration capabilities	Inconsistent performance in perturbation prediction
LangCell	5	Innovative tokenization approaches	Emerging model, requires further validation
scCello	6	Specialized architecture	Lower general performance across multiple tasks
Simple Baselines (e.g., HVGs, Seurat)	-	Highly efficient on specific datasets with limited resources	Lacks generalizability; no zero-shot capability

Table 2: Task-Specific Model Performance and Key Factors

Task Category	Top Performing Models	Key Evaluation Metrics	Decisive Factors for Selection
Batch Integration	scGPT, Harmony (Baseline), scVI (Baseline)	iLISI, kBET, scGraph-OntoRWR	Dataset size, biological complexity [1]
Cell Type Annotation	scBERT, scGPT	Accuracy, F1-score, Lowest Common Ancestor Distance (LCAD)	Presence of novel cell types, need for ontological consistency [1]
Clinical Prediction (e.g., Drug Sensitivity)	scGPT, scFoundation	AUC-ROC, Precision-Recall	Dataset size, task complexity, model's clinical relevance [4]
Gene-Level Tasks (e.g., Function Prediction)	Geneformer, scGPT	AUPRC (GO term prediction), Mean Rank (Tissue specificity)	Need for capturing functional gene relationships [1]
Perturbation Effect Prediction	Specialized models recommended	RMSE, Pearson correlation	Limited zero-shot performance of general scFMs [77]

Experimental Protocols for Key Tasks

To ensure reproducibility and provide a clear methodology for researchers, this section outlines the detailed experimental protocols for the core tasks used in the benchmark analysis.

Protocol for Cell Type Annotation with Ontological Evaluation

Objective: To evaluate the accuracy and biological relevance of cell type annotations generated by scFMs, including the severity of misclassifications based on ontological relationships.

- **Input Data Preparation:**
  - **Data Source:** Utilize high-quality, manually annotated scRNA-seq datasets from sources like the Asian Immune Diversity Atlas (AIDA) v2 available through CellxGene [1].
  - **Preprocessing:** Apply standard normalization and log-transformation to the raw count matrix. The dataset should be split into a training set (with known labels) and a hold-out test set.

- **Feature Extraction (Zero-Shot Setting):**
  - Extract cell embeddings from the pretrained scFM without any further fine-tuning (zero-shot). This is typically a dedicated cell-level embedding vector produced by the model's forward pass [1].
- **Cell Type Classification:**
  - Train a simple classifier (e.g., logistic regression, k-nearest neighbors) on the training set using the scFM-derived cell embeddings as features.
  - Use the trained classifier to predict cell type labels for the hold-out test set.
- **Performance Evaluation:**
  - **Standard Metrics:** Calculate accuracy, F1-score, and other multiclass classification metrics.
  - **Ontology-Informed Metric (LCAD):** For each misclassified cell, calculate the Lowest Common Ancestor Distance (LCAD). This metric measures the ontological proximity between the true cell type and the predicted cell type in a structured cell ontology (e.g., Cell Ontology). A smaller LCAD indicates a less severe error (e.g., confusing two types of T cells vs. confusing a T cell with a neuron) [1].

Protocol for Biological Insight Evaluation via scGraph-OntoRWR

Objective: To assess whether the cell-type relationship structure captured by an scFM's embedding space is consistent with established biological knowledge.

- **Embedding Space Construction:**
  - Obtain cell embeddings for a diverse dataset containing multiple known cell types using the scFM in a zero-shot manner.
- **Cell-Cell Similarity Graph:**
  - Construct a k-nearest neighbor (k-NN) graph from the cell embeddings, where nodes represent cells and edges connect each cell to its k most similar neighbors based on cosine distance in the embedding space.
- **Random Walk with Restart (RWR):**
  - For a given query cell type (e.g., "CD4+ T cell"), initiate RWR from all cells belonging to that type on the k-NN graph.
  - The RWR algorithm propagates proximity scores across the graph, resulting in a "learned" similarity profile for the query cell type.
- **Comparison with Prior Knowledge:**
  - Compare the RWR-derived similarity profile against a "gold-standard" similarity profile derived from the Cell Ontology. The gold-standard profile defines the relatedness of cell types based on their ontological distance.
  - The final scGraph-OntoRWR score is the correlation (e.g., Spearman's rank) between the learned and the gold-standard profiles. A higher score indicates the scFM has captured more biologically meaningful relationships [1].

Visualizing the Model Selection Workflow

The following diagram illustrates the logical decision process for selecting an appropriate scFM based on the specific research task and data characteristics, as derived from the benchmark findings.

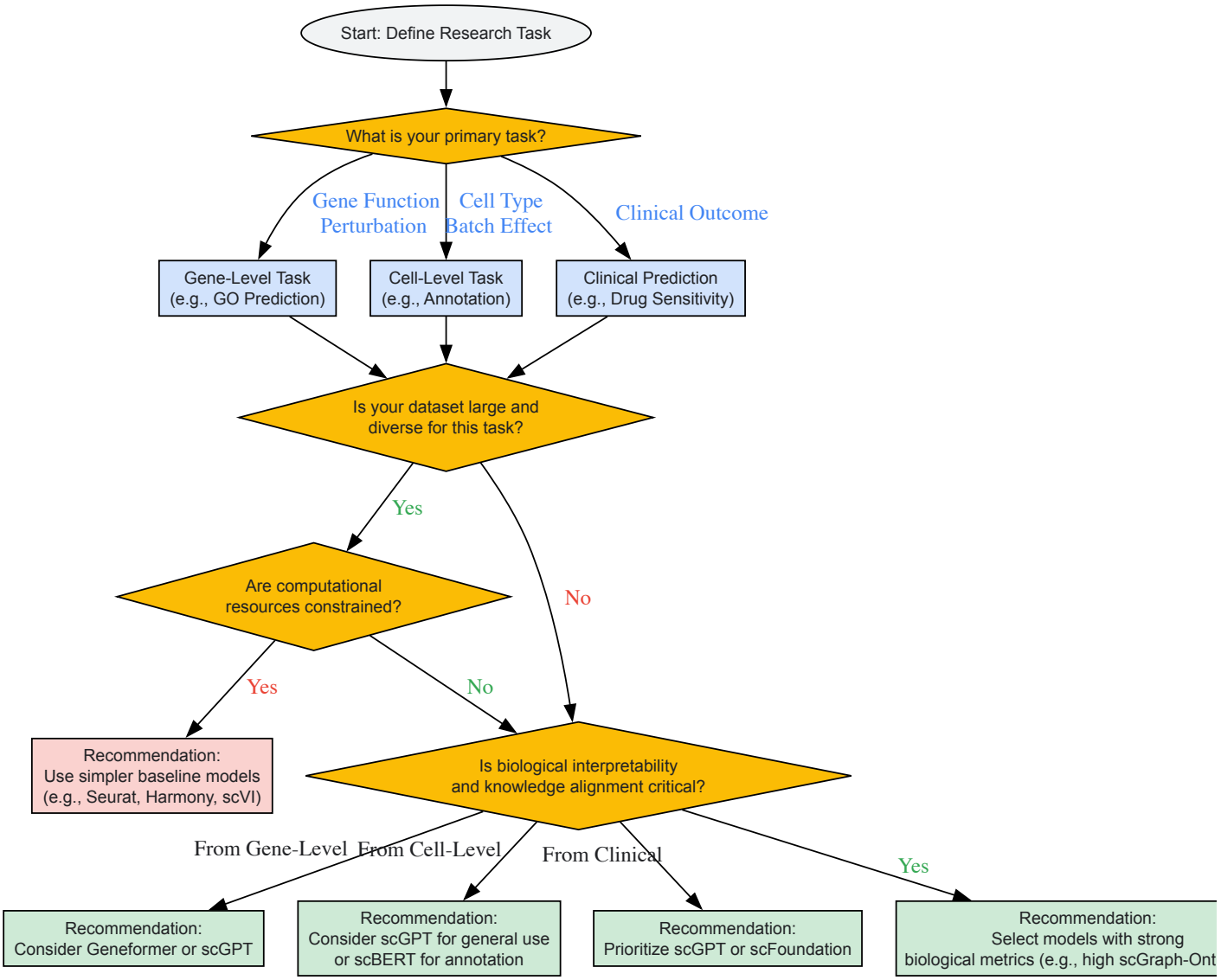


Figure 1: A decision workflow for selecting single-cell foundation models.

The Scientist's Toolkit: Essential Research Reagents

The following table details key resources and their functions, as utilized in the benchmark studies, for researchers aiming to implement or evaluate scFMs in their own work.

Table 3: Key Research Reagent Solutions for scFM Implementation

Item / Resource	Function / Description	Example Sources / Tools
Pretrained scFM Models	Provides the foundational model weights for generating embeddings or fine-tuning on new data.	scGPT, Geneformer, scFoundation, UCE, LangCell, scCello [4] [33]
Benchmarking Datasets	High-quality, annotated datasets used for rigorous evaluation and validation of model performance.	Asian Immune Diversity Atlas (AIDA) v2 from CellxGene; datasets spanning 7 cancer types and 4 drugs [4] [1]
Cell Ontology	A structured, controlled vocabulary for cell types. Serves as the "gold standard" for evaluating the biological relevance of model outputs.	Cell Ontology (from OBO Foundry); used for metrics like LCAD and scGraph-OntoRWR [1]
Integration & Annotation Tools (Baselines)	Established, non-foundation model methods used as performance baselines. Critical for determining if an scFM is necessary.	Seurat (anchor-based), Harmony (clustering-based), scVI (generative model) [4] [1]
Evaluation Metrics Suite	A collection of standardized metrics to holistically assess model performance across different axes.	Includes iLISI/kBET (integration), Accuracy/F1 (annotation), AUPRC (gene function), and novel metrics

Item / Resource	Function / Description	Example Sources / Tools
		like scGraph-OntoRWR and LCAD [4] [1]

The deployment of single-cell foundation models marks a significant evolution in computational biology, shifting the paradigm from building task-specific models to leveraging and adapting powerful, general-purpose tools. However, their power is not realized through a one-size-fits-all application. As the comprehensive data presented in this guide demonstrates, the key to unlocking the potential of scFMs lies in a deliberate, task-specific selection process.

Researchers must weigh the nature of their biological question, the scale and quality of their data, the imperative for biological interpretability, and their computational constraints. The benchmarks clearly show that simpler, traditional models remain formidable and often more efficient choices for well-defined problems with limited data. Conversely, for complex, multifaceted tasks like building a comprehensive cell atlas or predicting clinical outcomes from heterogeneous data, more versatile scFMs like scGPT show distinct advantages. By adopting the structured framework, protocols, and toolkit provided herein, scientists and drug developers can make informed, strategic decisions, ensuring that the right model is deployed for the job and accelerating the translation of single-cell genomics into meaningful biological insights and therapeutic breakthroughs.

The emergence of single-cell foundation models (scFMs) has revolutionized the interpretation of single-cell transcriptomics data by providing a unified framework for analyzing cellular heterogeneity. These models, trained on millions of single-cell transcriptomes, learn latent representations of genes and cells that can be adapted to various downstream tasks such as cell type annotation, batch integration, and perturbation prediction [6]. However, as the complexity and scale of these models grow, a critical challenge has emerged: how to effectively evaluate whether the representations learned by scFMs capture biologically meaningful patterns beyond technical artifacts [4] [1].

Traditional evaluation metrics for single-cell analysis often focus on technical aspects like clustering performance or batch correction efficiency, but they fail to assess whether the model's outputs align with established biological knowledge [1]. This limitation has prompted the development of novel biology-aware evaluation frameworks, among which scGraph-OntoRWR has emerged as a groundbreaking metric specifically designed to quantify the biological relevance of scFM embeddings [4] [1]. This metric represents a paradigm shift in model assessment by directly measuring the consistency between computational representations and prior biological knowledge encoded in structured ontologies.

This technical guide examines the role of scGraph-OntoRWR within the broader context of scFM research, providing researchers with a comprehensive framework for implementing this metric in their evaluation pipelines. We detail the methodological foundations, experimental protocols, and practical applications of this innovative approach to model assessment.

The Need for Biology-Aware Evaluation in scFM Research

Limitations of Conventional Evaluation Metrics

Standard evaluation approaches for scFMs primarily rely on performance-based metrics that measure task-specific accuracy, such as cell type classification accuracy or batch integration scores. While these metrics provide valuable insights into model utility, they suffer from significant limitations:

- **Task-specific focus**: Traditional metrics evaluate performance on narrow tasks but fail to assess general biological plausibility.
- **Knowledge blindness**: They cannot determine whether learned representations reflect established biological relationships.
- **Context independence**: They treat all misclassifications equally, ignoring the biological severity of errors [1].

The Emergence of Ontology-Informed Metrics

Recognition of these limitations has spurred the development of biology-aware evaluation frameworks. The scGraph-OntoRWR metric was introduced alongside another ontology-informed metric called Lowest Common Ancestor Distance (LCAD), which measures the ontological proximity between misclassified cell types to assess the biological severity of annotation errors [1]. Together, these metrics introduce a biologically grounded perspective that was previously overlooked in scFM benchmarking.

Methodological Foundations of scGraph-OntoRWR

Conceptual Framework

The scGraph-OntoRWR metric is designed to quantitatively evaluate the consistency between the relational structure of cell types captured by scFM embeddings and the known biological relationships encoded in cell ontologies [4] [1]. The core premise is that an effective scFM should organize cells in its latent space such that the proximity between cell types reflects their established biological relationships.

Algorithmic Implementation

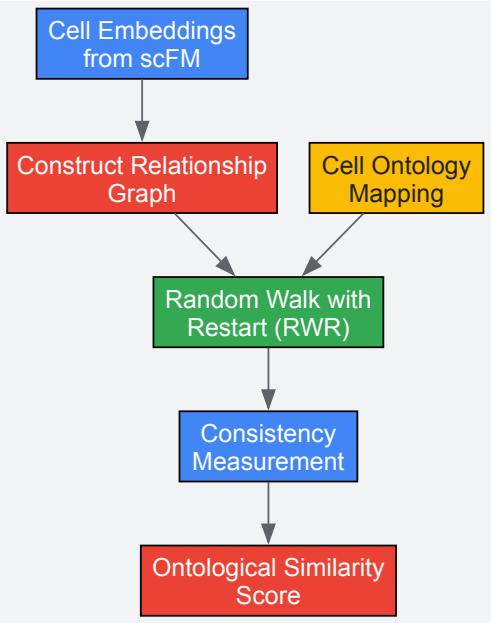
The metric operates through a multi-stage process that combines graph analysis and random walk with restart (RWR) algorithms:

- **Embedding Extraction**: Cell embeddings are obtained from scFMs in a zero-shot manner without task-specific fine-tuning.
- **Relationship Graph Construction**: A graph is built where nodes represent cell types, and edges are derived from proximity relationships in the embedding space.
- **Ontology Mapping**: Cell types are mapped to their corresponding terms in the Cell Ontology.
- **Random Walk with Restart**: The RWR algorithm propagates similarity scores through the ontology structure to quantify biological relatedness.
- **Consistency Measurement**: The algorithm measures the alignment between embedding-derived relationships and ontology-derived relationships.

Table 1: Key Computational Components of scGraph-OntoRWR

Component	Function	Implementation Notes
Cell Embeddings	Latent representations from scFMs	Extracted in zero-shot setting without fine-tuning
Relationship Graph	Captures proximity in embedding space	Graph structure varies by similarity threshold
Cell Ontology	Provides ground truth biological relationships	Standardized ontology ensures consistency
RWR Algorithm	Propagates biological similarity scores	Handles complex ontological relationships

The following diagram illustrates the core computational workflow of the scGraph-OntoRWR metric:



Experimental Protocols for scGraph-OntoRWR Implementation

Data Preparation and Preprocessing

Implementing scGraph-OntoRWR requires careful data preparation to ensure biologically meaningful evaluation:

- **Dataset Selection**: Curate diverse single-cell datasets with well-annotated cell types across different tissues and conditions. The original benchmark used five high-quality datasets with manual annotations varying in size and diversity, containing multiple sources of batch effects [1].

- **Cell Ontology Alignment**: Map each cell type to standardized Cell Ontology terms, ensuring consistent biological interpretation across datasets.

- **Embedding Extraction**: Extract cell embeddings from scFMs using zero-shot protocols to assess intrinsic model knowledge without task-specific adaptation [83].

Metric Calculation Procedure

The step-by-step protocol for calculating scGraph-OntoRWR scores:

- **Input Processing**:

    - Input: Cell embeddings from scFM (`embeddings`), cell type labels (`labels`), Cell Ontology structure (`ontology`)
    - Normalize embeddings using L2 normalization
    - Map cell type labels to ontology terms

- **Graph Construction**:
    - For each cell type, compute centroid in embedding space
    - Calculate pairwise cosine distances between centroids
    - Construct k-nearest neighbor graph (k=5) based on centroid distances

- **Ontological Similarity Calculation**:
    - Initialize RWR parameters: restart probability $r = 0.7$
    - For each cell type pair, execute RWR on ontology graph
    - Extract steady-state probabilities as similarity scores

- **Consistency Measurement**:
    - Compute Spearman correlation between graph distances and ontological similarities
    - Calculate alignment score using weighted Jaccard index
    - Derive final scGraph-OntoRWR score (higher values indicate better biological alignment)

Table 2: Evaluation Tasks and Datasets for scGraph-OntoRWR Validation

Task Category	Specific Tasks	Datasets Used	Evaluation Focus
Gene-Level Tasks	Gene function prediction, Tissue specificity	Multiple human tissue datasets	Functional consistency of gene embeddings
Cell-Level Tasks	Batch integration, Cell type annotation	Five datasets with diverse biological conditions	Biological structure preservation
Clinical Tasks	Cancer cell identification, Drug sensitivity	Seven cancer types, four drugs	Translational relevance

Integration into scFM Benchmarking Frameworks

Comprehensive Evaluation Strategy

scGraph-OntoRWR functions as part of a holistic benchmarking framework that incorporates multiple evaluation perspectives:

- **Unsupervised Metrics**: Assess intrinsic embedding quality without external labels.
- **Supervised Metrics**: Evaluate performance on specific tasks like classification.
- **Knowledge-Based Metrics**: scGraph-OntoRWR and LCAD that measure biological alignment [1].

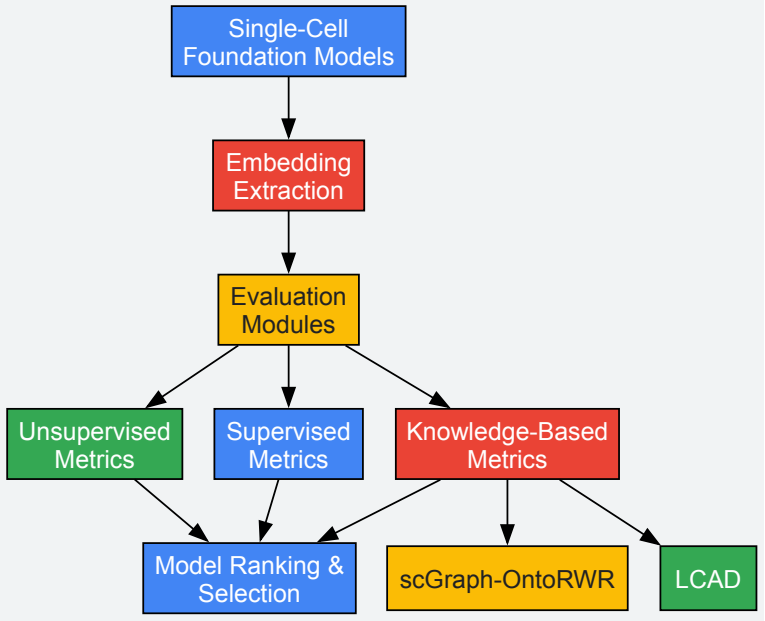
The integration of these perspectives provides a multidimensional view of scFM performance that balances technical capability with biological relevance.

Benchmarking Results and Insights

Application of scGraph-OntoRWR in large-scale benchmarks has revealed crucial insights about current scFMs:

- **No universal leader**: No single scFM consistently outperforms others across all tasks when biological relevance is considered [4] [1].
- **Task-model specificity**: The optimal model selection depends on specific task requirements and dataset characteristics.
- **Biological smoothness**: Performance improvements correlate with smoother cell-property landscapes in latent space, as quantified by the Roughness Index (ROGI) [1].

The following diagram illustrates how scGraph-OntoRWR integrates into a comprehensive scFM evaluation workflow:



Successful implementation of scGraph-OntoRWR requires access to specific computational resources and biological databases. The following table details essential components for establishing this evaluation framework:

Table 3: Essential Research Reagents and Resources for scGraph-OntoRWR Implementation

Resource Category	Specific Resources	Function/Purpose	Access Method
Single-Cell Data Repositories	CZ CELLxGENE, Human Cell Atlas, PanglaoDB	Provide diverse training and benchmarking data	Public download portals
Biological Ontologies	Cell Ontology, Gene Ontology	Standardized biological relationships for metric calculation	OBO Foundry, EMBL-EBI
scFM Implementations	scGPT, Geneformer, scBERT, UCE, scFoundation	Models to evaluate using scGraph-OntoRWR	GitHub repositories, model hubs
Benchmarking Frameworks	scFM-Bench [83]	Infrastructure for standardized evaluation	GitHub repository with implementation guidelines
Computational Environments	Python, PyTorch, TensorFlow	Execution environment for metric calculation	Conda environments, Docker containers

Advancing Biological Evaluation Metrics

While scGraph-OntoRWR represents a significant advancement in biological evaluation of scFMs, several directions for future development remain:

- **Multimodal Extensions**: Adapting the metric to evaluate multimodal single-cell data incorporating epigenomic, proteomic, and spatial information.
- **Dynamic Ontologies**: Incorporating temporal relationships and developmental trajectories into the evaluation framework.
- **Species-Agnostic Applications**: Extending the approach to cross-species evaluations requiring orthology mapping.

scGraph-OntoRWR has emerged as a critical tool for addressing one of the most pressing challenges in single-cell foundation model development: ensuring that computational advances translate to biologically meaningful insights. By providing a quantitative framework for measuring the alignment between learned representations and established biological knowledge, this metric moves the field beyond purely task-based evaluation toward more fundamental assessment of biological relevance.

As scFMs continue to grow in complexity and scale, ontology-informed metrics like scGraph-OntoRWR will play an increasingly vital role in guiding model selection, optimizing architectural decisions, and ultimately building computational tools that genuinely enhance our understanding of cellular biology and disease mechanisms. The integration of these biology-aware evaluation frameworks represents an essential step toward realizing the full potential of single-cell foundation models in both basic research and therapeutic development.

Selecting the right single-cell foundation model (scFM) is crucial for the success of downstream research and drug development projects. A comprehensive 2025 benchmark study of six prominent scFMs against established baselines reveals a key insight: no single scFM consistently outperforms others across all tasks. The decision to use a complex scFM or a simpler alternative depends on factors like **dataset size, task**

complexity, the need for biological interpretability, and available computational resources [4] [1]. This guide provides a structured approach to this selection process, synthesizing recent benchmarking results into actionable protocols.

Defining Your Project's Profile

The first step is to characterize your own project based on two primary axes: the scale of your dataset and the complexity of your biological question. The following table outlines recommended approaches for different scenarios.

Table 1: Model Selection Guide Based on Project Profile

Dataset Size	Task Complexity / Goal	Recommended Approach	Examples & Rationale
Small (≤ 10k cells)	**Simple Cell Type Annotation**	Standard ML baseline (e.g., Seurat)	Simple models adapt more efficiently to small, specific datasets with limited resources [4] [1].
	Batch Integration	Generative models (e.g., scVI) or baselines (Harmony)	These models are effective and computationally efficient for this core task on smaller sets [1].
Medium (10k - 100k cells)	**Exploring Novel Biological Insights**	Zero-shot embeddings from scFMs (e.g., scGPT, Geneformer)	Leverages biological knowledge pre-trained into scFMs, providing robustness [1].
	Clinically Relevant Prediction (e.g., cancer cell ID)	Fine-tuned scFMs	scFMs show strong performance on complex clinical tasks across diverse cancer types [4] [1].
Large (> 100k cells)	**Cell Atlas Construction**	Large-scale scFMs (e.g., scFoundation, CellFM)	Designed and pre-trained on millions to hundreds of millions of cells for broad generalization [3] [1].
	Cross-Species/Modality Analysis	Multimodal scFMs (e.g., PAST, SCARF, scGPT)	Models trained on diverse data types (RNA, ATAC, histology) can bridge modalities and species [3] [84].

Experimental Protocols for Evaluation

Once a candidate model is selected, a rigorous evaluation protocol is essential. The benchmark study employs a methodology focused on zero-shot embeddings to assess the intrinsic biological knowledge of a model before task-specific fine-tuning [1].

Protocol 1: Evaluating Zero-Shot Cell Embeddings for Data Integration

This protocol assesses how well a model's pre-trained cell embeddings can integrate datasets and remove technical noise without further training.

- **Feature Extraction**: Input your raw, unnormalized count matrix for each batch into the pre-trained scFM to extract cell embeddings without fine-tuning the model.
- **Neighborhood Graph Construction**: Construct a shared nearest neighbor (SNN) graph based on the cell embeddings in this latent space.
- **Metric Calculation**:
  - **Batch ASW (Batch Adjusted Rand Index)**: Measures batch mixing. Values range from 0 (poor) to 1 (good). Calculate the silhouette width of batches within the SNN graph. A higher score indicates better batch integration.
  - **Cell-type ASW (Cell-type Adjusted Rand Index)**: Measures biological conservation. Calculate the silhouette width of cell types within the SNN graph. A higher score indicates that biological variance is better preserved [1].
- **Biological Validation with scGraph-OntoRWR**: This novel metric evaluates whether the *relationships between cell types* in the latent space are consistent with established biological knowledge from cell ontologies. A higher score confirms the model is capturing biologically meaningful patterns [1].

Protocol 2: Evaluating Gene Embeddings for Functional Insight

This protocol evaluates the quality of a model's gene embeddings, which is crucial for tasks like perturbation prediction.

- **Gene Embedding Extraction**: Obtain the model's pre-trained gene vector for each gene from its input embedding layer.
- **Task Design**:
  - **Gene Ontology (GO) Term Prediction**: Frame it as a retrieval task. For a given GO term, the model should rank genes known to be associated with it higher than random genes.
  - **Tissue Specificity Prediction**: Evaluate if the embeddings can predict which genes are highly specific to certain tissues.
- **Metric Calculation**: Use metrics like **Area Under the Precision-Recall Curve (AUPRC)** to quantitatively evaluate the prediction performance, reflecting the functional relevance of the gene embeddings [1].

The Scientist's Toolkit: Research Reagent Solutions

The following table details key computational and data resources essential for working with single-cell foundation models.

Table 2: Essential Research Tools and Resources for scFM Workflows

Item Name	Function / Application	Specifications & Notes
Seurat (v5+)	R toolkit for single-cell analysis; often used as a baseline for integration and annotation.	Provides flexible, scalable workflows for large datasets and serves as a standard for benchmarking [85].
Harmony	Algorithm for dataset integration.	A strong, clustering-based baseline method for comparing against scFM integration performance [1].
scVI	Generative deep learning model for single-cell data.	Used for comparative benchmarking in integration and representation learning tasks [1].
CellxGene	Platform providing curated single-cell datasets.	A source of high-quality, independent datasets like the Asian Immune Diversity Atlas (AIDA) for unbiased validation and testing [1].
Gene Ontology (GO) Database	Repository of structured biological knowledge.	Serves as a ground truth for validating the biological relevance of gene and cell embeddings from scFMs [1].
Protein Data Bank (PDB)	Database of 3D protein structures.	Critical for structure-based drug discovery when scFM insights are translated into target identification [86].

Implementation Workflow and Decision Logic

The following diagram visualizes the end-to-end decision-making process for selecting and evaluating a single-cell foundation model, from problem definition to final deployment.

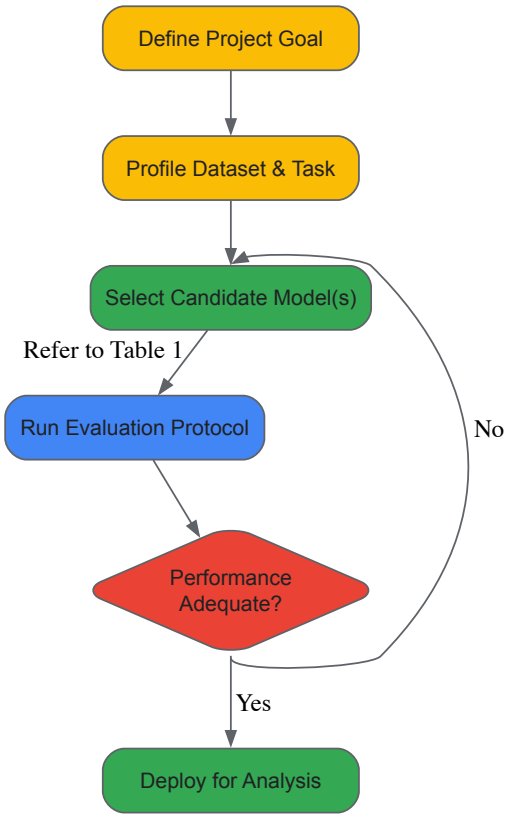


Diagram 1: scFM Selection and Evaluation Workflow.

Advanced Evaluation: The Role of Biological Metrics

Beyond standard performance metrics, the roughness index (ROGI) can serve as a powerful proxy for model selection. ROGI measures the smoothness of the cell-property landscape in a model's latent space. A **lower roughness index** indicates a smoother landscape, which makes it easier for a downstream classifier to learn and generally predicts better task-specific performance [1]. Calculating ROGI for candidate models on your dataset can provide a data-driven way to choose the most appropriate one.

Furthermore, for cell type annotation tasks, the **Lowest Common Ancestor Distance (LCAD)** metric is invaluable. Instead of treating all misclassifications equally, LCAD measures the ontological proximity between a misclassified cell and its true type. A lower LCAD score indicates a less severe error (e.g., mistaking two T-cell subtypes) versus a high LCAD error (e.g., mistaking a T-cell for a neuron). This provides a biologically-informed assessment of model performance [1].

Conclusion

Single-cell foundation models stand at a pivotal juncture, offering immense promise for unifying biological insight across vast datasets but facing significant challenges in reliability and interpretability. The key takeaway from current research is that no single scFM consistently outperforms all others; model selection must be tailored to specific tasks, dataset sizes, and available computational resources. While foundational models like scGPT demonstrate robust all-around capabilities, simpler methods can still be more effective for specific, narrow tasks, especially in zero-shot settings. The future of scFMs hinges on overcoming current limitations through improved pretraining strategies, enhanced biological grounding, and the development of standardized evaluation frameworks. Success in these areas will pave the way for scFMs to become indispensable tools in clinical research, enabling deeper insights into disease mechanisms, tumor microenvironments, and the development of personalized therapeutic strategies, ultimately bringing us closer to the vision of a comprehensive 'Virtual Cell'.

References

- 1. Biology-driven insights into the power of single-cell foundation models [https://genomebiology.biomedcentral.com/articles/10.1186/s13059-025-03781-6]
- 2. Transcoder-based Circuit Analysis for Interpretable Single-Cell... [https://arxiv.org/html/2509.14723v1]
- 3. GitHub - OmicsML/awesome- foundation - model - single - cell - papers [https://github.com/OmicsML/awesome-foundation-model-single-cell-papers]
- 4. Biology-driven insights into the power of single - cell foundation models [https://pubmed.ncbi.nlm.nih.gov/41044630/]
- 5. the Transformers of modern AI | Medium Architecture backbone [https://medium.com/expertminds/transformers-architecture-the-backbone-of-modern-ai-b4ad482202c0]
- 6. - Single : bringing artificial intelligence into cell... cell foundation models [https://www.nature.com/articles/s12276-025-01547-5?error=cookies_not_supported&code=2a219ee1-fa1b-4c94-a09d-2076346b9fd9]
- 7. Understanding Transformers : A Deep Dive into Architecture ... [https://atalupadhyay.wordpress.com/2025/02/12/understanding-transformers-a-deep-dive-into-architecture-attention-mechanisms-and-implementation/]
- 8. New Insights on Transformers and Feed-Forward Networks - Simple Science [https://scisimple.com/en/articles/2025-11-13-new-insights-on-transformers-and-feed-forward-networks--a9ny8gl]
- 9. Introduction to scRNA-Seq with R (Seurat) - Getting Started with scRNA-Seq Seminar Series [https://bioinformatics.ccr.cancer.gov/docs/getting-started-with-scrna-seq/IntroToR_Seurat/]
- 10. Single-cell foundation models: bringing artificial intelligence into cell biology | Experimental & Molecular Medicine [https://www.nature.com/articles/s12276-025-01547-5]
- 11. CZ CELLxGENE Discover Census | 2024-07-01 | Virtual Cells Platform [https://virtualcellmodels.cziscience.com/dataset/census]
- 12. CZ CELLxGENE Discover: a single - cell platform for scalable... data [https://pmc.ncbi.nlm.nih.gov/articles/PMC11701654/]
- 13. HCA Data Portal [https://data.humancellatlas.org/]
- 14. Public RNA-Seq and scRNA-Seq Databases: a Comparative Review [https://bigomics.ch/blog/ultimate-guide-to-public-rnaseq-and-sc-rna-seq-databases/]
- 15. Leveraging Public scRNA-seq Data: A Guide to Repositories and Resources [https://www.biomage.net/blog/public-scrna-seq-data]
- 16. - Registry of Human on AWS Cell Atlas Open Data [https://registry.opendata.aws/humancellatlas/]
- 17. A benchmark of batch - effect correction methods for single - cell ... RNA [https://genomebiology.biomedcentral.com/articles/10.1186/s13059-019-1850-9]
- 18. MASI enables fast model-free standardization and integration of... [https://pmc.ncbi.nlm.nih.gov/articles/PMC10144903/]
- 19. A benchmark of batch - effect correction methods for single - cell ... RNA [https://pubmed.ncbi.nlm.nih.gov/31948481/]
- 20. in Batch - effects - single ... | Nature Biotechnology cell RNA sequencing [https://www.nature.com/articles/nbt.4091?error=cookies_not_supported&code=58a993b1-9076-4056-a3ad-83de64ceee8c]
- 21. 6 Quality — Control - Single best practices cell [https://www.sc-best-practices.org/preprocessing_visualization/quality_control.html]
- 22. Current best practices in single-cell RNA-seq analysis: a tutorial - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC6582955/]
- 23. Common Considerations for Quality Filters for Control ... Single Cell [https://www.10xgenomics.com/analysis-guides/common-considerations-for-quality-control-filters-for-single-cell-rna-seq-data]
- 24. Scale-free and unbiased transformer with tokenization for cell type annotation from single-cell RNA-seq data - ScienceDirect [https://www.sciencedirect.com/science/article/abs/pii/S003132032500384X]
- 25. scGPT: toward building a foundation model for single-cell multi-omics using generative AI - PubMed [https://pubmed.ncbi.nlm.nih.gov/38409223/]

- 26. Frontiers | Single-cell technology for drug discovery and development [https://www.frontiersin.org/journals/drug-discovery/articles/10.3389/fddsv.2024.1459962/full]
- 27. scDrugMap: Benchmarking Large Foundation Models for Drug... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12083700/]
- 28. Robust self - supervised learning strategy to tackle the inherent... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11568879/]
- 29. GitHub - bowang-lab/scGPT [https://github.com/bowang-lab/scGPT]
- 30. GitHub - devvirani13/Benchmarking-Study-of-Single-Cell... [https://github.com/devvirani13/Benchmarking-Study-of-Single-Cell-Foundation-Models-for-Cell-Annotation-and-Batch-Integration]
- 31. Identification of drug-resistant individual cells within tumors by semi-supervised transfer learning from bulk to single-cell transcriptome | Communications Biology [https://www.nature.com/articles/s42003-025-07959-3]
- 32. Predicting drug responses of unseen cell types through transfer learning with foundation models - PubMed [https://pubmed.ncbi.nlm.nih.gov/41044387/]
- 33. - Single : bringing artificial intelligence into cell... cell foundation models [https://www.nature.com/articles/s12276-025-01547-5?error=cookies_not_supported&code=a5069175-7780-4c9c-9dd1-1958ed8f1041]
- 34. Batch alignment of single-cell transcriptomics data using deep metric learning | Nature Communications [https://www.nature.com/articles/s41467-023-36635-5]
- 35. scMamba: A Scalable Foundation Model for Single-Cell Multi-Omics Integration Beyond Highly Variable Feature Selection [https://arxiv.org/html/2506.20697v1]
- 36. New machine learning-based single-cell search engine makes cell annotation faster, more efficient | Broad Institute [https://www.broadinstitute.org/news/new-machine-learning-based-single-cell-search-engine-makes-cell-annotation-faster-more]
- 37. Recovery of biological signals lost in single-cell batch integration with CellANOVA | Nature Biotechnology [https://www.nature.com/articles/s41587-024-02463-1]
- 38. Preprocessing and clustering — scanpy [https://scanpy.readthedocs.io/en/stable/tutorials/basics/clustering.html]
- 39. of unseen... | Nature Computational Science Predicting drug responses [https://www.nature.com/articles/s43588-025-00887-6?error=cookies_not_supported&code=e1ab3046-4e4c-4dce-bf93-89ab5a19ad54]
- 40. Efficient Fine-Tuning of Single-Cell Foundation Models Enables Zero-Shot Molecular Perturbation Prediction [https://arxiv.org/html/2412.13478v2]
- 41. Nicheformer: a foundation model for single-cell and spatial omics | Nature Methods [https://www.nature.com/articles/s41592-025-02814-z]
- 42. GitHub - theislab/nicheformer: Repository for Nicheformer: a foundation model for single-cell and spatial omics [https://github.com/theislab/nicheformer]
- 43. GitHub - saeyslab/nichenetr: NicheNet : predict active ligand-target... [https://github.com/saeyslab/nichenetr]
- 44. Unraveling cell–cell communication with NicheNet ... | Nature Protocols [https://www.nature.com/articles/s41596-024-01121-9?error=cookies_not_supported&code=e13cc82b-a1ba-459f-a059-b1938cc842aa]
- 45. : modeling intercellular communication by linking ligands to... NicheNet [https://pubmed.ncbi.nlm.nih.gov/31819264/]
- 46. Applications of single-cell technologies in drug discovery for tumor treatment - ScienceDirect [https://www.sciencedirect.com/science/article/pii/S2589004224017115]
- 47. - Zero reveals limitations of shot - evaluation ... single cell foundation [https://genomebiology.biomedcentral.com/articles/10.1186/s13059-025-03574-x]
- 48. PertEval-scFM: Benchmarking Single-Cell Foundation... | OpenReview [https://openreview.net/forum?id=t04D9bkKUq]
- 49. - Zero reveals limitations of shot in... evaluation foundation models [https://www.microsoft.com/en-us/research/articles/zero-shot-evaluation-reveals-limitations-of-foundation-models-in-single-cell-biology/]
- 50. One-Embedding-Fits-All: Efficient Zero - Shot Time Series Forecasting... [https://arxiv.org/html/2509.04208v1]
- 51. in scRNA-seq based on biological- Batch ... effects correction noise [https://pubmed.ncbi.nlm.nih.gov/39461162/]
- 52. Comprehensive evaluation of noise reduction methods for single-cell RNA sequencing data - PubMed [https://pubmed.ncbi.nlm.nih.gov/35048125/]
- 53. Comprehensive evaluation of noise reduction methods for single-cell... [https://pmc.ncbi.nlm.nih.gov/articles/PMC8921632/]
- 54. - Single Centric Biomedical Cell for Cancer « DBIS Foundation Models [https://dbis.rwth-aachen.de/dbis/index.php/2024/single-cell-centric-biomedical-foundation-models-for-cancer/]
- 55. Transformation, Normalization , and Batch Removal Effect [https://bio-protocol.org/exchange/protocoldetail?id=4462&type=1]
- 56. Accelerating Single-Cell Deep Learning with scDataset and Tahoe-100M [https://blog.tahoebio.ai/p/accelerating-single-cell-deep-learning]
- 57. Teaching machines the language of biology: Scaling large language models for next-generation single-cell analysis [https://research.google/blog/teaching-machines-the-language-of-biology-scaling-large-language-models-for-next-generation-single-cell-analysis/]
- 58. A Study of Optimizations for Fine-tuning Large Language Models [https://arxiv.org/html/2406.02290v2]
- 59. Supervised Fine-Tuning: What It Is and Key Techniques [https://www.sapien.io/blog/what-is-supervised-fine-tuning-overview-and-techniques]
- 60. [2510.25807] Discovering Interpretable Biological Concepts in... [https://arxiv.org/abs/2510.25807]

- 61. Interpretability research of deep learning: A literature survey - ScienceDirect [https://www.sciencedirect.com/science/article/abs/pii/S1566253524004998]
- 62. LLMs for Explainable AI: A Comprehensive Survey [https://arxiv.org/html/2504.00125v1]
- 63. Interpretable deep of learning - single and epigenetic data reveals... cell [https://www.nature.com/articles/s41598-025-89646-1?error=cookies_not_supported&code=f4c989e7-7819-42a5-ac90-9cf03855acf1]
- 64. scDrugMap: Benchmarking Large Foundation Models for Drug Response Prediction - PubMed [https://pubmed.ncbi.nlm.nih.gov/40386575/]
- 65. Generating colorblind-friendly scatter plots for single-cell data | eLife [https://elifesciences.org/articles/82128]
- 66. : A BioLLM for integrating and benchmarking... standardized framework [https://pubmed.ncbi.nlm.nih.gov/40843339/]
- 67. GitHub - BGIResearch/ BioLLM [https://github.com/BGIResearch/BioLLM]
- 68. BioLLM: A standardized framework for integrating and benchmarking single-cell foundation models - ScienceDirect [https://www.sciencedirect.com/science/article/pii/S2666389925001746]
- 69. Welcome to BioLLM 's documentation! — BioLLM 0.1.1 documentation [https://biollm.readthedocs.io/]
- 70. Parameter-Efficient Fine-Tuning Enhances Adaptation of Single Cell Large Language Model for Cell Type Identification - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC10862733/]
- 71. [2412.13478] Efficient Fine - Tuning of Single - Cell ... Foundation Models [https://arxiv.org/abs/2412.13478]
- 72. Reusability report: Learning the transcriptional grammar in single-cell RNA-sequencing data using transformers | Nature Machine Intelligence [https://www.nature.com/articles/s42256-023-00757-8]
- 73. A Survey on Foundation Language Models for Single - cell Biology [https://aclanthology.org/2025.acl-long.26/]
- 74. scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data | Nature Machine Intelligence [https://www.nature.com/articles/s42256-022-00534-z]
- 75. GitHub - bowang-lab/ scGPT -spatial [https://github.com/bowang-lab/scGPT-spatial]
- 76. Benchmarking Geneformer — NVIDIA BioNeMo Framework [https://docs.nvidia.com/bionemo-framework/1.10/notebooks/Geneformer-celltype-classification-example.html]
- 77. Benchmarking Single - Cell for Perturbation Effect... Foundation Models [https://icml.cc/virtual/2025/poster/43799]
- 78. The art of seeing the elephant in the room: 2D embeddings of... [https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1012403]
- 79. Evaluating Single : A New Approach - Simple Science Cell Embeddings [https://scisimple.com/en/articles/2025-11-02-evaluating-single-cell-embeddings-a-new-approach--ak4qr88]
- 80. 12. Data integration — Single-cell best practices [https://www.sc-best-practices.org/cellular_structure/integration.html]
- 81. scE2TM: Toward Interpretable Single - Cell via Topic... Embedding [https://arxiv.org/html/2507.08355v1]
- 82. Signal recovery in single cell batch integration - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC10197537/]
- 83. GitHub - wujialu/scFM-Bench [https://github.com/wujialu/scFM-Bench]
- 84. [2507.06418] PAST: A multimodal single - cell for... foundation model [https://arxiv.org/abs/2507.06418]
- 85. Tools for Single Cell Genomics • Seurat [https://satijalab.org/seurat/]
- 86. 7 Limitations of Molecular Docking & Computer Aided Drug Design and Discovery [https://amitray.com/7-limitations-of-molecular-docking-computer-aided-drug-design-and-discovery/]